DRAWING LIST FOR TEST PACKAGE

H2SO4-502

PROJECT NO.: 22328D

PROJECT TITLE : M15X Ph2 Project

No	PKG No	UNIT	LINE NUMBER			P&ID		REMARK
			ISO DRAWING No	SHEET	REV	P&ID No	REV	
1	502	4FA	25A-H2SO4-ST4-N-6514-01	1 OF 1	0	CJ-M15W14-PR-A20-000145		
2	502	4FA	25A-H2SO4-ST4-N-6515-01	1 OF 1	0	CJ-M15W14-PR-A20-000145		
3	502	4FA	25A-H2SO4-ST4-N-6516-01	1 OF 1	0	CJ-M15W14-PR-A20-000145		
4	502	4FA	25A-H2SO4-ST4-N-6517-01	1 OF 1	0	CJ-M15W14-PR-A20-000145		
5	502	4FA	25A-H2SO4-ST4-N-6518-01	1 OF 1	0	CJ-M15W14-PR-A20-000145		
6	502	4FA	25A-H2SO4-ST4-N-6519-01	1 OF 1	0	CJ-M15W14-PR-A20-000145		
7	502	4FA	25A-H2SO4-ST4-N-6520-01	1 OF 1	0	CJ-M15W14-PR-A20-000144		
8	502	4FA	25A-H2SO4-ST4-N-6521-01	1 OF 1	1	CJ-M15W14-PR-A20-000144		
9	502	4FA	25A-H2SO4-ST4-N-6538-01	1 OF 1	1	CJ-M15W14-PR-A20-000145		
10	502	4FA	25A-H2SO4-ST4-N-6539-01	1 OF 1	0	CJ-M15W14-PR-A20-000145		
11	502	4FA	25A-H2SO4-ST4-N-6541-01	1 OF 1	0	CJ-M15W14-PR-A20-000145		
12	502	4FA	25A-H2SO4-ST4-N-6542-01	1 OF 1	0	CJ-M15W14-PR-A20-000144		
13	502	4FA	25A-H2SO4-ST4-N-6543-01	1 OF 1	0	CJ-M15W14-PR-A20-000144		
14	502	4FA	25A-H2SO4-ST4-N-6544-01	1 OF 1	0	CJ-M15W14-PR-A20-000144		
15	502	4FA	25A-H2SO4-ST4-N-6551-01	1 OF 1	0	CJ-M15W14-PR-A20-000144		
16	502	4FA	25A-H2SO4-ST4-N-6552-01	1 OF 1	0	CJ-M15W14-PR-A20-000144		
17	502	4FA	25A-H2SO4-ST4-N-6553-01	1 OF 1	0	CJ-M15W14-PR-A20-000144		
18	502	4FA	25A-H2SO4-ST4-N-6554-01	1 OF 1	0	CJ-M15W14-PR-A20-000146		
19	502	4FA	25A-H2SO4-ST4-N-6555-01	1 OF 1	0	CJ-M15W14-PR-A20-000146		
20	502	4FA	25A-H2SO4-ST4-N-6556-01	1 OF 1	0	CJ-M15W14-PR-A20-000146		

DRAWING LIST FOR TEST PACKAGE

H2SO4-502

PROJECT NO.: 22328D

PROJECT TITLE : M15X Ph2 Project

No	PKG No	UNIT	LINE NUMBER			P&ID		REMARK
			ISO DRAWING No	SHEET	REV	P&ID No	REV	
21	502	4FA	25A-H2SO4-ST4-N-6559-01	1 OF 5	0	CJ-M15W14-PR-A20-000144/145/146/147		
22	502	4FA	25A-H2SO4-ST4-N-6559-02	2 OF 5	0	CJ-M15W14-PR-A20-000144/145/146/147		
23	502	4FA	25A-H2SO4-ST4-N-6559-03	3 OF 5	0	CJ-M15W14-PR-A20-000144/145/146/147		
24	502	4FA	25A-H2SO4-ST4-N-6559-04	4 OF 5	0	CJ-M15W14-PR-A20-000144/145/146/147		
25	502	4FA	25A-H2SO4-ST4-N-6559-05	5 OF 5	0	CJ-M15W14-PR-A20-000144/145/146/147		
26	502	4FA	25A-H2SO4-ST4-N-6570-01	1 OF 1	0	CJ-M15W14-PR-A20-000147		
27	502	4FA	25A-H2SO4-ST4-N-6571-01	1 OF 1	0	CJ-M15W14-PR-A20-000147		
28	502	4FA	25A-H2SO4-ST4-N-6572-01	1 OF 1	0	CJ-M15W14-PR-A20-000147		
29	502	4FA	25A-H2SO4-PF1-N-6514-01	1 OF 2	0	CJ-M15W14-PR-A20-000145		
30	502	4FA	25A-H2SO4-PF1-N-6515-01	1 OF 2	0	CJ-M15W14-PR-A20-000145		
31	502	4FA	25A-H2SO4-PF1-N-6517-01	1 OF 2	0	CJ-M15W14-PR-A20-000145		
32	502	4FA	25A-H2SO4-PF1-N-6518-01	1 OF 2	0	CJ-M15W14-PR-A20-000145		
33	502	4FA	25A-H2SO4-PF1-N-6538-01	1 OF 2	0	CJ-M15W14-PR-A20-000145		
34	502	4FA	25A-H2SO4-PF1-N-6539-01	1 OF 2	0	CJ-M15W14-PR-A20-000145		
35	502	4FA	25A-H2SO4-PF1-N-6542-01	1 OF 2	0	CJ-M15W14-PR-A20-000144		
36	502	4FA	25A-H2SO4-PF1-N-6543-01	1 OF 2	0	CJ-M15W14-PR-A20-000144		
37	502	4FA	25A-H2SO4-PF1-N-6551-01	1 OF 2	0	CJ-M15W14-PR-A20-000144		
38	502	4FA	25A-H2SO4-PF1-N-6552-01	1 OF 2	0	CJ-M15W14-PR-A20-000144		
39	502	4FA	25A-H2SO4-PF1-N-6554-01	1 OF 2	0	CJ-M15W14-PR-A20-000146		
40	502	4FA	25A-H2SO4-PF1-N-6555-01	1 OF 2	0	CJ-M15W14-PR-A20-000146		

DRAWING LIST FOR TEST PACKAGE

H2SO4-502

PROJECT NO.: 22328D

PROJECT TITLE : M15X Ph2 Project

No	PKG No	UNIT	LINE NUMBER			P&ID		REMARK
			ISO DRAWING No	SHEET	REV	P&ID No	REV	
41	502	4FA	25A-H2SO4-PF1-N-6570-01	1 OF 2	0	CJ-M15W14-PR-A20-000147		
42	502	4FA	25A-H2SO4-PF1-N-6571-01	1 OF 2	0	CJ-M15W14-PR-A20-000147		

ITEM NO.		W14-T-H2SO4-4602 A/B
SERVICE		H2SO4 STORAGE TANK
Q'TY		2
TYPE		VERTICAL & CYLINDRICAL
SIZE	ID x H m	3.4 x 4.5 (EH(4.2)) 2A
CAPA.	m3	41 (EH(38)) 2A
MAT'L		FRP + TEFLON

대외비

KEY PLAN (4F)

H2SO4-502

NOTE

1. 온전 시 CLOSE 되는 배관에는 LOTO 설치.
2. DIAPHRAGM PUMP MAT'L 표기 : HEAD/DIAPHRAGM/CHECK BALL
3. 약품종류에 따라 LINE 색상을 구분하여야 한다.
4. CHEMICAL UNLOADING STATION에서 LEVEL을 볼 수 있도록 한다.
5. CHEMICAL STORAGE TANK & PUMP CURB에서 발생되는 DRAIN은 인근의 TRENCH로 집수되어, DRAIN HEADER로 이송된다.
6. PUMP VENDOR 공급.
7. 황산 LINE에 위치한 VALVE에 LOTO 적용.
8. H2SO4 STORAGE TANK VENT는 DESICCANT (SILICAGEL POT 등)을 설치할 것.
9. HS-W1202, W1203 중 선택운전.
10. HS-W1001A, W1001B 중 선택운전.
11. INTERLOCK NO. I-4655 (W13-CTP-H2SO4-4672 AUTO STOP) : REMOTE STATUS에서 W14-T-H2SO4-4602 A/B LEVEL (LI-W6053A/B) HIGH HIGH STOP.
12. INTERLOCK NO. I-4615 (W14-CDP-H2SO4-4610 A/B/C, W14-CDP-H2SO4-4611 A/B/C, W14-CDP-H2SO4-4612 A/B/C, W14-CDP-H2SO4-4613 A/B/C, W14-CDP-H2SO4-4614 A/B/C, W14-CDP-H2SO4-4615 A/B/C, W14-CDP-H2SO4-4616 A/B/C, W14-CDP-H2SO4-4618 A/B/C, W14-CDP-H2SO4-4623 A/B/C AUTO STOP) : REMOTE STATUS에서 W14-T-H2SO4-4602 LEVEL (LI-W6054A/B) LOW LOW STOP. ALL SPARE PUMP는 pH VALUE LOW STOP.
13. 모든 황산 배관의 LEAK POINT에 대해서는 비산 방지 SHIELD 및 SENSING TAPE 설치할 것.
14. 황산 펌프 AREA에는 비산방지막을 설치한다.
15. PF1 LINE의 CLEAR PVC(이중관)은 40A로 적용.
16. DELETED.
17. OPERATING CAPACITY : 5 LPM OPERATING PRESSURE : 0.6 MPa
18. OPERATING CAPACITY : 10 LPM OPERATING PRESSURE : 0.6 MPa

REV.	DESCRIPTION	DATE
2A	REVISED AFC	25.08.14
2	REVISED AFC	25.01.17
1	REVISED AFC	24.10.31

제 도 DRAWN BY	이 인 학
검 토 CHECKED BY	한 진 희
승 인 APPROVED BY	박 동 영
제 작 사	SK 에코플랜트
공 사 명 PROJECT TITLE	M15X Ph4 WWT Project
도면명 DWG. TITLE	PIPING & INSTRUMENT DIAGRAM H2SO4 STORAGE TANK & FEED PUMP (1/4)
축 척 SCALE	NONE
날 짜 DATE	2025. 08. 14
PROJECT NO.	22330D
도면번호 SHEET NO.	CJ-M15W14-PR-A20-000144
REV.	2A

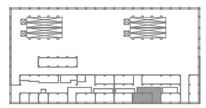
ITEM NO.	W14-CDP-H2SO4-4613 A/B/C
SERVICE	HF 2'nd H2SO4 FEED PUMP
Q'TY	2 + 1
TYPE	DIAPHRAGM
CAPA. l/min.ea x MPa	(MAX. 11.2 x MAX 0.7 NOTE 17) 2A
MAT'L	PVDF/EPDM+PTFE/CERAMIC
POWER kW	(0.75) 2A

ITEM NO.	W14-CDP-H2SO4-4614 A/B/C
SERVICE	CIF 2'nd H2SO4 FEED PUMP
Q'TY	2 + 1
TYPE	DIAPHRAGM
CAPA. l/min.ea x MPa	(MAX. 11.2 x MAX 0.7 NOTE 18) 2A
MAT'L	PVDF/EPDM+PTFE/CERAMIC
POWER kW	(0.75) 2A

C:₩Users₩TOPW₩Desktop₩P&ID251013₩[M15X Ph-4 for 입찰가]CAD FILE_0910₩CJ-M15W14-PR-A20-000144 R2 P&ID FOR H2SO

H2SO4-502

대외비



KEY PLAN (4F)

NOTE

1. 온전 시 CLOSE되는 배관에는 LOTO 실시.
2. DIAPHRAGM PUMP
   MAT'L 표기 : HEAD / DIAPHRAGM/ CHECK BALL
   약품종류에 따라 LINE 색상을 구분하여야 한다.
   CHEMICAL STORAGE TANK & PUMP CURB에서 발생되는 DRAIN은 인근의 TRENCH로 집수되어, DRAIN HEADER로 이송된다.
5. PUMP VENDOR 공급.
6. 황산 LINE에 위치한 VALVE에 LOTO 적용.
7. AIC-W1054A/B 중 선택운전.
8. AIC-W1257, W1265 중 선택운전.
9. N-STRIPPER PACKAGE TRAIN A/B 중 선택운전.
10. 모든 황산 배관의 LEAK POINT에 대해서는 비산 방지 SHIELD 및 SENSING TAPE 설치할 것.
11. 황산 펌프 AREA에는 비산방지막을 설치한다.
12. PF1 LINE의 CLEAR PVC(이중관)은 40A로 적용.
13. OPERATING CAPACITY : 5 LPM
    OPERATING PRESSURE : 0.6 MPa
14. OPERATING CAPACITY : 10 LPM
    OPERATING PRESSURE : 0.6 MPa

ITEM NO.	W14-CDP-H2SO4-4610 A/B/C
SERVICE	HF 1'st H2SO4 FEED PUMP
Q'TY	2 + 1
TYPE	DIAPHRAGM
CAPA. l/min x MPa	1A (MAX. 11.2 x MAX 0.7 NOTE 13)
MAT'L	PVDF/EPDM+PTFE/CERAMIC
POWER kW	0.75 1A

ITEM NO.	W14-CDP-H2SO4-4611 A/B/C
SERVICE	CIF 1'st H2SO4 FEED PUMP
Q'TY	2 + 1
TYPE	DIAPHRAGM
CAPA. l/min x MPa	1A (MAX. 11.2 x MAX 0.7 NOTE 14)
MAT'L	PVDF/EPDM+PTFE/CERAMIC
POWER kW	0.75 1A

ITEM NO.	W14-CDP-H2SO4-4612 A/B/C
SERVICE	AMMONIA ABSORBER TOWER H2SO4 FEED PUMP
Q'TY	2 + 1
TYPE	DIAPHRAGM
CAPA. l/min.ea x MPa	1A (MAX. 11.2 x MAX 0.7 NOTE 14)
MAT'L	PVDF/EPDM+PTFE/CERAMIC
POWER kW	0.75 1A

REV.	DESCRIPTION	DATE
1A	REVISED AFC	25.08.14
1	REVISED AFC	25.01.17
0	APPROVAL FOR CONSTRUCTION	24.09.30

제 도 DRAWN BY	이 인 학
검 토 CHECKED BY	한 진 희
승 인 APPROVED BY	박 동 영
제 작 사	SK 에코플랜트
공 사 명 PROJECT TITLE	M15X Ph4 WWT Project
도면명 DWG. TITLE	PIPING & INSTRUMENT DIAGRAM H2SO4 STORAGE TANK & FEED PUMP (2/4)
축 척 SCALE	NONE
날 짜 DATE	2025. 08. 14
PROJECT NO.	22330D
도면번호 SHEET NO.	CJ-M15W14-PR-A20-000145 REV. 1A

C:₩Users₩TOPWDesktop₩P&ID251013₩[M15X Ph-4 for 인허가]CAD FILE_0910₩CJ-M15W14-PR-A20-000145 R1 P&ID FOR H2SO

H2SO4-502

대외비

KEY PLAN (4F)

NOTE
1. 운전 시 CLOSE 되는 배관에는 LOTO 설치.
2. DIAPHRAGM PUMP
MAT'L 표기 : HEAD / DIAPHRAGM/ CHECK BALL
3. 약품종류에 따라 LINE 색상을 구분하여야 한다.
4. CHEMICAL STORAGE TANK & PUMP CURB에서 발생되는 DRAIN은 인근의 TRENCH로 집수되어, DRAIN HEADER로 이송된다.
5. PUMP VENDOR 공급.
6. 황산 LINE에 위치한 VALVE에 LOTO 적용.
7. AIC-W2251, W2261 중 선택운전.
8. 모든 황산 배관의 LEAK POINT에 대해서는 비산 방지 SHIELD 및 SENSING TAPE 설치할 것.
9. 황산 펌프 AREA에는 비산방지막을 설치한다.
10. PF1 LINE의 CLEAR PVC(이중관)은 40A로 적용.
11. OPERATING CAPACITY : 0.2 LPM
OPERATING PRESSURE : 0.6 MPa (2A)

CONTROL
M15W14-PR-A20-000145
FROM I-4615

503 H2SO4 M15W14-PR-A20-000145 FROM H2SO4 HEADER
100A-H2SO4-ST4-N-6522

503 H2SO4 M15W14-PR-A20-000144 FROM H2SO4 STORAGE TANK W14-T-H2SO4-4602 B
100A-H2SO4-ST4-N-6507

503 H2SO4 M15W14-PR-A20-000147 TO H2SO4 HEADER
100A-H2SO4-ST4-N-6507

503 H2SO4 M15W14-PR-A20-000147 TO H2SO4 HEADER
100A-H2SO4-ST4-N-6522

CONTROL
M15W14-PR-A20-000147
TO H2SO4 FEED PUMP

502 H2SO4 M15W14-PR-A20-000147 TO H2SO4 STORAGE TANK
25A-H2SO4-ST4-N-6559

623 H2SO4 M15W14-PR-A20-000112 TO CIF MBR NEUTRALIZATION TANK W14-T-4222A
OD25-H2SO4-PF1-N-6554 NOTE 10

CONTROL
M15W14-PR-A20-000112
FROM AIC-W2251

624 H2SO4 M15W14-PR-A20-000114 TO CIF MBR NEUTRALIZATION TANK W14-T-4222B
OD25-H2SO4-PF1-N-6555 NOTE 10

CONTROL
M15W14-PR-A20-000114
FROM AIC-W2261

506 H2SO4 M15W14-PR-A20-000317 FROM REUSE H2SO4 FEED PUMP W14-CDP-H2SO4-4755/4756/4757
25A-H2SO4-ST4-N-7012

W14-CDP-H2SO4-4615 A
W14-CDP-H2SO4-4615 B
W14-CDP-H2SO4-4615 C

PG W46958 / PG W46959 / PG W46950
SET P. : 0.66 MPa.g NOTE 5
PSV W46958 / PSV W46959 / PSV W46950
25A-H2SO4-ST4-N-6554
25A-H2SO4-ST4-N-6555
25A-H2SO4-ST4-N-6556
25A-H2SO4-ST4-N-6548
25A-H2SO4-ST4-N-6549
25A-H2SO4-ST4-N-6550
20A
25A
B / A
M
VFD
H : START
L : STOP
NOTE 7
HS W46150
ST4 PF1
PF1 ST4
PH4 REUSE

LDA W64711
LDS W64710
DIKE
FRP LINING
NOTE 4
CURB

ITEM NO.	W14-CDP-H2SO4-4615 A/B/C
SERVICE	CIF ORGANIC H2SO4 FEED PUMP
Q'TY	2 + 1
TYPE	DIAPHRAGM
CAPA. l/min.ea x MPa	MAX. 0.24 x MAX 1.0 NOTE 11 (2A)
MAT'L	PVDF/EPDM+PTFE/CERAMIC
POWER kW	(0.4 / 0.55) (2A)

REV.	수정 DESCRIPTION	날짜 DATE
2A	REVISED AFC	25.08.14
2	REVISED AFC	25.01.17
1	REVISED AFC	24.10.31

제도 DRAWN BY	이 인 학
검토 CHECKED BY	한 진 희
승인 APPROVED BY	박 동 영
제작사	SK 에코플랜트
공사명 PROJECT TITLE	M15X Ph4 WWT Project
도면명 DWG. TITLE	PIPING & INSTRUMENT DIAGRAM H2SO4 STORAGE TANK & FEED PUMP (3/4)
축척 SCALE	NONE
날짜 DATE	2025. 08. 14
PROJECT NO.	22330D
도면번호 SHEET NO.	CJ-M15W14-PR-A20-000146
REV.	2A

A1(594x841mm ±2mm)

C:₩Users₩TOPWDesktop₩P&ID251013₩[M15X Ph-4 for 인허가]CAD FILE_0910₩CJ-M15W14-PR-A20-000146 R2 P&ID FOR H2SO
본 도면의 저작권은 SK HHI/E&C 가 보유하며, 사전 동의 없이 무단 복제, 배포, 수정, 제3자에게 제공 또는 상업적 목적으로 사용할 수 없으며, 위반 시 민·형사상 책임을 질 수 있습니다.

H2SO4-502

H2SO4-502

502 H2SO4 M15W14-PR-A20-000146 FROM H2SO4 FEED PUMP — 25A-H2SO4-ST4-N-6559

CONTROL M15W14-PR-A20-000146 FROM I-4615

625 H2SO4 M15W14-PR-A20-000120 TO W.W COAGULATION TANK W14-T-4193 A — OD25-H2SO4-PF1-N-6570 NOTE 11

CONTROL M15W14-PR-A20-000120 FROM HS-W1902

626 H2SO4 M15W14-PR-A20-000121 TO W.W COAGULATION TANK W14-T-4193 B — OD25-H2SO4-PF1-N-6571 NOTE 11

CONTROL M15W14-PR-A20-000121 FROM HS-W1903

ST4 PF1 / PF1 ST4 / PF1 ST4 / PF1 ST4

25A / 20A

PG W46986 2A

25A-H2SO4-ST4-N-6570

25A-H2SO4-ST4-N-6571

25A-H2SO4-ST4-N-6572

SET P. : 0.66 MPa.g 1A NOTE 5 PSV W46984

SET P. : 0.66 MPa.g 1A NOTE 5 PSV W46985

SET P. : 0.66 MPa.g 1A NOTE 5 PSV W46987

B VFD M

A M

NOTE 8 HS W46187 H : START L : STOP

25A-H2SO4-ST4-N-6564 — W14-CDP-H2SO4-4618 A

25A-H2SO4-ST4-N-6565 — W14-CDP-H2SO4-4618 B

25A-H2SO4-ST4-N-6566 — W14-CDP-H2SO4-4618 C

LDA W6468 / LDS W6468

DIKE

FRP LINING NOTE 4 CURB

ITEM NO.	W14-CDP-H2SO4-4618 A/B/C
SERVICE	W.W H2SO4 FEED PUMP
Q'TY	2 + 1
TYPE	DIAPHRAGM
CAPA. l/min.ea x MPa	(MAX. 0.52 x MAX 1.0 NOTE 12) 1A
MAT'L	PVDF/EPDM+PTFE/CERAMIC
POWER kW	(0.4 / 0.55) 1A

H2SO4-502

H2SO4-502

25A-H2SO4-ST4-N-6559 — 502 H2SO4 M15W14-PR-A20-000145 TO H2SO4 STORAGE TANK

100A-H2SO4-ST4-N-6522 — 503 H2SO4 M15W14-PR-A20-000146 FROM H2SO4 HEADER

100A-H2SO4-ST4-N-6507 — 503 H2SO4 M15W14-PR-A20-000146 FROM H2SO4 HEADER

25A / 25A / 25A

대외비

KEY PLAN (4F)

NOTE

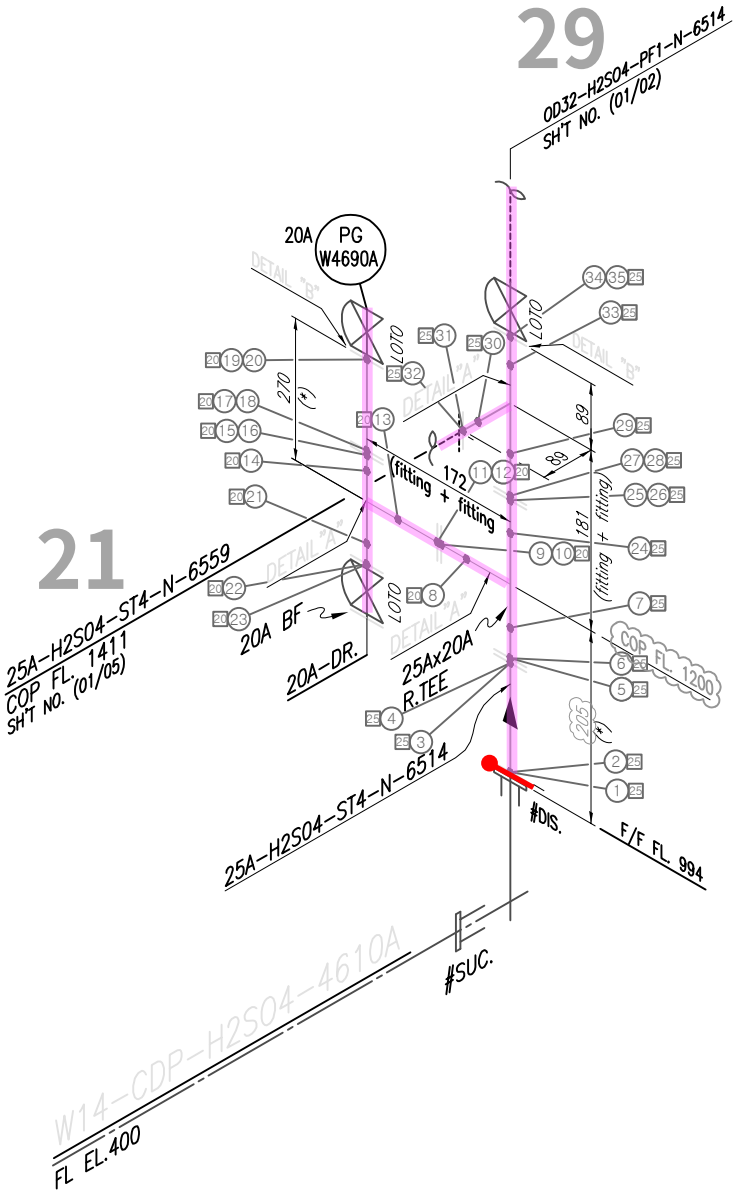
1. 온전 시 CLOSE 되는 배관에는 LOTO 설치.
2. DIAPHRAGM PUMP
   MAT'L 표기 : HEAD / DIAPHRAGM/ CHECK BALL
3. 약품종류에 따라 LINE 색상을 구분하여야 한다.
4. CHEMICAL STORAGE TANK & PUMP CURB에서 발생되는 DRAIN은 인근의 TRENCH로 집수되어, DRAIN HEADER로 이송된다.
5. PUMP VENDOR 공급.
6. 황산 LINE에 위치한 VALVE에 LOTO 적용.
7. AIC-W8062, W8064 중 선택운전.
8. HS-W1902, W1903 중 선택운전.
9. 모든 황산 배관의 LEAK POINT에 대해서는 비산 방지 SHIELD 및 SENSING TAPE 설치할 것.
10. 황산 펌프 AREA에는 비산방지막을 설치한다.
11. PF1 LINE의 CLEAR PVC(이중관)은 40A로 적용.
12. OPERATING CAPACITY : 0.5 LPM
    OPERATING PRESSURE : 0.6 MPa 1A

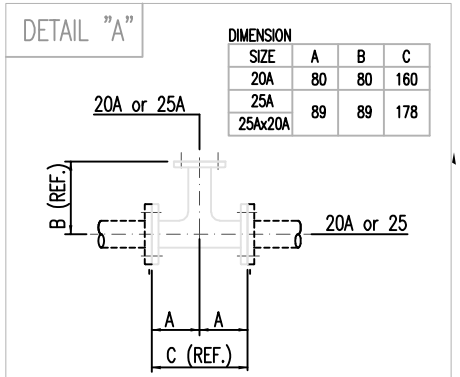
REV.	DESCRIPTION	DATE
1A	REVISED AFC	25.08.14
1	REVISED AFC	25.01.17
0	APPROVAL FOR CONSTRUCTION	24.09.30

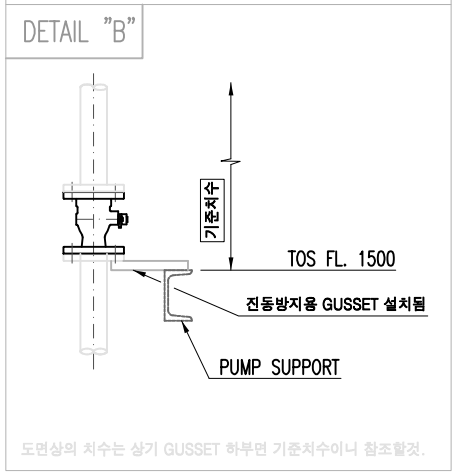
제 도 DRAWN BY	이 인 학
검 토 CHECKED BY	한 진 희
승 인 APPROVED BY	박 동 영
제 작 사	SK 에코플랜트
공 사 명 PROJECT TITLE	M15X Ph4 WWT Project
도면명 DWG. TITLE	PIPING & INSTRUMENT DIAGRAM H2SO4 STORAGE TANK & FEED PUMP (4/4)
축 척 SCALE	NONE
날 짜 DATE	2025. 08. 14
PROJECT NO.	22330D
도면번호 SHEET NO.	CJ-M15W14-PR-A20-000147
REV.	1A

A1(594x841mm ±2mm)

C:₩Users₩TOPW₩Desktop₩P&ID251013₩[M15X Ph-4 for 외허가]CAD FILE_0910₩CJ-M15W14-PR-A20-000147 R1 P&ID FOR H2SO







<NOTE>
1. (*) FIELD ADJUST

ITEM	SIZE	Q'TY	DESCRIPTION
		ST4	
P	25A	0.2M	PIPE, STS 304TP+PTFE Lined, ERW – SCH 10S
P	20A	0.2M	PIPE, STS 304TP+PTFE Lined, ERW – SCH 10S
T	25A	1EA	TEE EQUAL, STSF304+PTFE Lined, ERW – SCH 10S
T	20A	1EA	TEE EQUAL, STSF304+PTFE Lined, ERW – SCH 10S
TR	25Ax20A	1EA	TEE RED, STSF304+PTFE Lined, ERW – SCH 10S
F	25A	7EA	FLG LAP JOINT, KS 10K, RF, STS304+PTFE Lined
F	20A	6EA	FLG LAP JOINT, KS 10K, RF, STS304+PTFE Lined
FB	20A	1EA	FLG BLIND, KS 10K, RF, STS304+PTFE Lined
G	25A	5EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
G	20A	5EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
B1	25A	5SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
B1	20A	5SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
VDLO	25A	1EA	KS 10K, RF, Teflon Lined SSC13, EPDM + PTFE Diaphragm, BB, HW
			+LOTO
VDLO	20A	2EA	KS 10K, RF, Teflon Lined SSC13, EPDM + PTFE Diaphragm, BB, HW
			+LOTO
SH	25A	5EA	SHIELD
SH	20A	5EA	SHIELD
PG	20A	1EA	PRESSURE GAUGE, KS 10K, RF

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14				
	THICK.	–	DESIGN	1.0	60	UNIT	4FA				
			OPERATING	ATM	AMB	REF. PLAN DWG.	CJ-M15W14-PI-A30-040012-A				
						REF. P & ID	CJ-M15W14-PR-A20-000145				

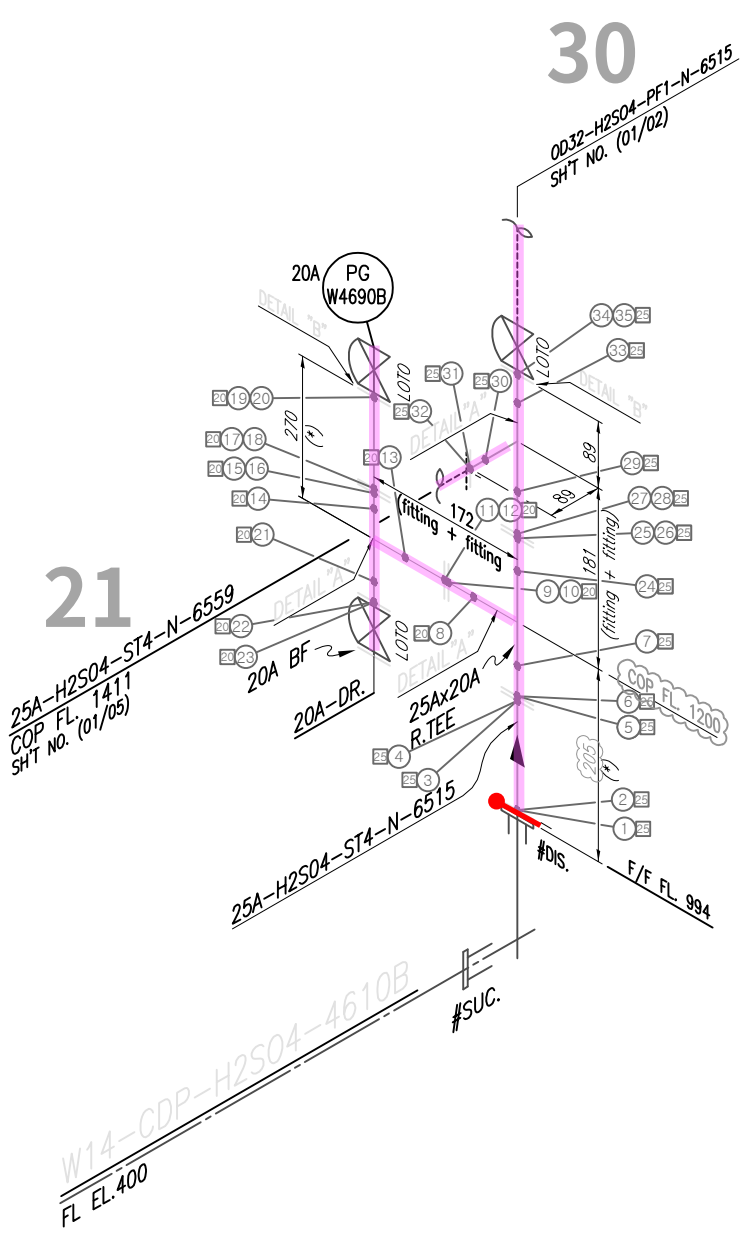
REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래종	안남후	박동영

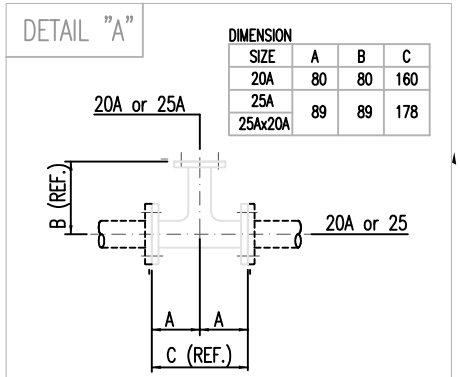
CONTRACTOR: SK ecoplant
COMMODITY: H2SO4
LINE NO.: 25A-H2SO4-ST4-N-6514
PROJECT NAME: M15X Ph4 Project
JOB NO.: 22330D
SH'T NO.: 01 / 01
REV.: 0

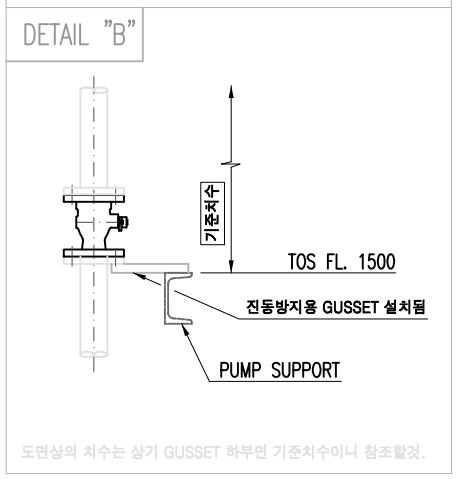
CLIENT: SK hynix

1

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.







<NOTE>
1. (*) FIELD ADJUST

SK ecoplant

ITEM	SIZE	Q'TY	DESCRIPTION
		ST4	
P	25A	0.2M	PIPE, STS 304TP+PTFE Lined, ERW – SCH 10S
	20A	0.2M	PIPE, STS 304TP+PTFE Lined, ERW – SCH 10S
T	25A	1EA	TEE EQUAL, STSF304+PTFE Lined, ERW – SCH 10S
T	20A	1EA	TEE EQUAL, STSF304+PTFE Lined, ERW – SCH 10S
TR	25Ax20A	1EA	TEE RED, STSF304+PTFE Lined, ERW – SCH 10S
F	25A	7EA	FLG LAP JOINT, KS 10K, RF, STS304+PTFE Lined
F	20A	6EA	FLG LAP JOINT, KS 10K, RF, STS304+PTFE Lined
FB	20A	1EA	FLG BLIND, KS 10K, RF, STS304+PTFE Lined
G	25A	5EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
G	20A	5EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
B1	25A	5SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
B1	20A	5SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
VDLO	25A	1EA	KS 10K, RF, Teflon Lined SSC13, EPDM + PTFE Diaphragm, BB, HW
			+LOTO
VDLO	20A	2EA	KS 10K, RF, Teflon Lined SSC13, EPDM + PTFE Diaphragm, BB, HW
			+LOTO
SH	25A	5EA	SHIELD
SH	20A	5EA	SHIELD
PG	20A	1EA	PRESSURE GAUGE, KS 10K, RF

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

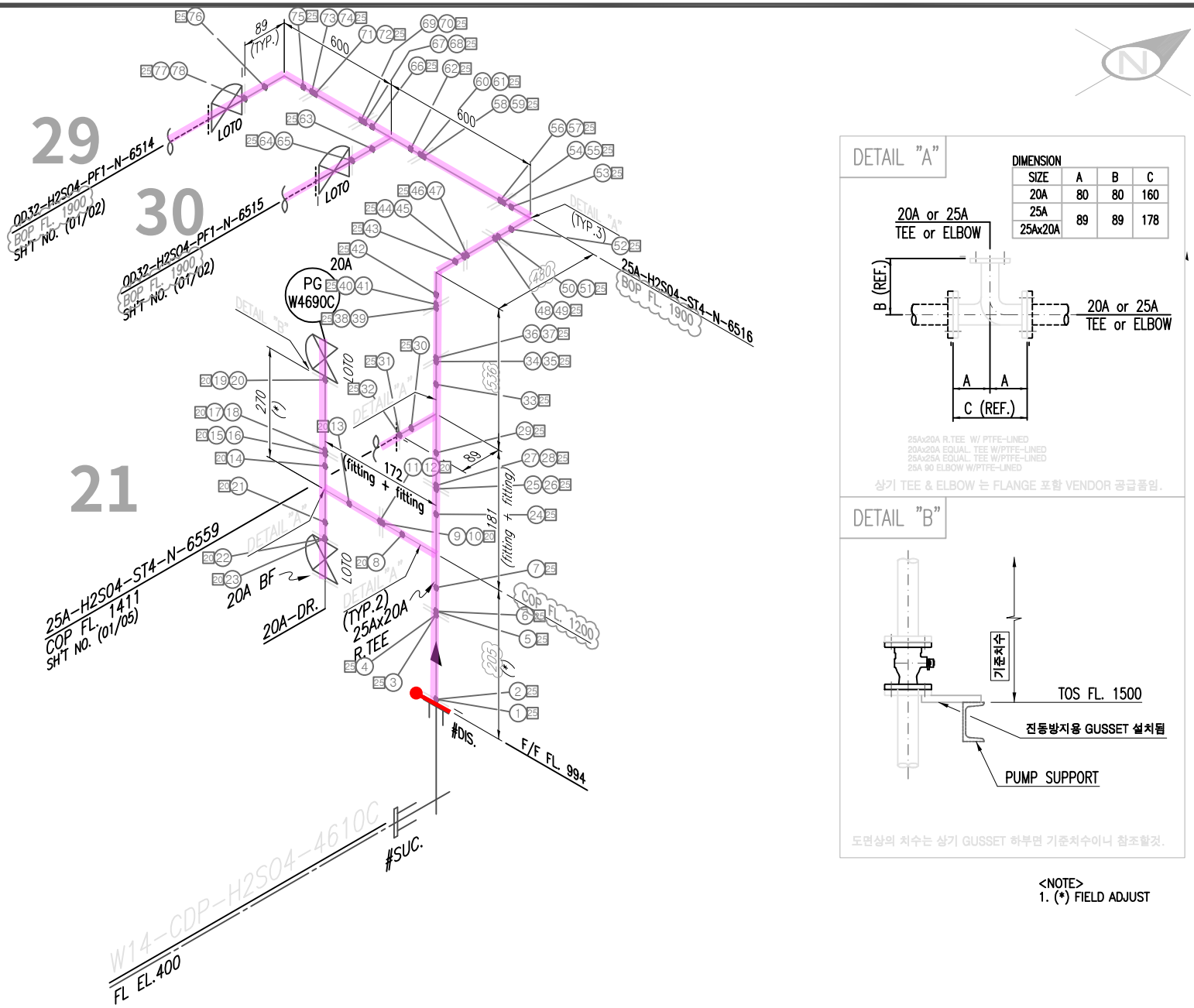
CLIENT: SK hynix 2

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14			
	THICK.	–	DESIGN	1.0	60	UNIT	4FA			
			OPERATING	ATM	AMB	REF. PLAN DWG.	CJ-M15W14-PI-A30-040012-A			
						REF. P & ID	CJ-M15W14-PR-A20-000145			

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래종	안남후	박동영

CONTRACTOR: SK ecoplant
PROJECT NAME: M15X Ph4 Project
COMMODITY: H2SO4
JOB NO.: 22330D
LINE NO.: 25A-H2SO4-ST4-N-6515
SH'T NO.: 01 / 01
REV.: 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



DETAIL "A"

DIMENSION

SIZE	A	B	C
20A	80	80	160
25A 25Ax20A	89	89	178

25Ax20A R.TEE W/ PTFE-LINED
20Ax20A EQUAL. TEE W/PTFE-LINED
25Ax25A EQUAL. TEE W/PTFE-LINED
25A 90 ELBOW W/PTFE-LINED

상기 TEE & ELBOW 는 FLANGE 포함 VENDOR 공급품임.

DETAIL "B"

도면상의 치수는 상기 GUSSET 하부면 기준치수이니 참조할것.

<NOTE>
1. (*) FIELD ADJUST

ITEM	SIZE	Q'TY	DESCRIPTION
		ST4	
P	25A	1.8M	PIPE, STS 304TP+PTFE Lined, ERW - SCH 10S
P	20A	0.2M	PIPE, STS 304TP+PTFE Lined, ERW - SCH 10S
E90	25A	3EA	ELBOW 90 DEG, STSF304+PTFE Lined, ERW - SCH 10S
T	25A	2EA	TEE EQUAL, STSF304+PTFE Lined, ERW - SCH 10S
T	20A	1EA	TEE EQUAL, STSF304+PTFE Lined, ERW - SCH 10S
TR	25Ax20A	1EA	TEE RED, STSF304+PTFE Lined, ERW - SCH 10S
F	25A	24EA	FLG LAP JOINT, KS 10K, RF, STS304+PTFE Lined
F	20A	6EA	FLG LAP JOINT, KS 10K, RF, STS304+PTFE Lined
FB	20A	1EA	FLG BLIND, KS 10K, RF, STS304+PTFE Lined
G	25A	14EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
G	20A	5EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
B1	25A	14SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
B1	20A	5SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
VDLO	25A	2EA	KS 10K, RF, Teflon Lined SSC13, EPDM + PTFE Diaphragm, BB, HW
			+LOTO
VDLO	20A	2EA	KS 10K, RF, Teflon Lined SSC13, EPDM + PTFE Diaphragm, BB, HW
			+LOTO
SH	25A	14EA	SHIELD
SH	20A	5EA	SHIELD
PG	20A	1EA	PRESSURE GAUGE, KS 10K, RF

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

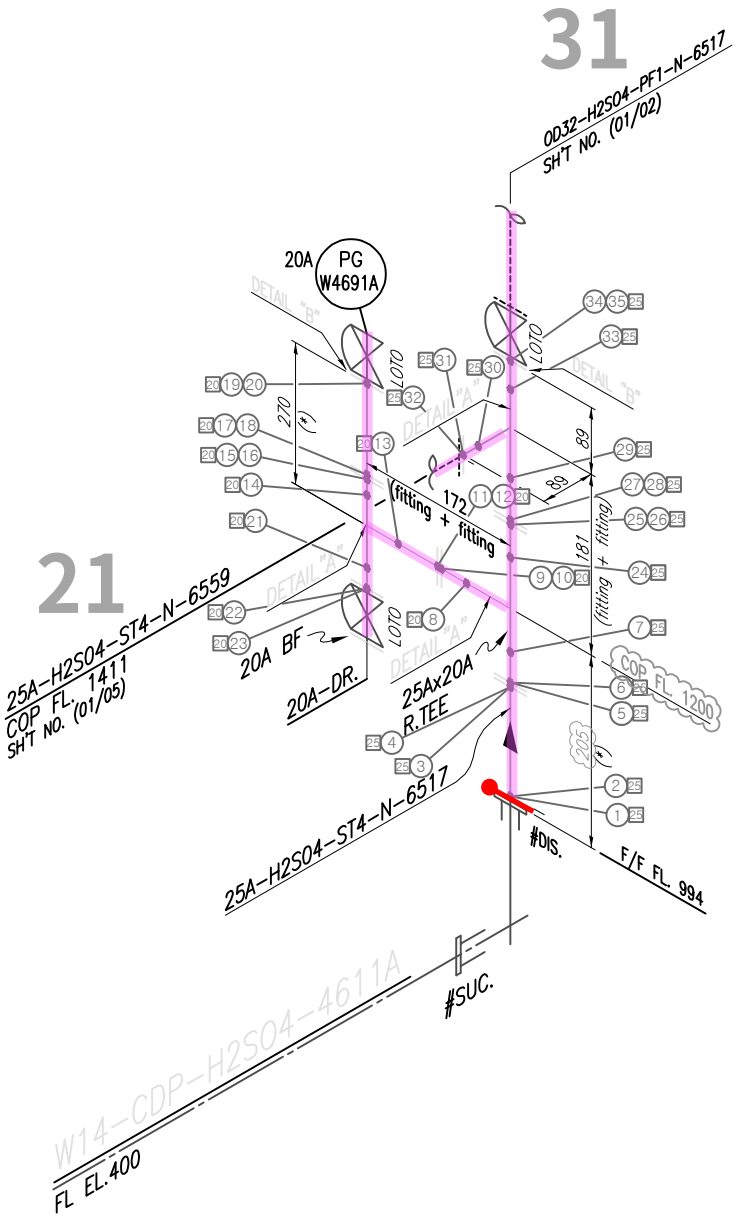
INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14				
	THICK.	-	DESIGN	6.7	60	UNIT	4FA				
			OPERATING	6.1	AMB	REF. PLAN DWG.	CJ-M15W14-PI-A30-040012-A				
						REF. P & ID	CJ-M15W14-PR-A20-000145				

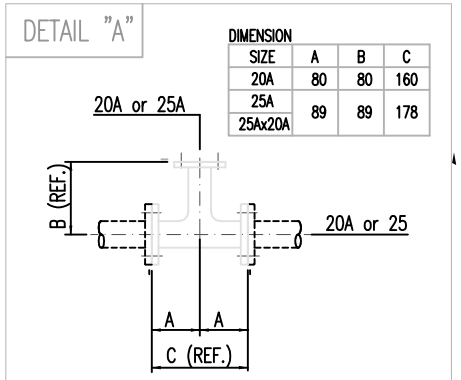
REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래종	안남후	박동영

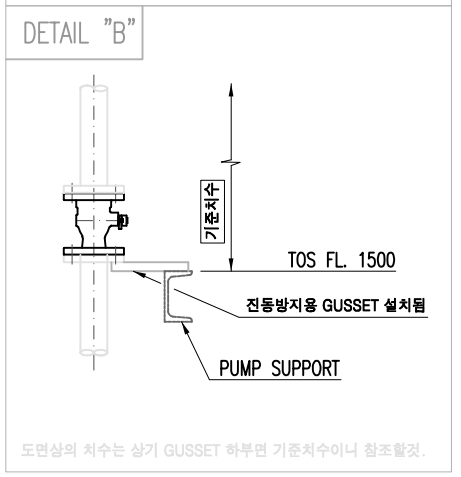
CONTRACTOR: SK ecoplant
COMMODITY: H2S04
LINE NO.: 25A-H2S04-ST4-N-6516
PROJECT NAME: M15X Ph4 Project
JOB NO.: 22330D
SH'T NO.: 01 / 01
REV.: 0

CLIENT: SK hynix 3

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.







<NOTE>
1. (*) FIELD ADJUST

ITEM	SIZE	Q'TY	DESCRIPTION
		ST4	
P	25A	0.2M	PIPE, STS 304TP+PTFE Lined, ERW – SCH 10S
	20A	0.2M	PIPE, STS 304TP+PTFE Lined, ERW – SCH 10S
T	25A	1EA	TEE EQUAL, STSF304+PTFE Lined, ERW – SCH 10S
T	20A	1EA	TEE EQUAL, STSF304+PTFE Lined, ERW – SCH 10S
TR	25Ax20A	1EA	TEE RED, STSF304+PTFE Lined, ERW – SCH 10S
F	25A	7EA	FLG LAP JOINT, KS 10K, RF, STS304+PTFE Lined
F	20A	6EA	FLG LAP JOINT, KS 10K, RF, STS304+PTFE Lined
FB	20A	1EA	FLG BLIND, KS 10K, RF, STS304+PTFE Lined
G	25A	5EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
G	20A	5EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
B1	25A	5SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
B1	20A	5SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
VDLO	25A	1EA	KS 10K, RF, Teflon Lined SSC13, EPDM + PTFE Diaphragm, BB, HW
			+LOTO
VDLO	20A	2EA	KS 10K, RF, Teflon Lined SSC13, EPDM + PTFE Diaphragm, BB, HW
			+LOTO
SH	25A	5EA	SHIELD
SH	20A	5EA	SHIELD
PG	20A	1EA	PRESSURE GAUGE, KS 10K, RF

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT: SK hynix

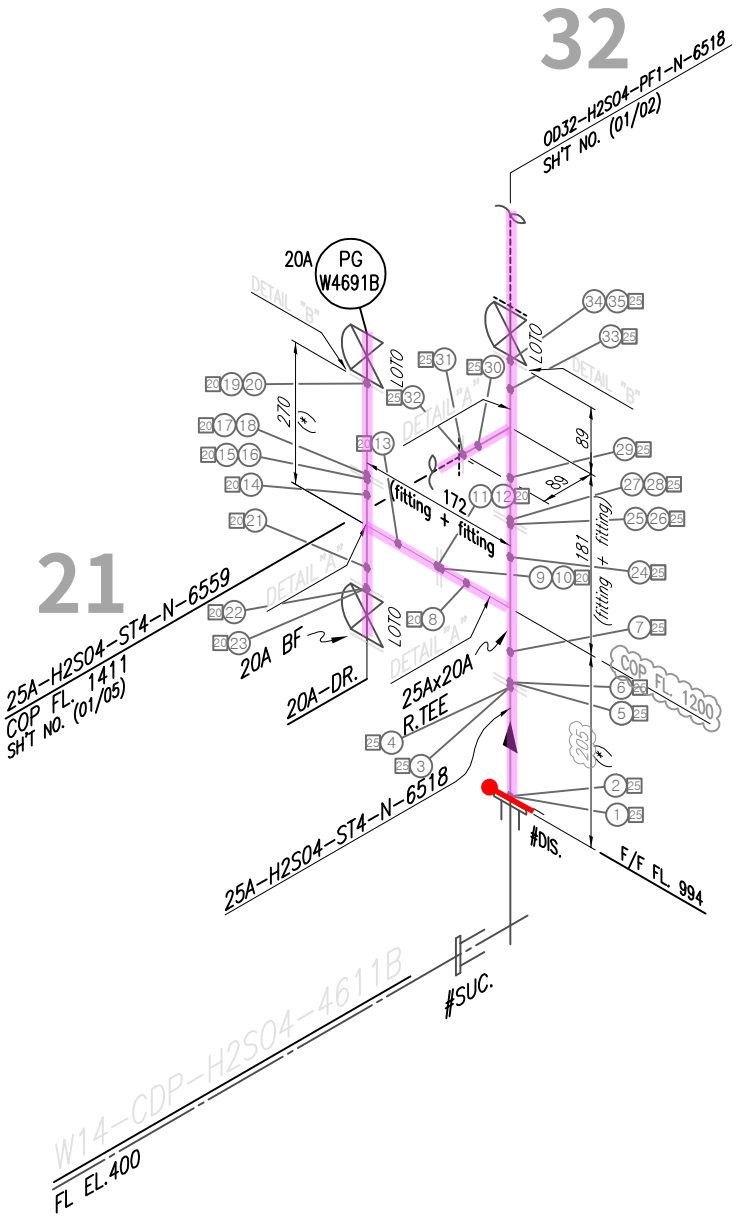
4

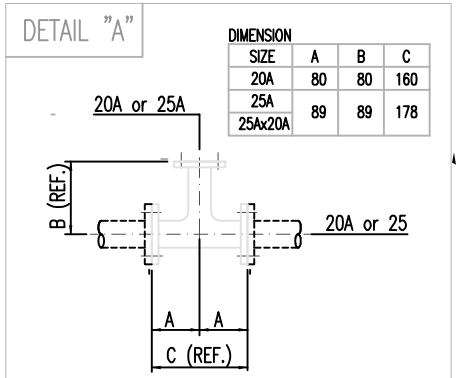
INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14			
	THICK.	–	DESIGN	1.0	60	UNIT	4FA			
			OPERATING	ATM	AMB	REF. PLAN DWG.	CJ-M15W14-PI-A30-040013-A			
						REF. P & ID	CJ-M15W14-PR-A20-000145			

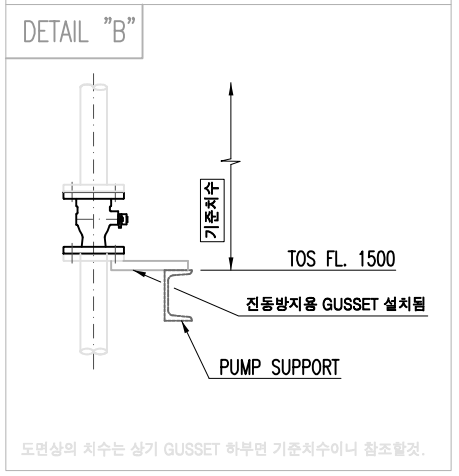
REV.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래종	안남후	박동영

CONTRACTOR: SK ecoplant
COMMODITY: H2S04
LINE NO.: 25A-H2S04-ST4-N-6517
PROJECT NAME: M15X Ph4 Project
JOB NO.: 22330D
SH'T NO.: 01 / 01
REV.: 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.







<NOTE>
1. (*) FIELD ADJUST

SK ecoplant

ITEM	SIZE	Q'TY	DESCRIPTION
		ST4	
P	25A	0.2M	PIPE, STS 304TP+PTFE Lined, ERW – SCH 10S
	20A	0.2M	PIPE, STS 304TP+PTFE Lined, ERW – SCH 10S
T	25A	1EA	TEE EQUAL, STSF304+PTFE Lined, ERW – SCH 10S
T	20A	1EA	TEE EQUAL, STSF304+PTFE Lined, ERW – SCH 10S
TR	25Ax20A	1EA	TEE RED, STSF304+PTFE Lined, ERW – SCH 10S
F	25A	7EA	FLG LAP JOINT, KS 10K, RF, STS304+PTFE Lined
F	20A	6EA	FLG LAP JOINT, KS 10K, RF, STS304+PTFE Lined
FB	20A	1EA	FLG BLIND, KS 10K, RF, STS304+PTFE Lined
G	25A	5EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
G	20A	5EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
B1	25A	5SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
B1	20A	5SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
VDLO	25A	1EA	KS 10K, RF, Teflon Lined SSC13, EPDM + PTFE Diaphragm, BB, HW
			+LOTO
VDLO	20A	2EA	KS 10K, RF, Teflon Lined SSC13, EPDM + PTFE Diaphragm, BB, HW
			+LOTO
SH	25A	5EA	SHIELD
SH	20A	5EA	SHIELD
PG	20A	1EA	PRESSURE GAUGE, KS 10K, RF

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT: SK hynix

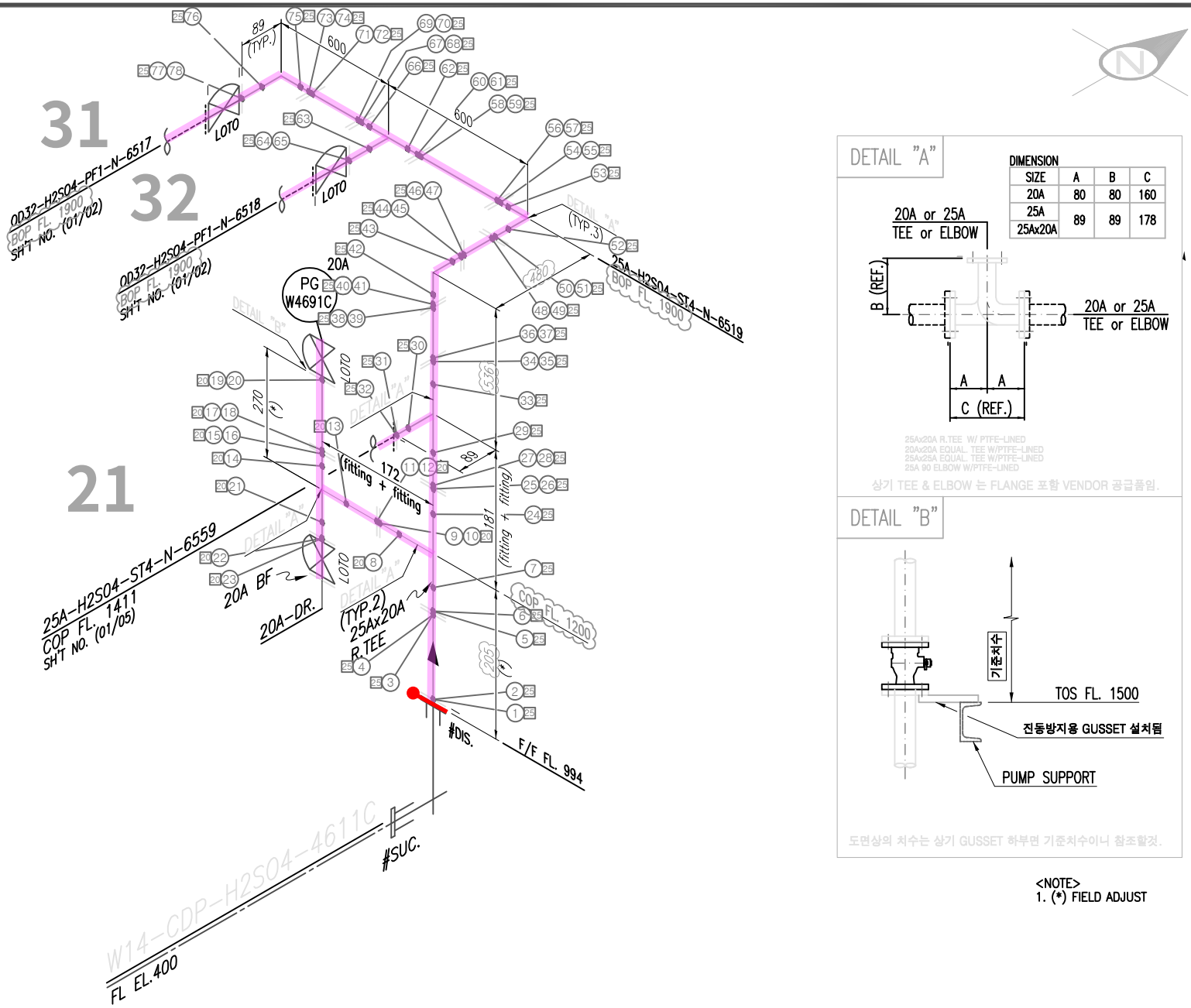
5

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14		
	THICK.	–	DESIGN	1.0	60	UNIT	4FA		
			OPERATING	ATM	AMB	REF. PLAN DWG.	CJ-M15W14-PI-A30-040013-A		
						REF. P & ID	CJ-M15W14-PR-A20-000145		

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래종	안남후	박동영

CONTRACTOR: SK ecoplant
PROJECT NAME: M15X Ph4 Project
COMMODITY: H2SO4
JOB NO.: 22330D
LINE NO.: 25A-H2SO4-ST4-N-6518
SH'T NO.: 01 / 01
REV.: 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



DETAIL "A"

DIMENSION

SIZE	A	B	C
20A	80	80	160
25A 25Ax20A	89	89	178

25Ax20A R.TEE W/ PTFE-LINED
20Ax20A EQUAL. TEE W/PTFE-LINED
25Ax25A EQUAL. TEE W/PTFE-LINED
25A 90 ELBOW W/PTFE-LINED

상기 TEE & ELBOW 는 FLANGE 포함 VENDOR 공급품임.

DETAIL "B"

도면상의 치수는 상기 GUSSET 하부면 기준치수이니 참조할것.

<NOTE>
1. (*) FIELD ADJUST

SK ecoplant

ITEM	SIZE	Q'TY	DESCRIPTION
		ST4	
P	25A	1.8M	PIPE, STS 304TP+PTFE Lined, ERW – SCH 10S
P	20A	0.2M	PIPE, STS 304TP+PTFE Lined, ERW – SCH 10S
E90	25A	3EA	ELBOW 90 DEG, STSF304+PTFE Lined, ERW – SCH 10S
T	25A	2EA	TEE EQUAL, STSF304+PTFE Lined, ERW – SCH 10S
T	20A	1EA	TEE EQUAL, STSF304+PTFE Lined, ERW – SCH 10S
TR	25Ax20A	1EA	TEE RED, STSF304+PTFE Lined, ERW – SCH 10S
F	25A	24EA	FLG LAP JOINT, KS 10K, RF, STS304+PTFE Lined
F	20A	6EA	FLG LAP JOINT, KS 10K, RF, STS304+PTFE Lined
FB	20A	1EA	FLG BLIND, KS 10K, RF, STS304+PTFE Lined
G	25A	14EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
G	20A	5EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
B1	25A	14SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
B1	20A	5SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
VDLO	25A	2EA	KS 10K, RF, Teflon Lined SSC13, EPDM + PTFE Diaphragm, BB, HW
			+LOTO
VDLO	20A	2EA	KS 10K, RF, Teflon Lined SSC13, EPDM + PTFE Diaphragm, BB, HW
			+LOTO
SH	25A	14EA	SHIELD
SH	20A	5EA	SHIELD
PG	20A	1EA	PRESSURE GAUGE, KS 10K, RF

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT: SK hynix

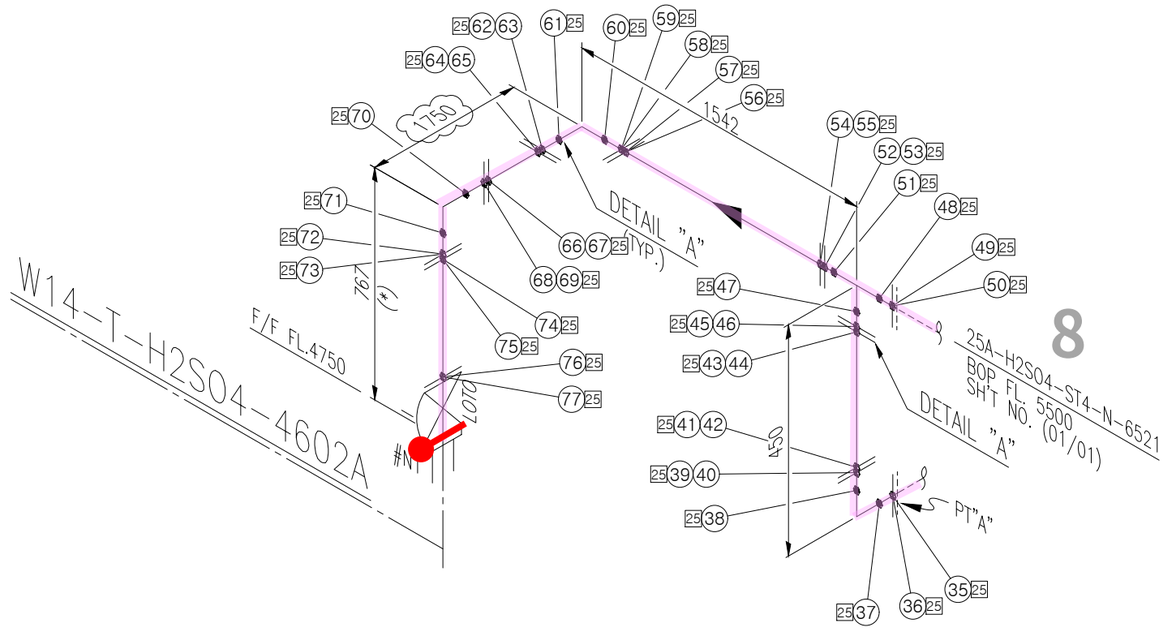
6

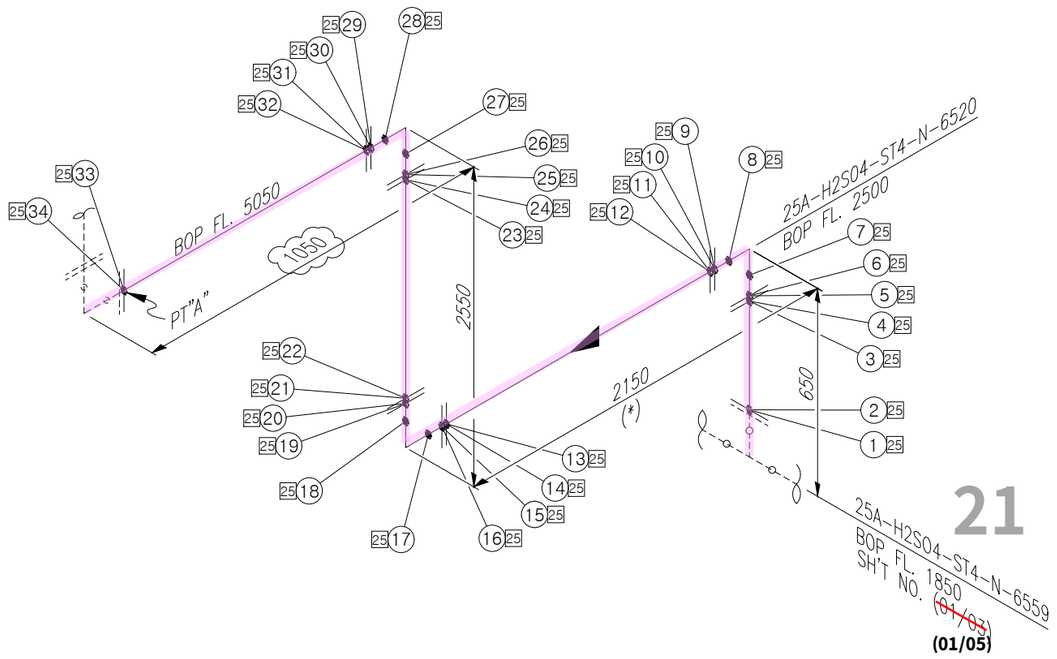
INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14
	THICK.	–	DESIGN	6.7	60	UNIT	4FA
			OPERATING	6.1	AMB	REF. PLAN DWG.	CJ-M15W14-PI-A30-040013-A
						REF. P & ID	CJ-M15W14-PR-A20-000145

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래홍	안남후	박동영

CONTRACTOR	SK ecoplant	PROJECT NAME	M15X Ph4 Project
COMMODITY	H2S04	JOB NO.	22330D
LINE NO.	25A-H2S04-ST4-N-6519	SH'T NO.	01 / 01
		REV.	0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.

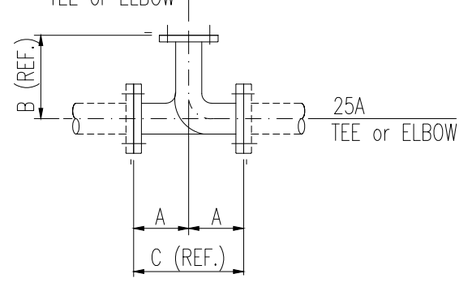




DETAIL "A"

DIMENSION

SIZE	A	B	C
25A	89	89	178



25Ax25A EQUAL. TEE W/PTFE-LINED
25A 90 ELBOW W/PTFE-LINED

상기 TEE & ELBOW 는 FLANGE 포함 VENDOR 공급품임.

<NOTE>
1. (*) FIELD ADJUST

ITEM	SIZE	Q'TY	DESCRIPTION
		ST4	
P	25A	9.7M	PIPE, STS 304TP+PTFE Lined, ERW – SCH 10S
E90	25A	6EA	ELBOW 90 DEG, STSF304+PTFE Lined, ERW – SCH 10S
T	25A	1EA	TEE EQUAL, STSF304+PTFE Lined, ERW – SCH 10S
F	25A	31EA	FLG LAP JOINT, KS 10K, RF, STS304+PTFE Lined
G	25A	17EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
B1	25A	17SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
VDLO	25A	1EA	KS 10K, RF, Teflon Lined SSC13, EPDM + PTFE Diaphragm, BB, HW
			+LOTO
SH	25A	17EA	SHIELD

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT SK hynix

7

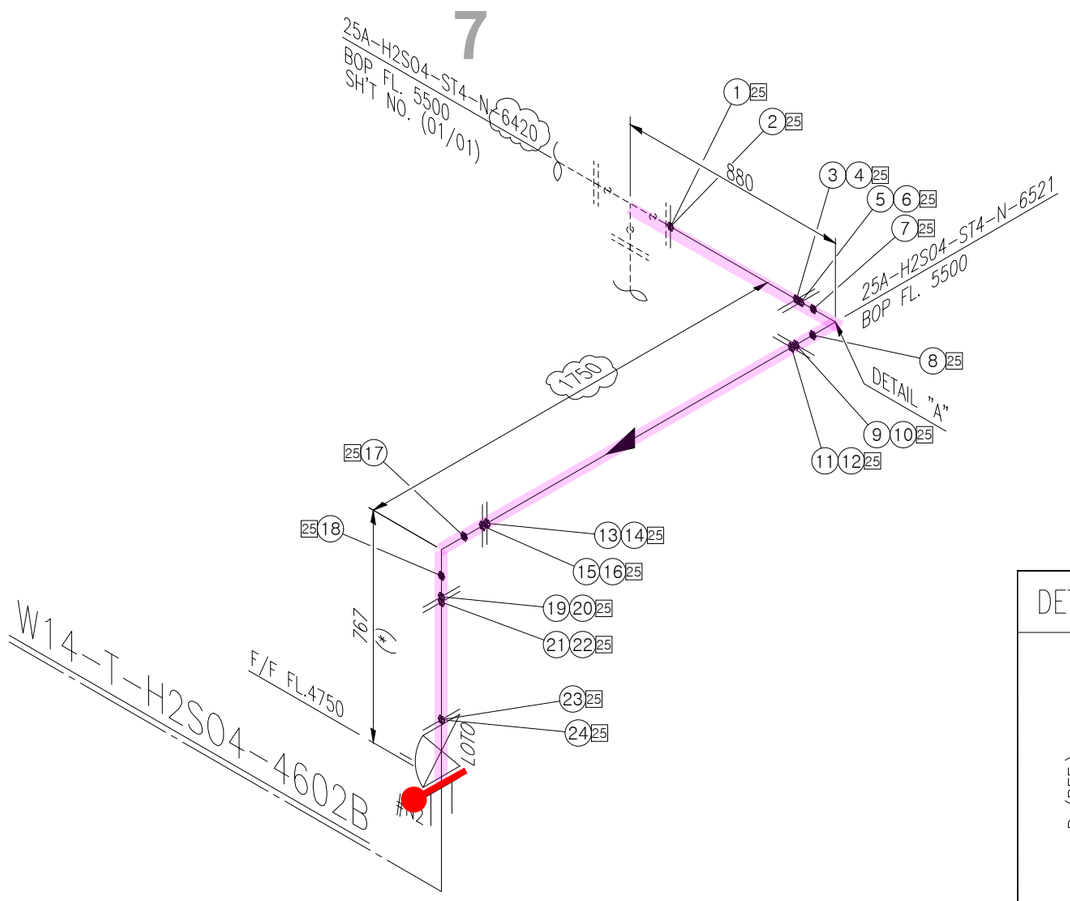
INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
	THICK.	–	DESIGN	6.7	60
			OPERATING	6.1	AMB

AREA W14
UNIT 4FA
REF. PLAN DWG. CJ-M15W14-PI-A30-040013-D
REF. P & ID CJ-M15W14-PR-A20-000144

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
1	25.03.31	REVISED AS MARKED	황래종	안남후	박동영
0	25.01.24	APPROVED FOR CONSTRUCTION	황래종	안남후	박동영

CONTRACTOR SK ecoplant
PROJECT NAME M15X Ph4 Project
COMMODITY H2S04
JOB NO. 22330D
LINE NO. 25A-H2S04-ST4-N-6520
SH'T NO. 01 / 01
REV. 1

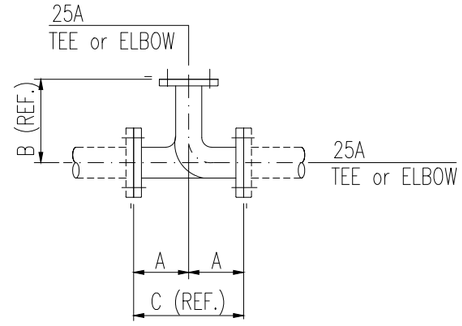
경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



DETAIL "A"

DIMENSION

SIZE	A	B	C
25A	89	89	178



25Ax25A EQUAL. TEE W/PTFE-LINED
25A 90 ELBOW W/PTFE-LINED

상기 TEE & ELBOW 는 FLANGE 포함 VENDOR 공급품임.

<NOTE>
1. (*) FIELD ADJUST

ITEM	SIZE	Q'TY	DESCRIPTION
		ST4	
P	25A	2.6M	PIPE, STS 304TP+PTFE Lined, ERW – SCH 10S
E90	25A	2EA	ELBOW 90 DEG, STSF304+PTFE Lined, ERW – SCH 10S
F	25A	10EA	FLG LAP JOINT, KS 10K, RF, STS304+PTFE Lined
G	25A	7EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
B1	25A	7EA	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
VDLO	25A	1EA	KS 10K, RF, Teflon Lined SSC13, EPDM + PTFE Diaphragm, BB, HW
			+LOTO
SH	25A	7EA	SHIELD

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT SK hynix

8

INSULATION	CODE	N
	THICK.	–

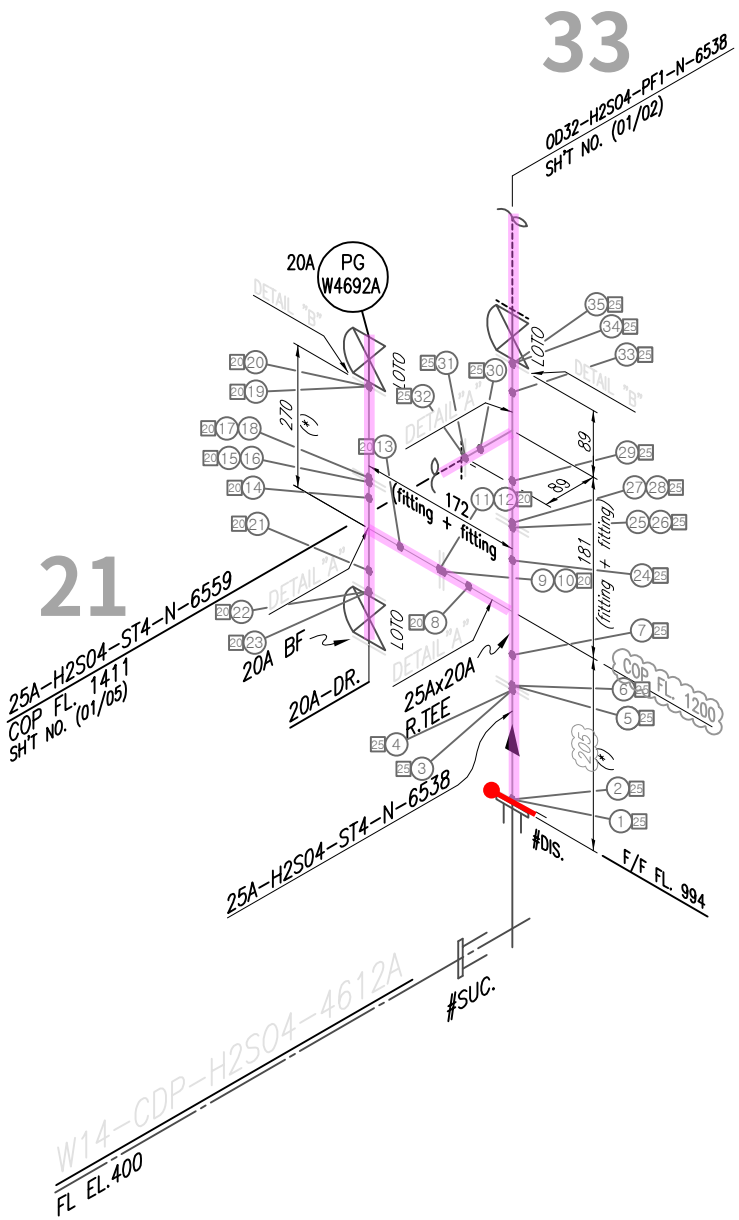
CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
DESIGN	6.7	60
OPERATING	6.1	AMB

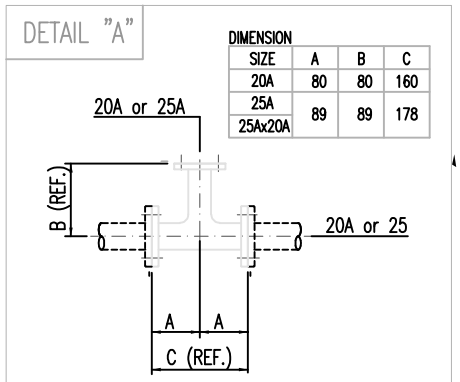
AREA W14
UNIT 4FA
REF. PLAN DWG. CJ-M15W14-PI-A30-040013-D
REF. P & ID CJ-M15W14-PR-A20-000144

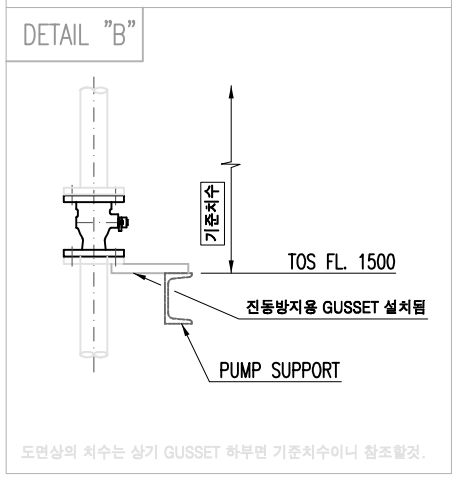
REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
1	25.03.31	REVISED AS MARKED	황래종	안남후	박동영
0	25.01.24	APPROVED FOR CONSTRUCTION	황래종	안남후	박동영

CONTRACTOR SK ecoplant
PROJECT NAME M15X Ph4 Project
COMMODITY H2S04
JOB NO. 22330D
LINE NO. 25A-H2S04-ST4-N-6521
SH'T NO. 01 / 01
REV. 1

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.







<NOTE>
1. (*) FIELD ADJUST

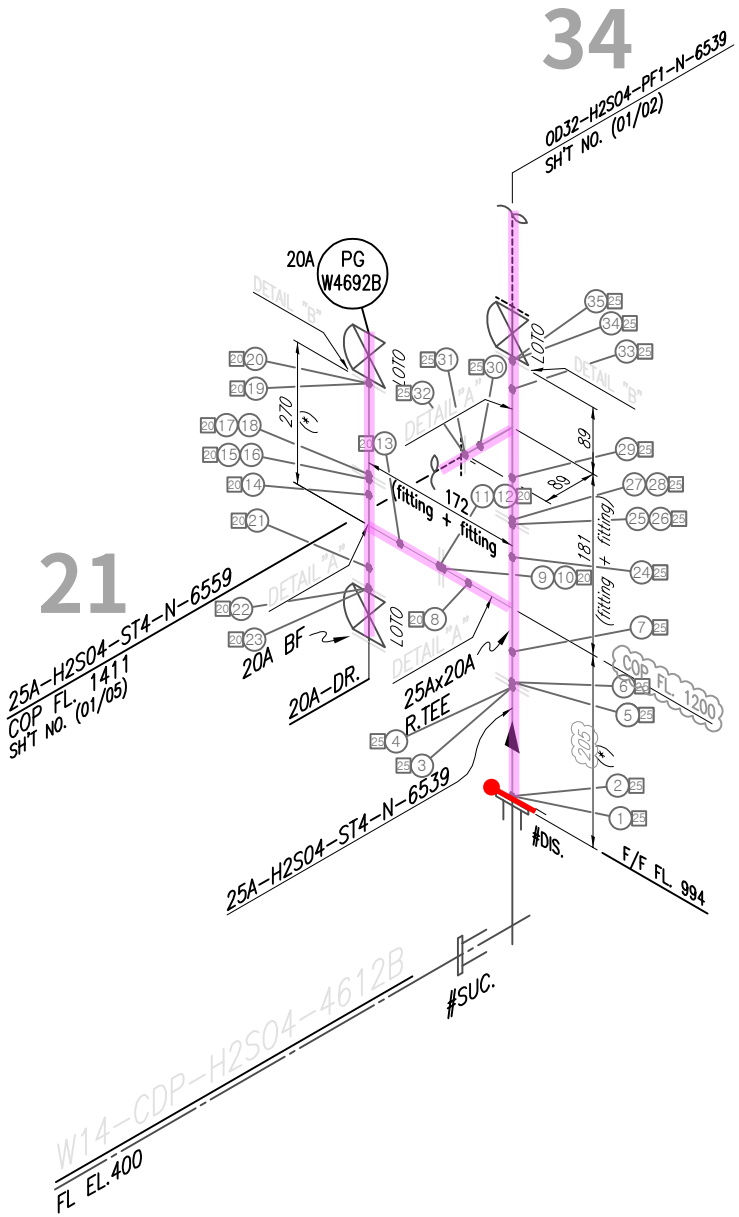
ITEM	SIZE	Q'TY	DESCRIPTION
		ST4	
P	25A	0.2M	PIPE, STS 304TP+PTFE Lined, ERW – SCH 10S
	20A	0.2M	PIPE, STS 304TP+PTFE Lined, ERW – SCH 10S
T	25A	1EA	TEE EQUAL, STSF304+PTFE Lined, ERW – SCH 10S
T	20A	1EA	TEE EQUAL, STSF304+PTFE Lined, ERW – SCH 10S
TR	25Ax20A	1EA	TEE RED, STSF304+PTFE Lined, ERW – SCH 10S
F	25A	7EA	FLG LAP JOINT, KS 10K, RF, STS304+PTFE Lined
F	20A	6EA	FLG LAP JOINT, KS 10K, RF, STS304+PTFE Lined
FB	20A	1EA	FLG BLIND, KS 10K, RF, STS304+PTFE Lined
G	25A	5EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
G	20A	5EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
B1	25A	5SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
B1	20A	5SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
VDLO	25A	1EA	KS 10K, RF, Teflon Lined SSC13, EPDM + PTFE Diaphragm, BB, HW
			+LOTO
VDLO	20A	2EA	KS 10K, RF, Teflon Lined SSC13, EPDM + PTFE Diaphragm, BB, HW
			+LOTO
SH	25A	5EA	SHIELD
SH	20A	5EA	SHIELD
PG	20A	1EA	PRESSURE GAUGE, KS 10K, RF

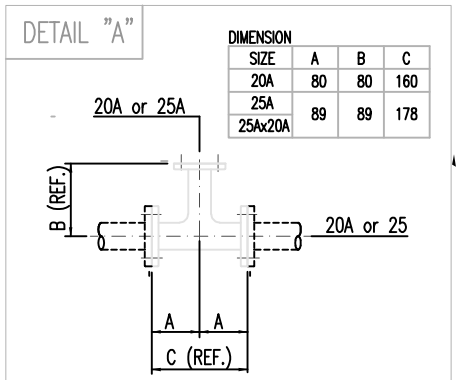
* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

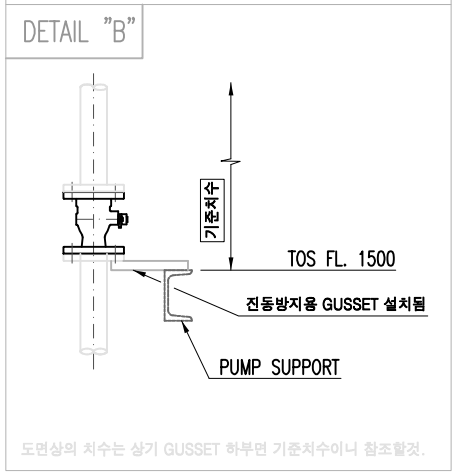
INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14				
	THICK.	–	DESIGN	1.0	60	UNIT	4FA				
			OPERATING	ATM	AMB	REF. PLAN DWG.	CJ-M15W14-PI-A30-040013-A				
						REF. P & ID	CJ-M15W14-PR-A20-000145	0	25.01.24	APPROVED FOR CONSTRUCTION	황래종 / 안남후 / 박동영
						REV. NO.			DATE	DESCRIPTION	BY / CHK / APP'D

CLIENT: SK hynix 9
CONTRACTOR: SK ecoplant
COMMODITY: H2S04
LINE NO.: 25A-H2S04-ST4-N-6538
PROJECT NAME: M15X Ph4 Project
JOB NO.: 22330D
SH'T NO.: 01 / 01
REV.: 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.







<NOTE>
1. (*) FIELD ADJUST

SK ecoplant

ITEM	SIZE	Q'TY	DESCRIPTION
		ST4	
P	25A	0.2M	PIPE, STS 304TP+PTFE Lined, ERW – SCH 10S
	20A	0.2M	PIPE, STS 304TP+PTFE Lined, ERW – SCH 10S
T	25A	1EA	TEE EQUAL, STSF304+PTFE Lined, ERW – SCH 10S
T	20A	1EA	TEE EQUAL, STSF304+PTFE Lined, ERW – SCH 10S
TR	25Ax20A	1EA	TEE RED, STSF304+PTFE Lined, ERW – SCH 10S
F	25A	7EA	FLG LAP JOINT, KS 10K, RF, STS304+PTFE Lined
F	20A	6EA	FLG LAP JOINT, KS 10K, RF, STS304+PTFE Lined
FB	20A	1EA	FLG BLIND, KS 10K, RF, STS304+PTFE Lined
G	25A	5EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
G	20A	5EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
B1	25A	5SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
B1	20A	5SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
VDLO	25A	1EA	KS 10K, RF, Teflon Lined SSC13, EPDM + PTFE Diaphragm, BB, HW
			+LOTO
VDLO	20A	2EA	KS 10K, RF, Teflon Lined SSC13, EPDM + PTFE Diaphragm, BB, HW
			+LOTO
SH	25A	5EA	SHIELD
SH	20A	5EA	SHIELD
PG	20A	1EA	PRESSURE GAUGE, KS 10K, RF

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT: SK hynix

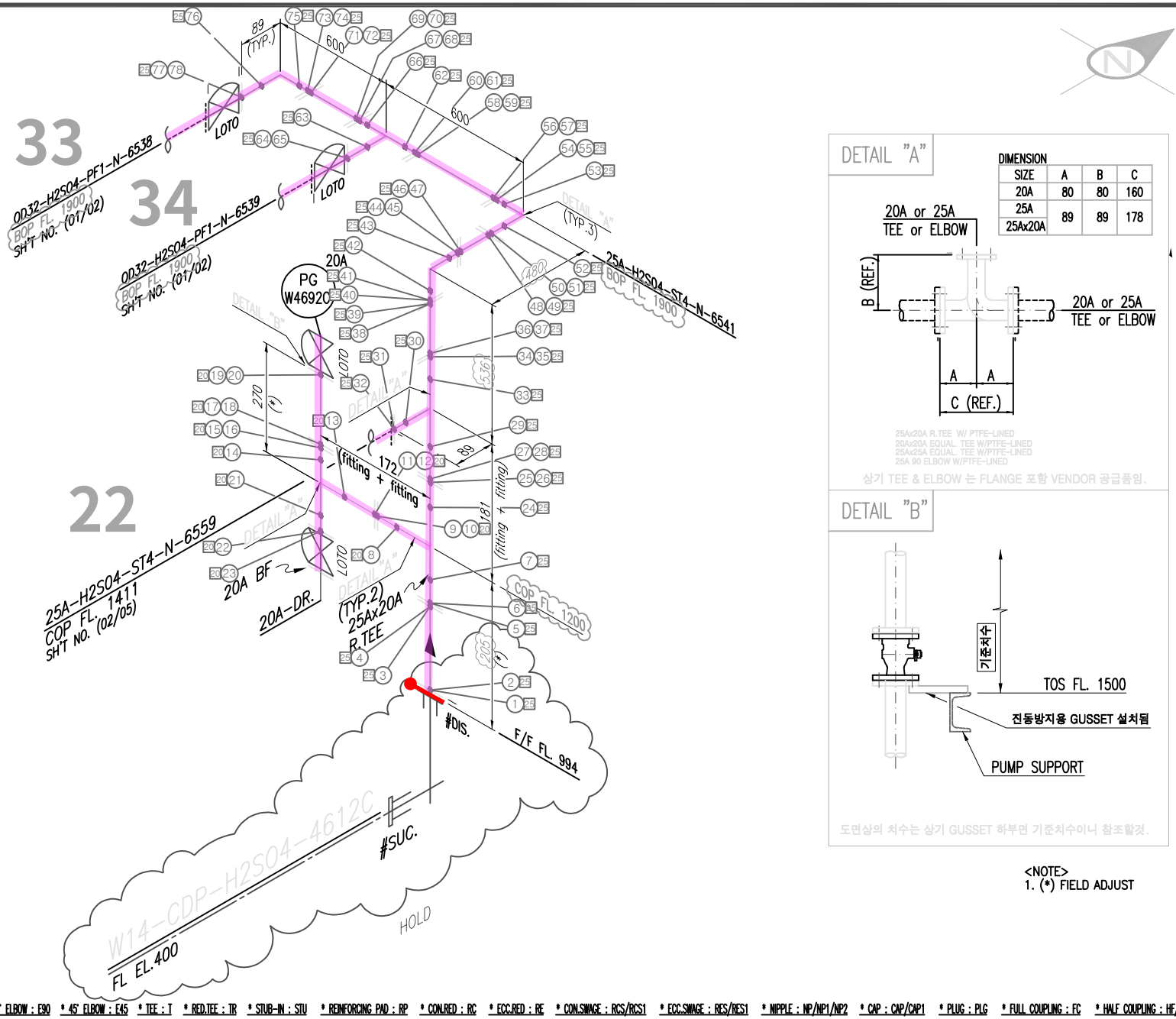
10

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14			
	THICK.	–	DESIGN	1.0	60	UNIT	4FA			
			OPERATING	ATM	AMB	REF. PLAN DWG.	CJ-M15W14-PI-A30-040013-A			
						REF. P & ID	CJ-M15W14-PR-A20-000145			

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래종	안남후	박동영

CONTRACTOR: SK ecoplant
PROJECT NAME: M15X Ph4 Project
COMMODITY: H2SO4
JOB NO.: 22330D
LINE NO.: 25A-H2SO4-ST4-N-6539
SH'T NO.: 01 / 01
REV.: 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



DETAIL "A"

DIMENSION

SIZE	A	B	C
20A	80	80	160
25A 25Ax20A	89	89	178

25Ax20A R.TEE W/ PTFE-LINED
20Ax20A EQUAL. TEE W/PTFE-LINED
25Ax25A EQUAL. TEE W/PTFE-LINED
25A 90 ELBOW W/PTFE-LINED

상기 TEE & ELBOW 는 FLANGE 포함 VENDOR 공급품임.

DETAIL "B"

도면상의 치수는 상기 GUSSET 하부면 기준치수이니 참조할것.

<NOTE>
1. (*) FIELD ADJUST

SK ecoplant

ITEM	SIZE	Q'TY	DESCRIPTION
		ST4	
P	25A	1.8M	PIPE, STS 304TP+PTFE Lined, ERW – SCH 10S
P	20A	0.2M	PIPE, STS 304TP+PTFE Lined, ERW – SCH 10S
E90	25A	3EA	ELBOW 90 DEG, STSF304+PTFE Lined, ERW – SCH 10S
T	25A	2EA	TEE EQUAL, STSF304+PTFE Lined, ERW – SCH 10S
T	20A	1EA	TEE EQUAL, STSF304+PTFE Lined, ERW – SCH 10S
TR	25Ax20A	1EA	TEE RED, STSF304+PTFE Lined, ERW – SCH 10S
F	25A	24EA	FLG LAP JOINT, KS 10K, RF, STS304+PTFE Lined
F	20A	6EA	FLG LAP JOINT, KS 10K, RF, STS304+PTFE Lined
FB	20A	1EA	FLG BLIND, KS 10K, RF, STS304+PTFE Lined
G	25A	14EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
G	20A	5EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
B1	25A	14SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
B1	20A	5SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
VDLO	25A	2EA	KS 10K, RF, Teflon Lined SSC13, EPDM + PTFE Diaphragm, BB, HW
			+LOTO
VDLO	20A	2EA	KS 10K, RF, Teflon Lined SSC13, EPDM + PTFE Diaphragm, BB, HW
			+LOTO
SH	25A	14EA	SHIELD
SH	20A	5EA	SHIELD
PG	20A	1EA	PRESSURE GAUGE, KS 10K, RF

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT: SK hynix

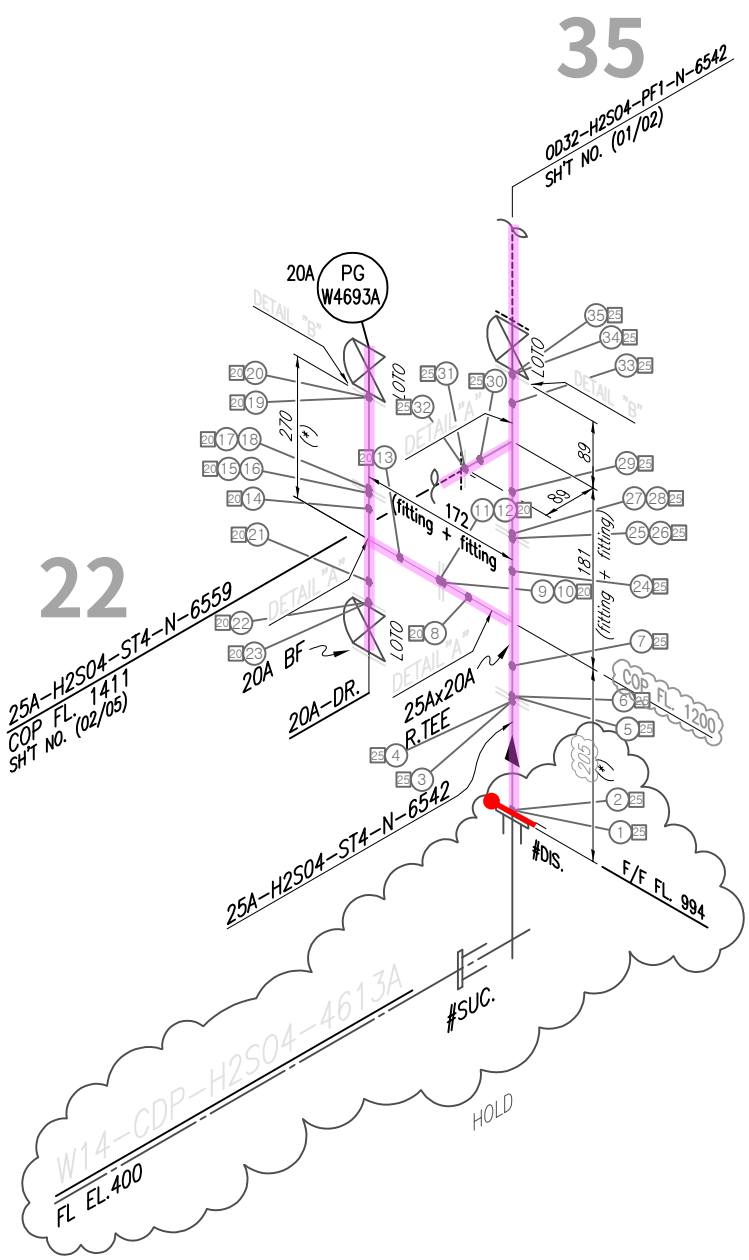
11

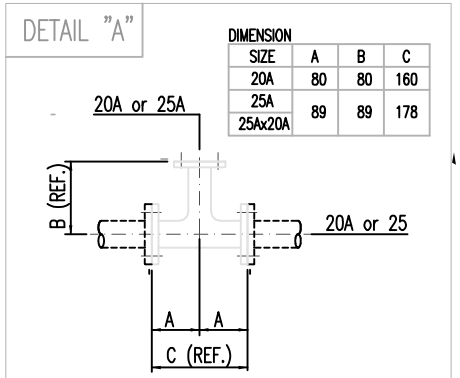
INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14	
	THICK.	–	DESIGN	6.7	60	UNIT	4FA	
			OPERATING	6.1	AMB	REF. PLAN DWG.	CJ-M15W14-PI-A30-040013-A	
						REF. P & ID	CJ-M15W14-PR-A20-000145	

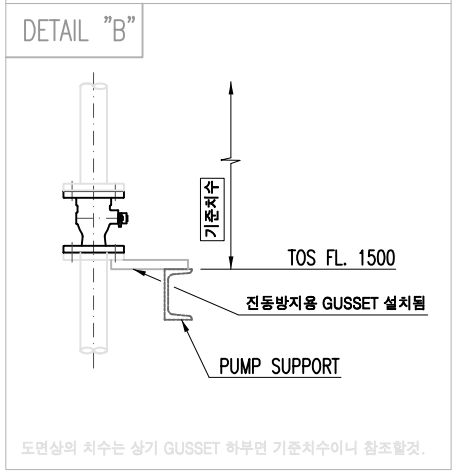
REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래종	안남후	박동영

CONTRACTOR: SK ecoplant
COMMODITY: H2S04
LINE NO.: 25A-H2S04-ST4-N-6541
PROJECT NAME: M15X Ph4 Project
JOB NO.: 22330D
SH'T NO.: 01 / 01
REV.: 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.







<NOTE>
1. (*) FIELD ADJUST

ITEM	SIZE	Q'TY	DESCRIPTION
		ST4	
P	25A	0.2M	PIPE, STS 304TP+PTFE Lined, ERW – SCH 10S
	20A	0.2M	PIPE, STS 304TP+PTFE Lined, ERW – SCH 10S
T	25A	1EA	TEE EQUAL, STSF304+PTFE Lined, ERW – SCH 10S
T	20A	1EA	TEE EQUAL, STSF304+PTFE Lined, ERW – SCH 10S
TR	25Ax20A	1EA	TEE RED, STSF304+PTFE Lined, ERW – SCH 10S
F	25A	7EA	FLG LAP JOINT, KS 10K, RF, STS304+PTFE Lined
F	20A	6EA	FLG LAP JOINT, KS 10K, RF, STS304+PTFE Lined
FB	20A	1EA	FLG BLIND, KS 10K, RF, STS304+PTFE Lined
G	25A	5EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
G	20A	5EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
B1	25A	5SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
B1	20A	5SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
VDLO	25A	1EA	KS 10K, RF, Teflon Lined SSC13, EPDM + PTFE Diaphragm, BB, HW
			+LOTO
VDLO	20A	2EA	KS 10K, RF, Teflon Lined SSC13, EPDM + PTFE Diaphragm, BB, HW
			+LOTO
SH	25A	5EA	SHIELD
SH	20A	5EA	SHIELD
PG	20A	1EA	PRESSURE GAUGE, KS 10K, RF

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

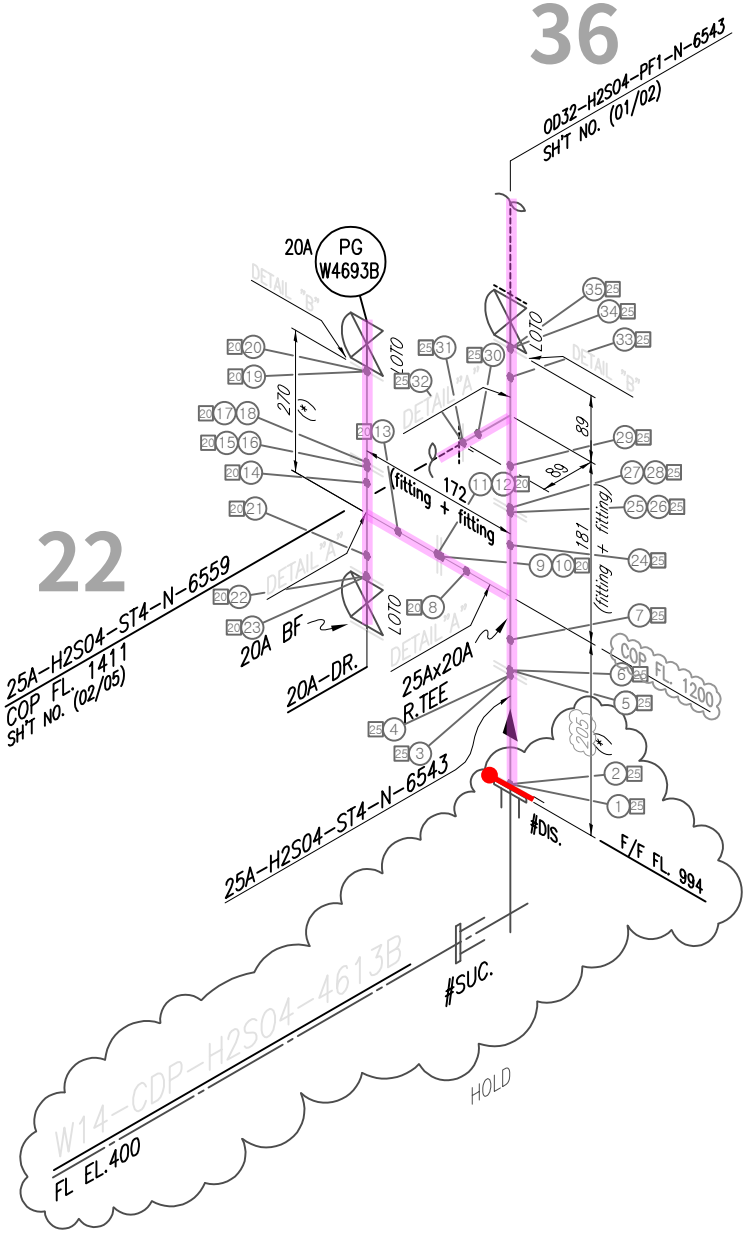
INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14				
	THICK.	–	DESIGN	6.7	60	UNIT	4FA				
			OPERATING	6.1	AMB	REF. PLAN DWG.	CJ-M15W14-PI-A30-040013-A				
						REF. P & ID	CJ-M15W14-PR-A20-000144	0	25.01.24	APPROVED FOR CONSTRUCTION	황래종

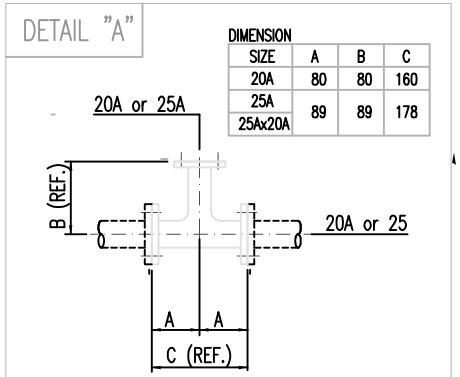
CLIENT: SK hynix

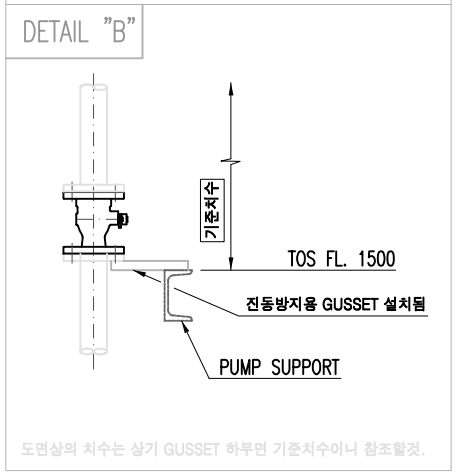
12

CONTRACTOR: SK ecoplant
COMMODITY: H2SO4
LINE NO.: 25A-H2SO4-ST4-N-6542
PROJECT NAME: M15X Ph4 Project
JOB NO.: 22330D
SH'T NO.: 01 / 01
REV.: 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.







<NOTE>
1. (*) FIELD ADJUST

SK ecoplant

ITEM	SIZE	Q'TY	DESCRIPTION
		ST4	
P	25A	0.2M	PIPE, STS 304TP+PTFE Lined, ERW – SCH 10S
	20A	0.2M	PIPE, STS 304TP+PTFE Lined, ERW – SCH 10S
T	25A	1EA	TEE EQUAL, STSF304+PTFE Lined, ERW – SCH 10S
T	20A	1EA	TEE EQUAL, STSF304+PTFE Lined, ERW – SCH 10S
TR	25Ax20A	1EA	TEE RED, STSF304+PTFE Lined, ERW – SCH 10S
F	25A	7EA	FLG LAP JOINT, KS 10K, RF, STS304+PTFE Lined
F	20A	6EA	FLG LAP JOINT, KS 10K, RF, STS304+PTFE Lined
FB	20A	1EA	FLG BLIND, KS 10K, RF, STS304+PTFE Lined
G	25A	5EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
G	20A	5EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
B1	25A	5SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
B1	20A	5SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
VDLO	25A	1EA	KS 10K, RF, Teflon Lined SSC13, EPDM + PTFE Diaphragm, BB, HW
			+LOTO
VDLO	20A	2EA	KS 10K, RF, Teflon Lined SSC13, EPDM + PTFE Diaphragm, BB, HW
			+LOTO
SH	25A	5EA	SHIELD
SH	20A	5EA	SHIELD
PG	20A	1EA	PRESSURE GAUGE, KS 10K, RF

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT: SK hynix

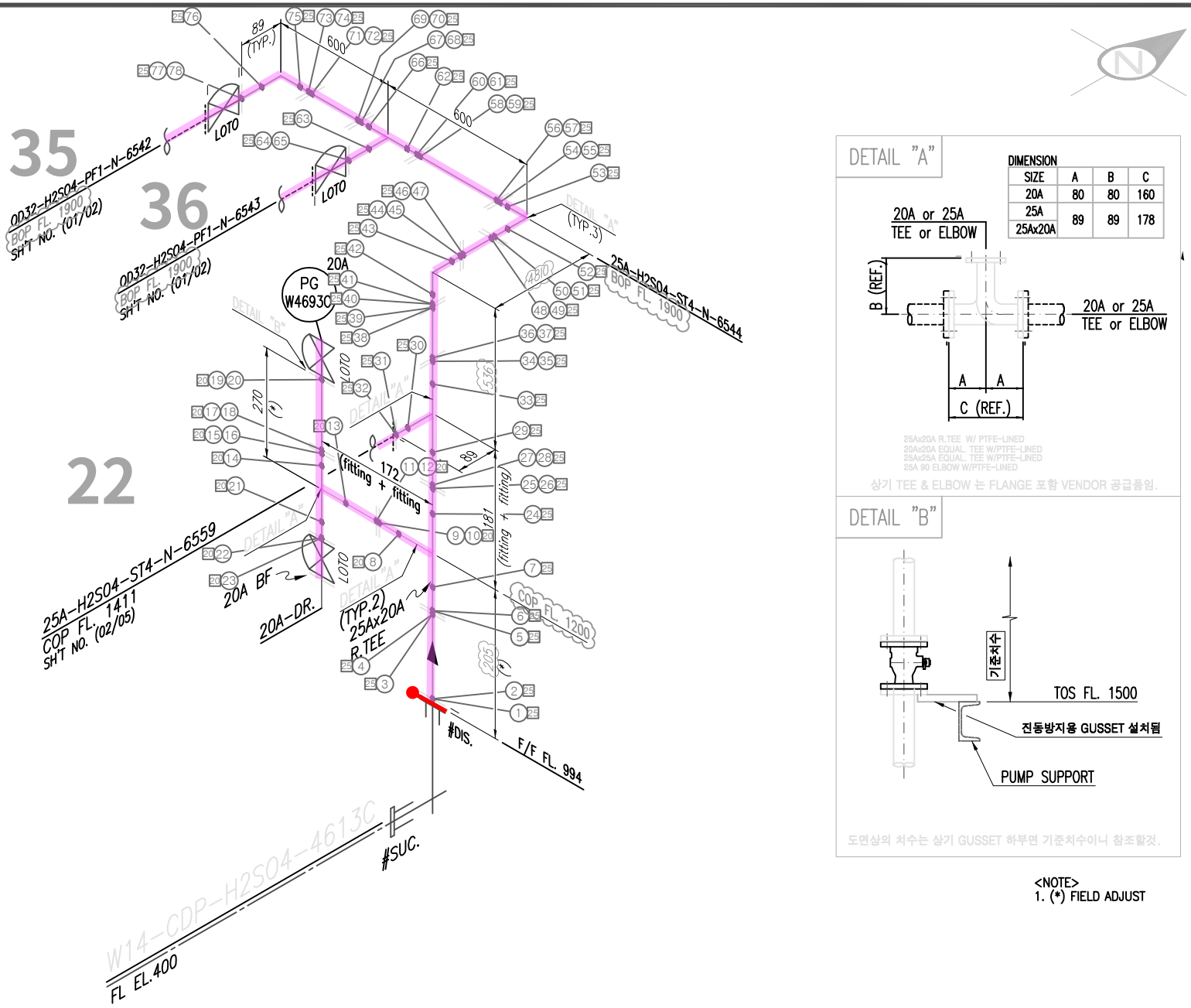
13

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14	CONTRACTOR	SK ecoplant
	THICK.	–	DESIGN	6.7	60	UNIT	4FA	PROJECT NAME	M15X Ph4 Project
			OPERATING	6.1	AMB	REF. PLAN DWG.	CJ-M15W14-PI-A30-040013-A	COMMODITY	H2SO4
						REF. P & ID	CJ-M15W14-PR-A20-000144	JOB NO.	22330D

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래종	안남후	박동영

LINE NO. 25A-H2SO4-ST4-N-6543 | SH'T NO. 01 / 01 | REV. 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



DETAIL "A"

DIMENSION

SIZE	A	B	C
20A	80	80	160
25A 25Ax20A	89	89	178

25Ax20A R.TEE W/ PTFE-LINED
20Ax20A EQUAL. TEE W/PTFE-LINED
25Ax25A EQUAL. TEE W/PTFE-LINED
25A 90 ELBOW W/PTFE-LINED

상기 TEE & ELBOW 는 FLANGE 포함 VENDOR 공급품임.

DETAIL "B"

도면상의 치수는 상기 GUSSET 하부면 기준치수이니 참조할것.

<NOTE>
1. (*) FIELD ADJUST

SK ecoplant

ITEM	SIZE	Q'TY	DESCRIPTION
		ST4	
P	25A	1.8M	PIPE, STS 304TP+PTFE Lined, ERW – SCH 10S
P	20A	0.2M	PIPE, STS 304TP+PTFE Lined, ERW – SCH 10S
E90	25A	3EA	ELBOW 90 DEG, STSF304+PTFE Lined, ERW – SCH 10S
T	25A	2EA	TEE EQUAL, STSF304+PTFE Lined, ERW – SCH 10S
T	20A	1EA	TEE EQUAL, STSF304+PTFE Lined, ERW – SCH 10S
TR	25Ax20A	1EA	TEE RED, STSF304+PTFE Lined, ERW – SCH 10S
F	25A	24EA	FLG LAP JOINT, KS 10K, RF, STS304+PTFE Lined
F	20A	6EA	FLG LAP JOINT, KS 10K, RF, STS304+PTFE Lined
FB	20A	1EA	FLG BLIND, KS 10K, RF, STS304+PTFE Lined
G	25A	14EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
G	20A	5EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
B1	25A	14SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
B1	20A	5SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
VDLO	25A	2EA	KS 10K, RF, Teflon Lined SSC13, EPDM + PTFE Diaphragm, BB, HW
			+LOTO
VDLO	20A	2EA	KS 10K, RF, Teflon Lined SSC13, EPDM + PTFE Diaphragm, BB, HW
			+LOTO
SH	25A	14EA	SHIELD
SH	20A	5EA	SHIELD
PG	20A	1EA	PRESSURE GAUGE, KS 10K, RF

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

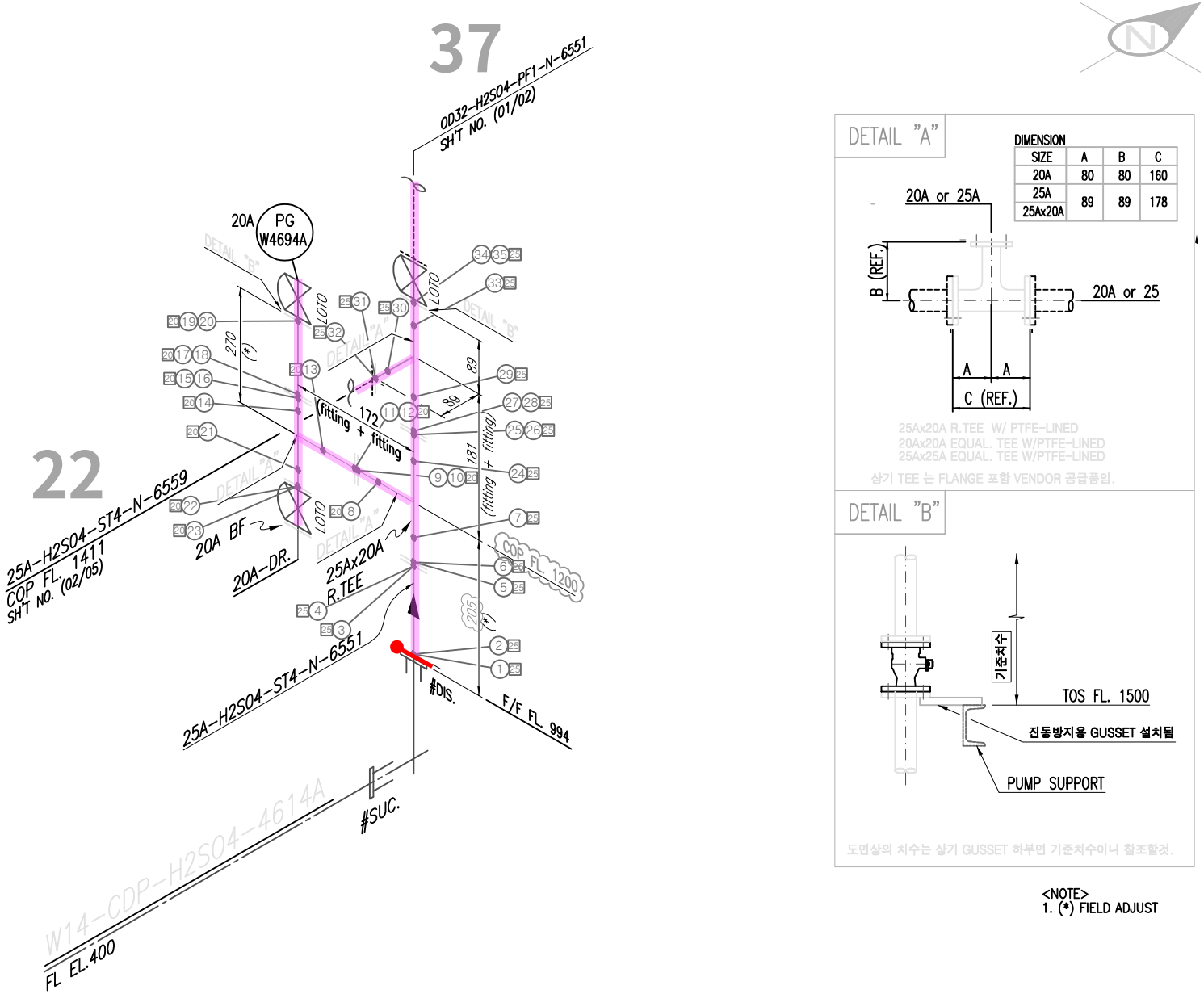
CLIENT: SK hynix

14

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14	CONTRACTOR: SK ecoplant	PROJECT NAME: M15X Ph4 Project
	THICK.	–	DESIGN	6.7	60	UNIT	4FA	COMMODITY: H2S04	JOB NO.: 22330D
			OPERATING	6.1	AMB	REF. PLAN DWG.: CJ-M15W14-PI-A30-040013-A		LINE NO.: 25A-H2S04-ST4-N-6544	SH'T NO.: 01 / 01 REV.: 0
						REF. P & ID: CJ-M15W14-PR-A20-000144			

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



<NOTE>
1. (*) FIELD ADJUST

SK ecoplant

ITEM	SIZE	Q'TY	DESCRIPTION
		ST4	
P	25A	0.2M	PIPE, STS 304TP+PTFE Lined, ERW – SCH 10S
	20A	0.2M	PIPE, STS 304TP+PTFE Lined, ERW – SCH 10S
T	25A	1EA	TEE EQUAL, STSF304+PTFE Lined, ERW – SCH 10S
T	20A	1EA	TEE EQUAL, STSF304+PTFE Lined, ERW – SCH 10S
TR	25Ax20A	1EA	TEE RED, STSF304+PTFE Lined, ERW – SCH 10S
F	25A	7EA	FLG LAP JOINT, KS 10K, RF, STS304+PTFE Lined
F	20A	6EA	FLG LAP JOINT, KS 10K, RF, STS304+PTFE Lined
FB	20A	1EA	FLG BLIND, KS 10K, RF, STS304+PTFE Lined
G	25A	5EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
G	20A	5EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
B1	25A	5SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
B1	20A	5SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
VDLO	25A	1EA	KS 10K, RF, Teflon Lined SSC13, EPDM + PTFE Diaphragm, BB, HW
			+LOTO
VDLO	20A	2EA	KS 10K, RF, Teflon Lined SSC13, EPDM + PTFE Diaphragm, BB, HW
			+LOTO
SH	25A	5EA	SHIELD
SH	20A	5EA	SHIELD
PG	20A	1EA	PRESSURE GAUGE, KS 10K, RF

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT: SK hynix

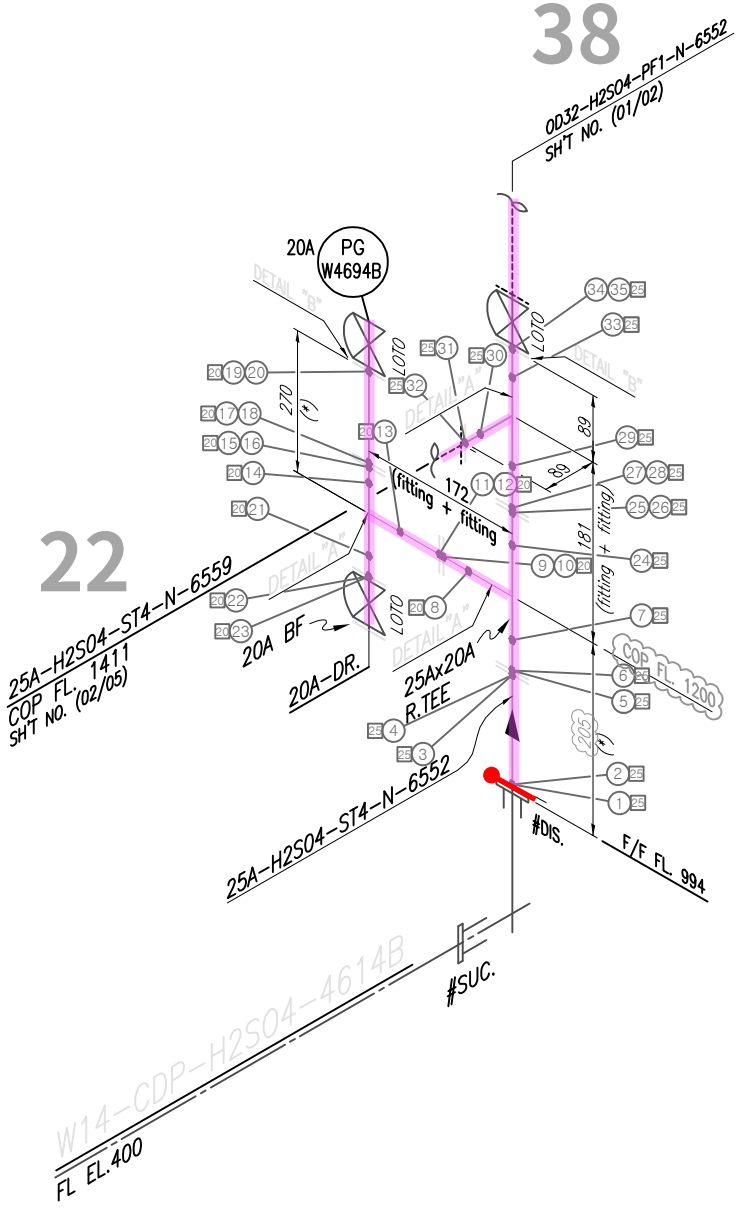
15

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14	CONTRACTOR	SK ecoplant
	THICK.	–	DESIGN	6.7	60	UNIT	4FA	PROJECT NAME	M15X Ph4 Project
			OPERATING	6.1	AMB	REF. PLAN DWG.	CJ-M15W14-PI-A30-040013-A	COMMODITY	H2SO4
						REF. P & ID	CJ-M15W14-PR-A20-000144	JOB NO.	22330D

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래종	안남후	박동영

LINE NO.: 25A-H2SO4-ST4-N-6551 | SH'T NO.: 01 / 01 | REV.: 0

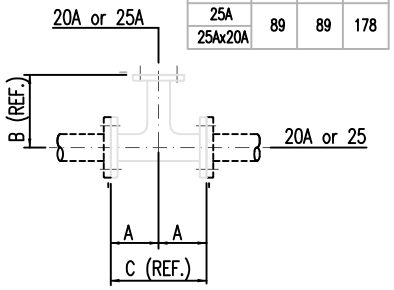
경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



DETAIL "A"

DIMENSION

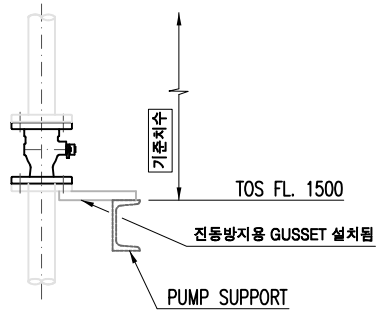
SIZE	A	B	C
20A	80	80	160
25A 25Ax20A	89	89	178



25Ax20A R.TEE W/ PTFE-LINED
20Ax20A EQUAL. TEE W/PTFE-LINED
25Ax25A EQUAL. TEE W/PTFE-LINED

상기 TEE 는 FLANGE 포함 VENDOR 공급품임.

DETAIL "B"



도면상의 치수는 상기 GUSSET 하부면 기준치수이니 참조할것.

<NOTE>
1. (*) FIELD ADJUST

SK ecoplant

ITEM	SIZE	Q'TY	DESCRIPTION
		ST4	
P	25A	0.2M	PIPE, STS 304TP+PTFE Lined, ERW – SCH 10S
	20A	0.2M	PIPE, STS 304TP+PTFE Lined, ERW – SCH 10S
T	25A	1EA	TEE EQUAL, STSF304+PTFE Lined, ERW – SCH 10S
T	20A	1EA	TEE EQUAL, STSF304+PTFE Lined, ERW – SCH 10S
TR	25Ax20A	1EA	TEE RED, STSF304+PTFE Lined, ERW – SCH 10S
F	25A	7EA	FLG LAP JOINT, KS 10K, RF, STS304+PTFE Lined
F	20A	6EA	FLG LAP JOINT, KS 10K, RF, STS304+PTFE Lined
FB	20A	1EA	FLG BLIND, KS 10K, RF, STS304+PTFE Lined
G	25A	5EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
G	20A	5EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
B1	25A	5SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
B1	20A	5SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
VDLO	25A	1EA	KS 10K, RF, Teflon Lined SSC13, EPDM + PTFE Diaphragm, BB, HW
			+LOTO
VDLO	20A	2EA	KS 10K, RF, Teflon Lined SSC13, EPDM + PTFE Diaphragm, BB, HW
			+LOTO
SH	25A	5EA	SHIELD
SH	20A	5EA	SHIELD
PG	20A	1EA	PRESSURE GAUGE, KS 10K, RF

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT: SK hynix

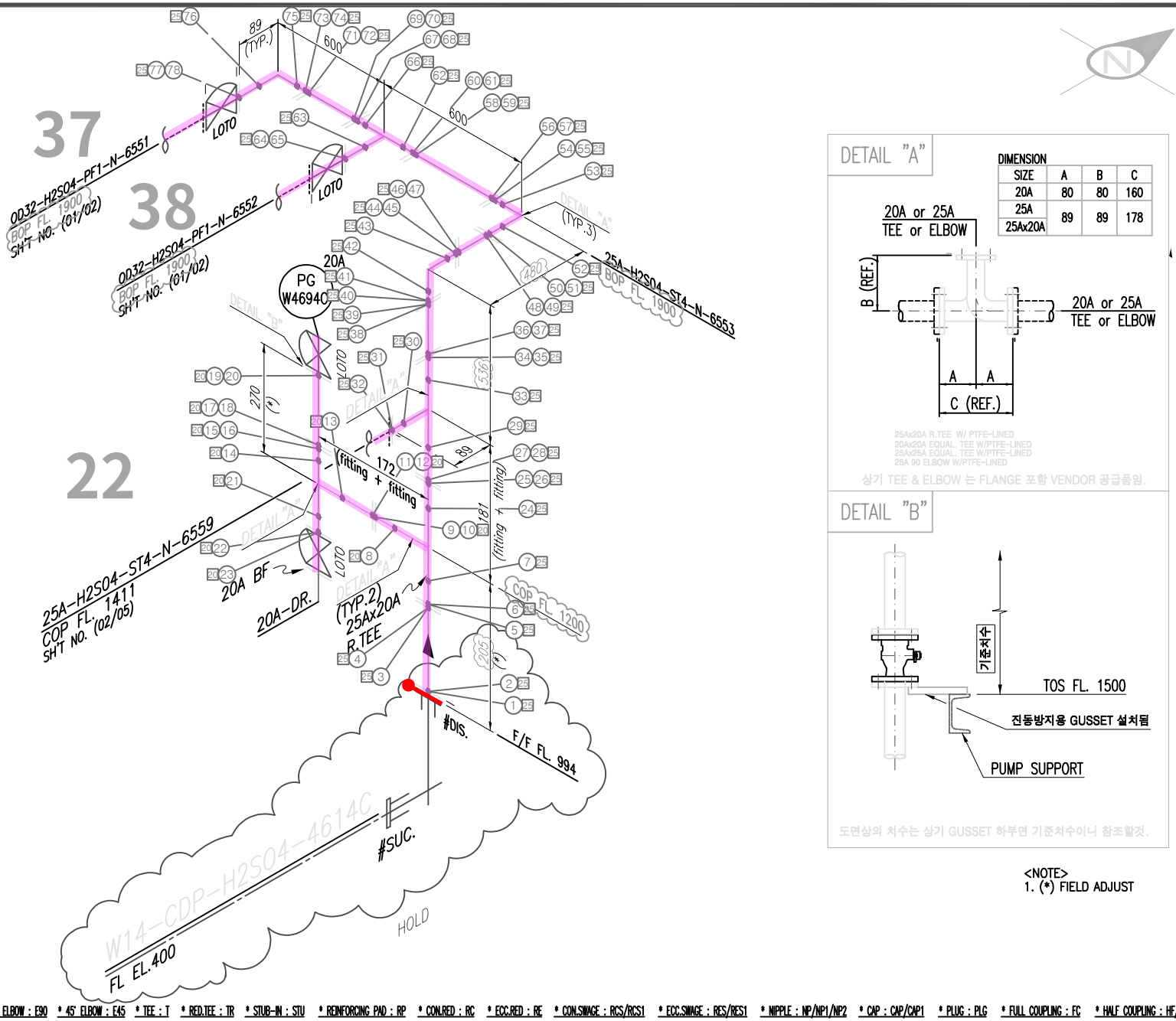
16

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14		
	THICK.	–	DESIGN	6.7	60	UNIT	4FA		
			OPERATING	6.1	AMB	REF. PLAN DWG.	CJ-M15W14-PI-A30-040013-A		
						REF. P & ID	CJ-M15W14-PR-A20-000144		

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래종	안남후	박동영

CONTRACTOR	SK ecoplant	PROJECT NAME	M15X Ph4 Project
COMMODITY	H2S04	JOB NO.	22330D
LINE NO.	25A-H2S04-ST4-N-6552	SH'T NO.	01 / 01
		REV.	0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



DETAIL "A"

DIMENSION

SIZE	A	B	C
20A	80	80	160
25A 25Ax20A	89	89	178

25Ax20A R.TEE W/ PTFE-LINED
20Ax20A EQUAL. TEE W/PTFE-LINED
25Ax25A EQUAL. TEE W/PTFE-LINED
25A 90 ELBOW W/PTFE-LINED

상기 TEE & ELBOW 는 FLANGE 포함 VENDOR 공급품임.

DETAIL "B"

도면상의 치수는 상기 GUSSET 하부면 기준치수이니 참조할것.

<NOTE>
1. (*) FIELD ADJUST

SK ecoplant

ITEM	SIZE	Q'TY	DESCRIPTION
		ST4	
P	25A	1.8M	PIPE, STS 304TP+PTFE Lined, ERW – SCH 10S
P	20A	0.2M	PIPE, STS 304TP+PTFE Lined, ERW – SCH 10S
E90	25A	3EA	ELBOW 90 DEG, STSF304+PTFE Lined, ERW – SCH 10S
T	25A	2EA	TEE EQUAL, STSF304+PTFE Lined, ERW – SCH 10S
T	20A	1EA	TEE EQUAL, STSF304+PTFE Lined, ERW – SCH 10S
TR	25Ax20A	1EA	TEE RED, STSF304+PTFE Lined, ERW – SCH 10S
F	25A	24EA	FLG LAP JOINT, KS 10K, RF, STS304+PTFE Lined
F	20A	6EA	FLG LAP JOINT, KS 10K, RF, STS304+PTFE Lined
FB	20A	1EA	FLG BLIND, KS 10K, RF, STS304+PTFE Lined
G	25A	14EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
G	20A	5EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
B1	25A	14SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
B1	20A	5SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
VDLO	25A	2EA	KS 10K, RF, Teflon Lined SSC13, EPDM + PTFE Diaphragm, BB, HW
			+LOTO
VDLO	20A	2EA	KS 10K, RF, Teflon Lined SSC13, EPDM + PTFE Diaphragm, BB, HW
			+LOTO
SH	25A	14EA	SHIELD
SH	20A	5EA	SHIELD
PG	20A	1EA	PRESSURE GAUGE, KS 10K, RF

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT: SK hynix

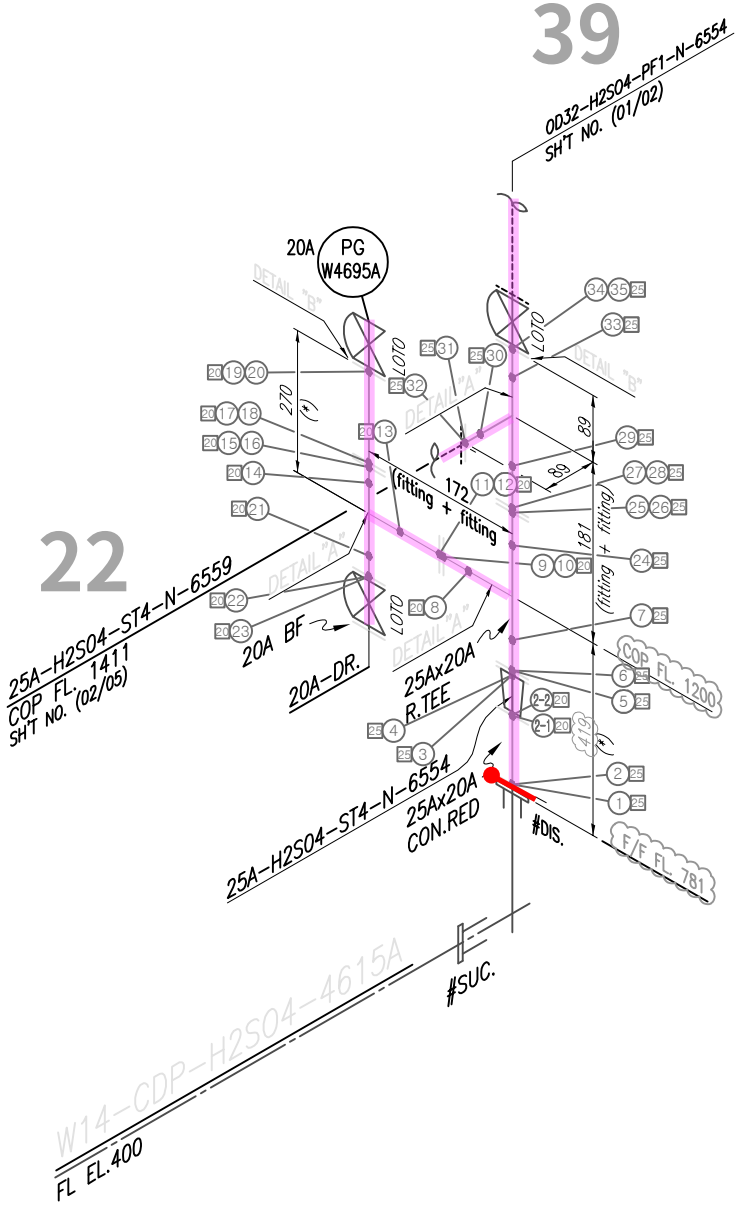
17

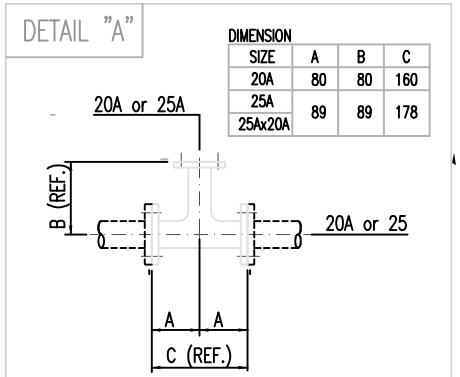
INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14			
	THICK.	–	DESIGN	6.7	60	UNIT	4FA			
			OPERATING	6.1	AMB	REF. PLAN DWG.	CJ-M15W14-PI-A30-040013-A			
						REF. P & ID	CJ-M15W14-PR-A20-000144			

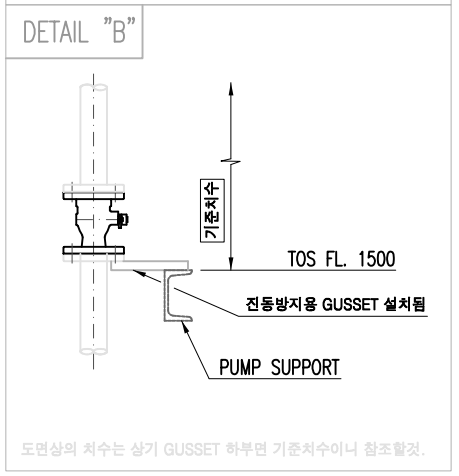
REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래종	안남후	박동영

CONTRACTOR	SK ecoplant	PROJECT NAME	M15X Ph4 Project
COMMODITY	H2S04	JOB NO.	22330D
LINE NO.	25A-H2S04-ST4-N-6553	SH'T NO.	01 / 01
		REV.	0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.







<NOTE>
1. (*) FIELD ADJUST

SK ecoplant

ITEM	SIZE	Q'TY	DESCRIPTION
		ST4	
P	25A	0.2M	PIPE, STS 304TP+PTFE Lined, ERW – SCH 10S
	20A	0.2M	PIPE, STS 304TP+PTFE Lined, ERW – SCH 10S
T	25A	1EA	TEE EQUAL, STSF304+PTFE Lined, ERW – SCH 10S
T	20A	1EA	TEE EQUAL, STSF304+PTFE Lined, ERW – SCH 10S
TR	25Ax20A	1EA	TEE RED, STSF304+PTFE Lined, ERW – SCH 10S
F	25A	7EA	FLG LAP JOINT, KS 10K, RF, STS304+PTFE Lined
F	20A	6EA	FLG LAP JOINT, KS 10K, RF, STS304+PTFE Lined
FB	20A	1EA	FLG BLIND, KS 10K, RF, STS304+PTFE Lined
G	25A	5EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
G	20A	5EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
B1	25A	5SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
B1	20A	5SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
VDLO	25A	1EA	KS 10K, RF, Teflon Lined SSC13, EPDM + PTFE Diaphragm, BB, HW
			+LOTO
VDLO	20A	2EA	KS 10K, RF, Teflon Lined SSC13, EPDM + PTFE Diaphragm, BB, HW
			+LOTO
SH	25A	5EA	SHIELD
SH	20A	5EA	SHIELD
PG	20A	1EA	PRESSURE GAUGE, KS 10K, RF

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT: SK hynix

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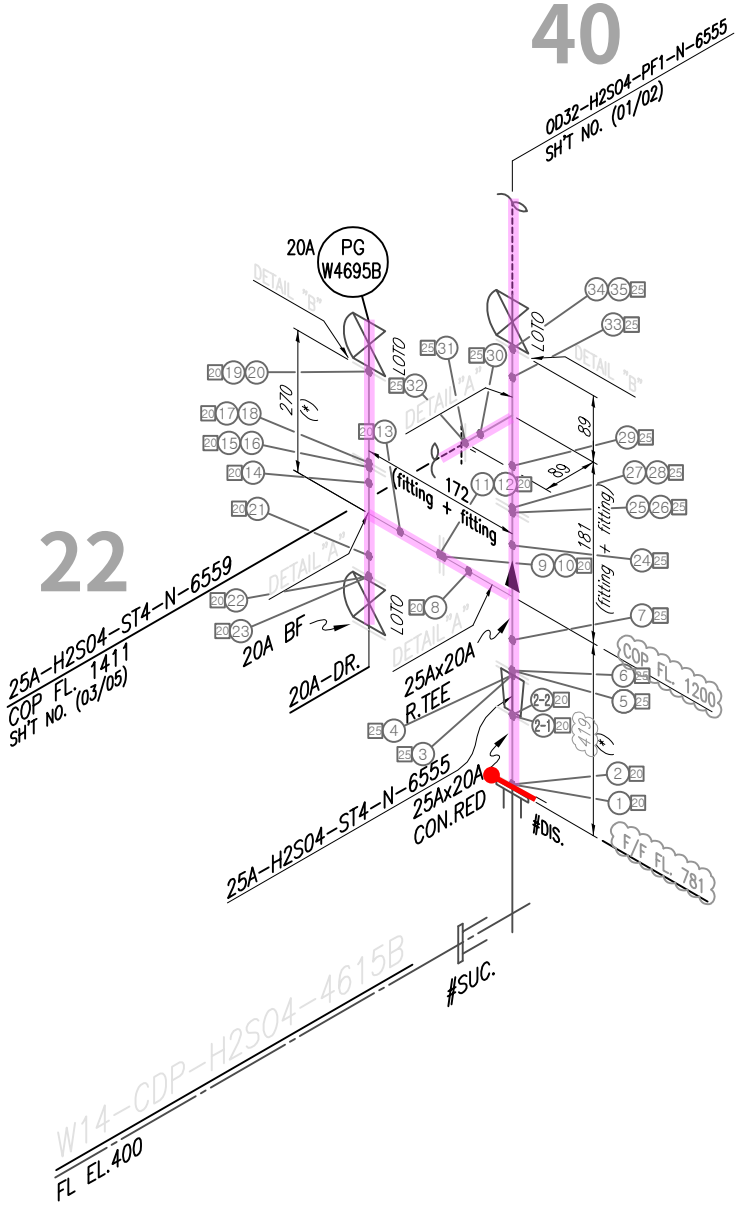
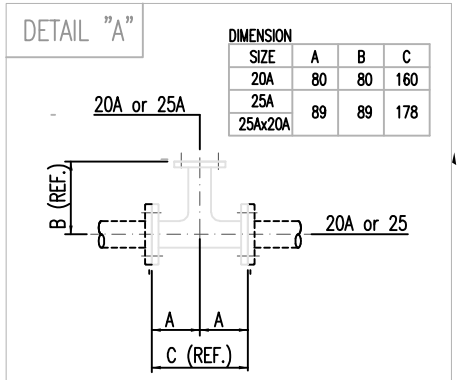
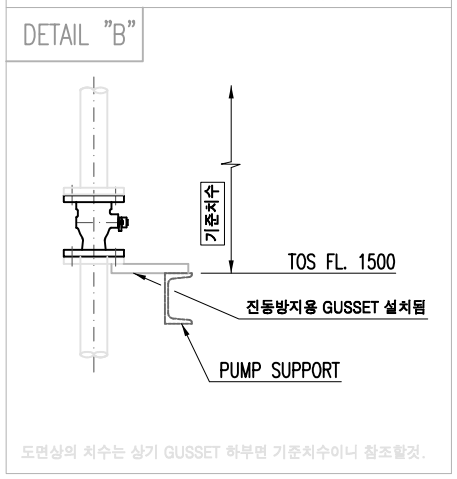
INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14	CONTRACTOR: SK ecoplant	PROJECT NAME: M15X Ph4 Project
	THICK.	–	DESIGN	6.7	60	UNIT	4FA	COMMODITY: H2S04	JOB NO.: 22330D
			OPERATING	6.1	AMB	REF. PLAN DWG.: CJ-M15W14-PI-A30-040013-A		LINE NO.: 25A-H2S04-ST4-N-6554	SH'T NO.: 01 / 01 REV.: 0
						REF. P & ID: CJ-M15W14-PR-A20-000146			

REV.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래종	안남후	박동영

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.







<NOTE>
1. (*) FIELD ADJUST

SK ecoplant

ITEM	SIZE	Q'TY	DESCRIPTION
		ST4	
P	25A	0.2M	PIPE, STS 304TP+PTFE Lined, ERW – SCH 10S
P	20A	0.2M	PIPE, STS 304TP+PTFE Lined, ERW – SCH 10S
T	25A	1EA	TEE EQUAL, STSF304+PTFE Lined, ERW – SCH 10S
T	20A	1EA	TEE EQUAL, STSF304+PTFE Lined, ERW – SCH 10S
TR	25Ax20A	1EA	TEE RED, STSF304+PTFE Lined, ERW – SCH 10S
F	25A	7EA	FLG LAP JOINT, KS 10K, RF, STS304+PTFE Lined
F	20A	6EA	FLG LAP JOINT, KS 10K, RF, STS304+PTFE Lined
FB	20A	1EA	FLG BLIND, KS 10K, RF, STS304+PTFE Lined
G	25A	5EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
G	20A	5EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
B1	25A	5SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
B1	20A	5SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
VDLO	25A	1EA	KS 10K, RF, Teflon Lined SSC13, EPDM + PTFE Diaphragm, BB, HW
			+LOTO
VDLO	20A	2EA	KS 10K, RF, Teflon Lined SSC13, EPDM + PTFE Diaphragm, BB, HW
			+LOTO
SH	25A	5EA	SHIELD
SH	20A	5EA	SHIELD
PG	20A	1EA	PRESSURE GAUGE, KS 10K, RF
RE	25Ax20A	1EA	CON. RED, STSF304+PTFE Lined, ERW – SCH 10S

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT: SK hynix

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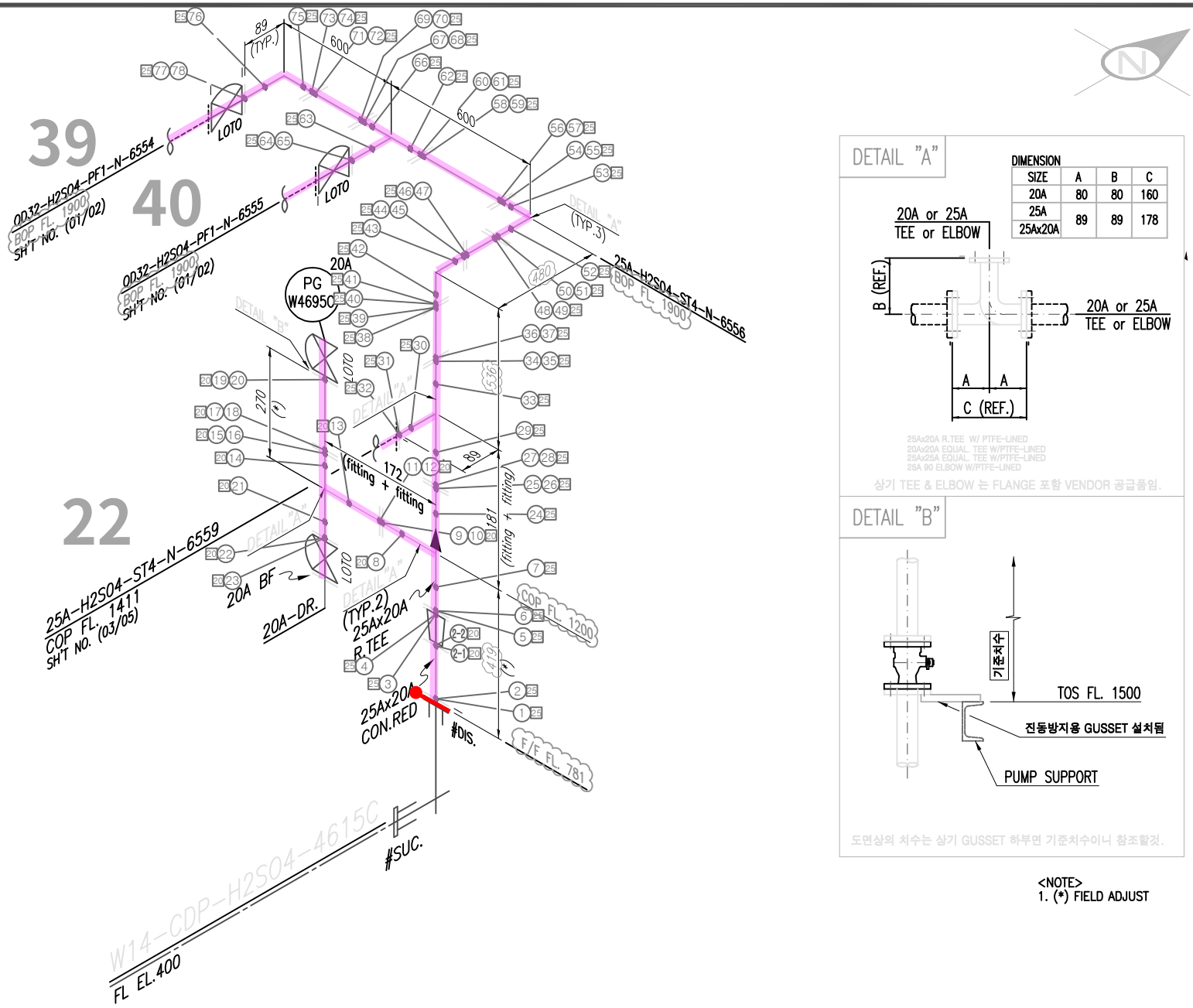
INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14			
	THICK.	–	DESIGN	6.7	60	UNIT	4FA			
			OPERATING	6.1	AMB	REF. PLAN DWG.	CJ-M15W14-PI-A30-040013-A			
						REF. P & ID	CJ-M15W14-PR-A20-000146			

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래종	안남후	박동영

CONTRACTOR: SK ecoplant
PROJECT NAME: M15X Ph4 Project
COMMODITY: H2S04
JOB NO.: 22330D
LINE NO.: 25A-H2S04-ST4-N-6555
SH'T NO.: 01 / 01
REV.: 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



<NOTE>
1. (*) FIELD ADJUST

ITEM	SIZE	Q'TY	DESCRIPTION
		ST4	
P	25A	1.8M	PIPE, STS 304TP+PTFE Lined, ERW – SCH 10S
P	20A	0.2M	PIPE, STS 304TP+PTFE Lined, ERW – SCH 10S
E90	25A	3EA	ELBOW 90 DEG, STSF304+PTFE Lined, ERW – SCH 10S
T	25A	2EA	TEE EQUAL, STSF304+PTFE Lined, ERW – SCH 10S
T	20A	1EA	TEE EQUAL, STSF304+PTFE Lined, ERW – SCH 10S
TR	25Ax20A	1EA	TEE RED, STSF304+PTFE Lined, ERW – SCH 10S
F	25A	24EA	FLG LAP JOINT, KS 10K, RF, STS304+PTFE Lined
F	20A	6EA	FLG LAP JOINT, KS 10K, RF, STS304+PTFE Lined
FB	20A	1EA	FLG BLIND, KS 10K, RF, STS304+PTFE Lined
G	25A	14EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
G	20A	5EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
B1	25A	14SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
B1	20A	5SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
VDLO	25A	2EA	KS 10K, RF, Teflon Lined SSC13, EPDM + PTFE Diaphragm, BB, HW
			+LOTO
VDLO	20A	2EA	KS 10K, RF, Teflon Lined SSC13, EPDM + PTFE Diaphragm, BB, HW
			+LOTO
SH	25A	14EA	SHIELD
SH	20A	5EA	SHIELD
PG	20A	1EA	PRESSURE GAUGE, KS 10K, RF
RE	25Ax20A	1EA	CON. RED, STSF304+PTFE Lined, ERW – SCH 10S

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT: SK hynix

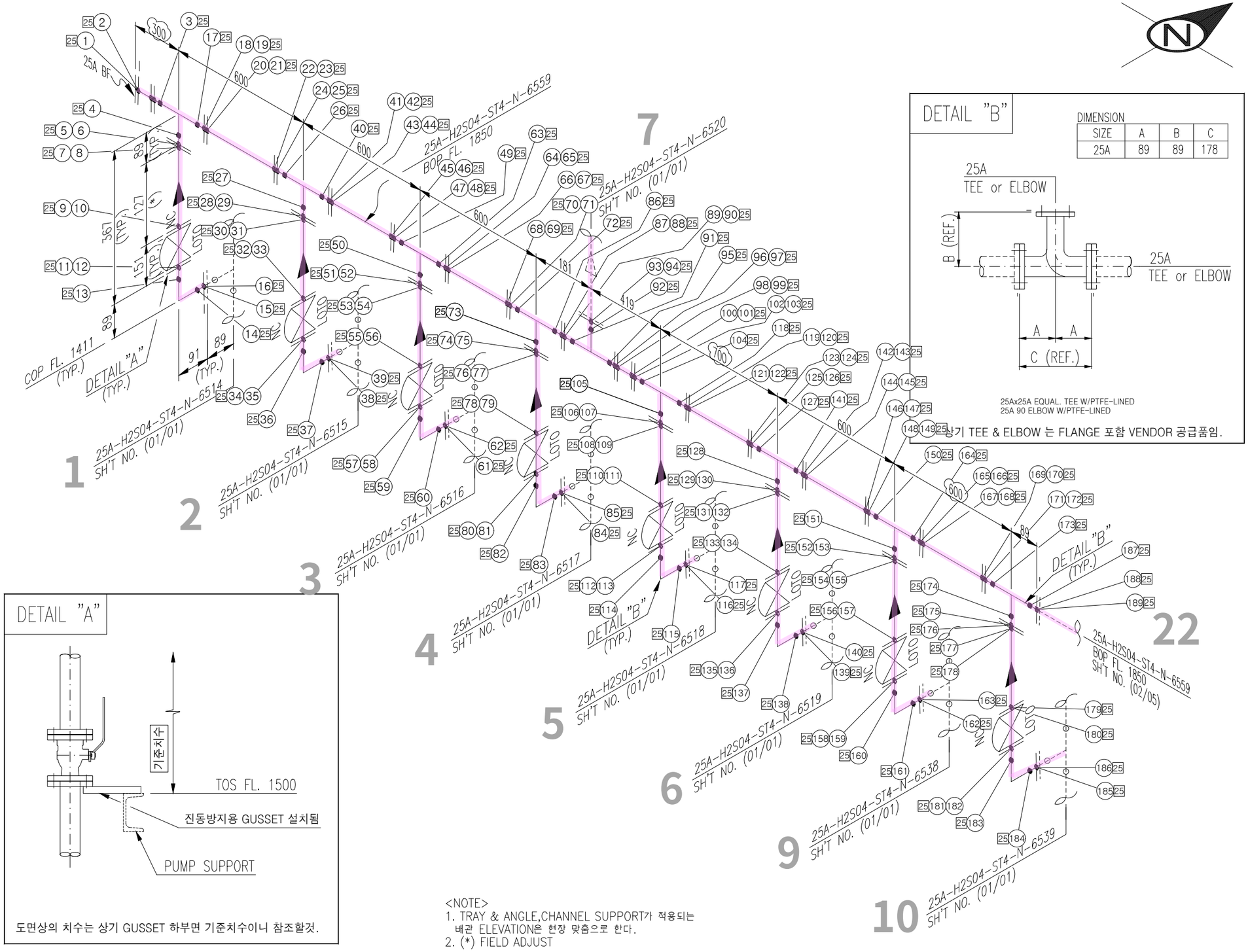
20

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14			
	THICK.	–	DESIGN	6.7	60	UNIT	4FA			
			OPERATING	6.1	AMB	REF. PLAN DWG.	CJ-M15W14-PI-A30-040013-A			
						REF. P & ID	CJ-M15W14-PR-A20-000146			

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래종	안남후	박동영

CONTRACTOR	SK ecoplant	PROJECT NAME	M15X Ph4 Project
COMMODITY	H2S04	JOB NO.	22330D
LINE NO.	25A-H2S04-ST4-N-6556	SH'T NO.	01 / 01
		REV.	0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



DETAIL "B"

DIMENSION

SIZE	A	B	C
25A	89	89	178

25Ax25A EQUAL. TEE W/PTFE-LINED
25A 90 ELBOW W/PTFE-LINED

상기 TEE & ELBOW 는 FLANGE 포함 VENDOR 공급품임.

DETAIL "A"

도면상의 치수는 상기 GUSSET 하부면 기준치수이니 참조할것.

<NOTE>
1. TRAY & ANGLE,CHANNEL SUPPORT가 적용되는 배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

ITEM	SIZE	Q'TY	DESCRIPTION
		ST4	
P	25A	5.6M	PIPE, STS 304TP+PTFE Lined, ERW – SCH 10S
E90	25A	8EA	ELBOW 90 DEG, STSF304+PTFE Lined, ERW – SCH 10S
T	25A	9EA	TEE EQUAL, STSF304+PTFE Lined, ERW – SCH 10S
F	25A	48EA	FLG SO, KS 10K, RF, STS304+PTFE Lined
FB	25A	1EA	FLG BLIND, KS 10K, RF, STS304+PTFE Lined
G	25A	49EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
B1	25A	49SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
VDLO	25A	8EA	KS 10K, RF, Teflon Lined SSC13, EPDM + PTFE Diaphragm, BB, HW
			+LOTO
SH	25A	49EA	SHIELD

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT SK hynix

21

INSULATION	CODE	N
	THICK.	–

CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
DESIGN	6.7	60
OPERATING	6.1	AMB

AREA	W14
UNIT	4FA
REF. PLAN DWG.	CJ-M15W14-PI-A30-040013-A
REF. P & ID	CJ-M15W14-PR-A20-000145

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
1	25.03.31	REVISED AS MARKED	황래종	안남후	박동영
0	25.01.24	APPROVED FOR CONSTRUCTION	황래종	안남후	박동영

CONTRACTOR SK ecoplant

PROJECT NAME: M15X Ph4 Project

COMMODITY: H2S04

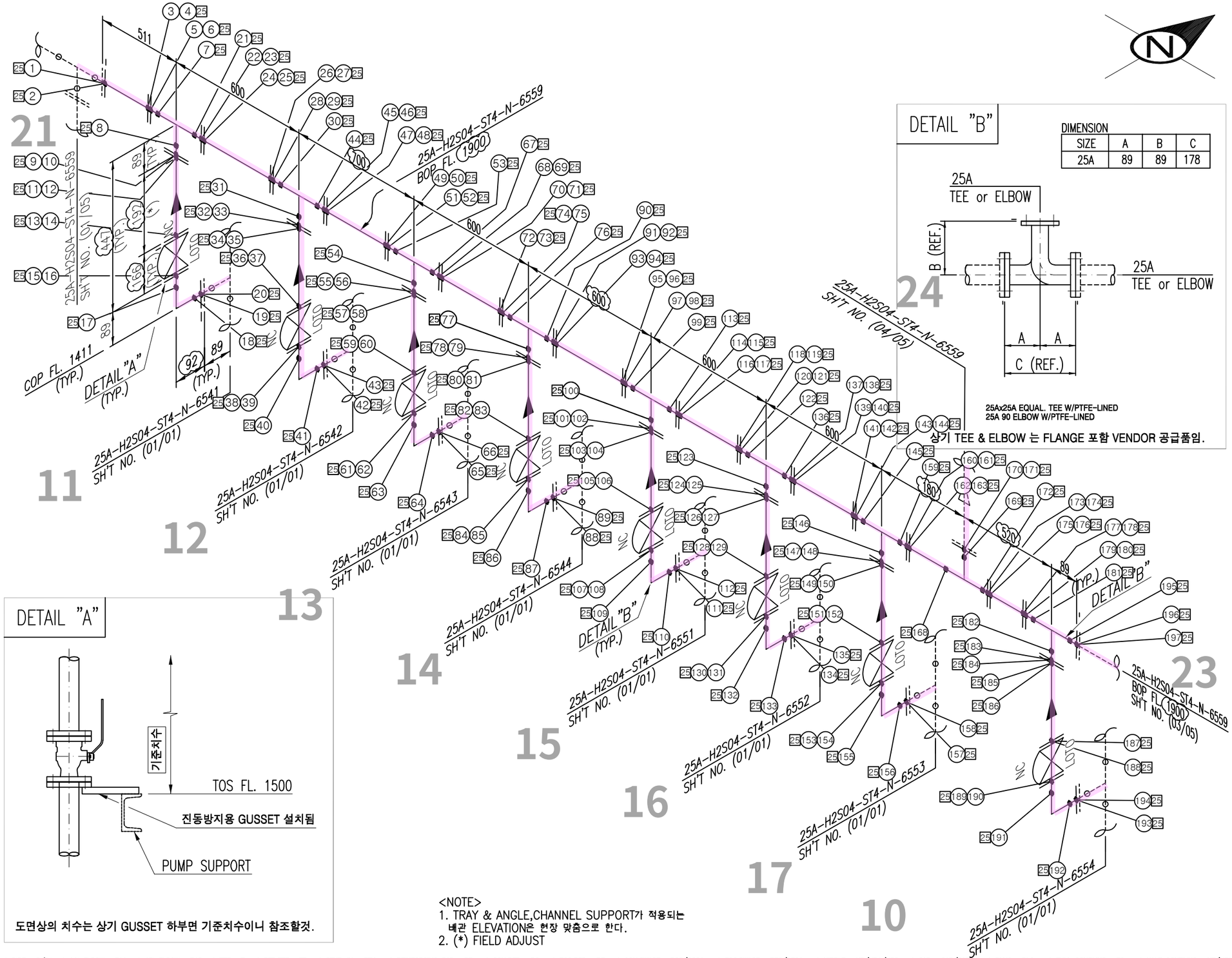
JOB NO.: 22330D

LINE NO.: 25A-H2S04-ST4-N-6559

SH'T NO.: 01 / 05

REV.: 1

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



DETAIL "B"

DIMENSION

SIZE	A	B	C
25A	89	89	178

25Ax25A EQUAL. TEE W/PTFE-LINED
25A 90 ELBOW W/PTFE-LINED

상기 TEE & ELBOW 는 FLANGE 포함 VENDOR 공급품임.

DETAIL "A"

도면상의 치수는 상기 GUSSET 하부면 기준치수이니 참조할것.

<NOTE>
1. TRAY & ANGLE,CHANNEL SUPPORT가 적용되는 배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

ITEM	SIZE	Q'TY	DESCRIPTION
	ST4		
P	25A	4.6M	PIPE, STS 304TP+PTFE Lined, ERW – SCH 10S
E90	25A	8EA	ELBOW 90 DEG, STSF304+PTFE Lined, ERW – SCH 10S
T	25A	9EA	TEE EQUAL, STSF304+PTFE Lined, ERW – SCH 10S
F	25A	77EA	FLG SO, KS 10K, RF, STS304+PTFE Lined
G	25A	42EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
B1	25A	42SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
VDLO	25A	8EA	KS 10K, RF, Teflon Lined SSC13, EPDM + PTFE Diaphragm, BB, HW
			+LOTO
SH	25A	42EA	SHIELD

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT SK hynix

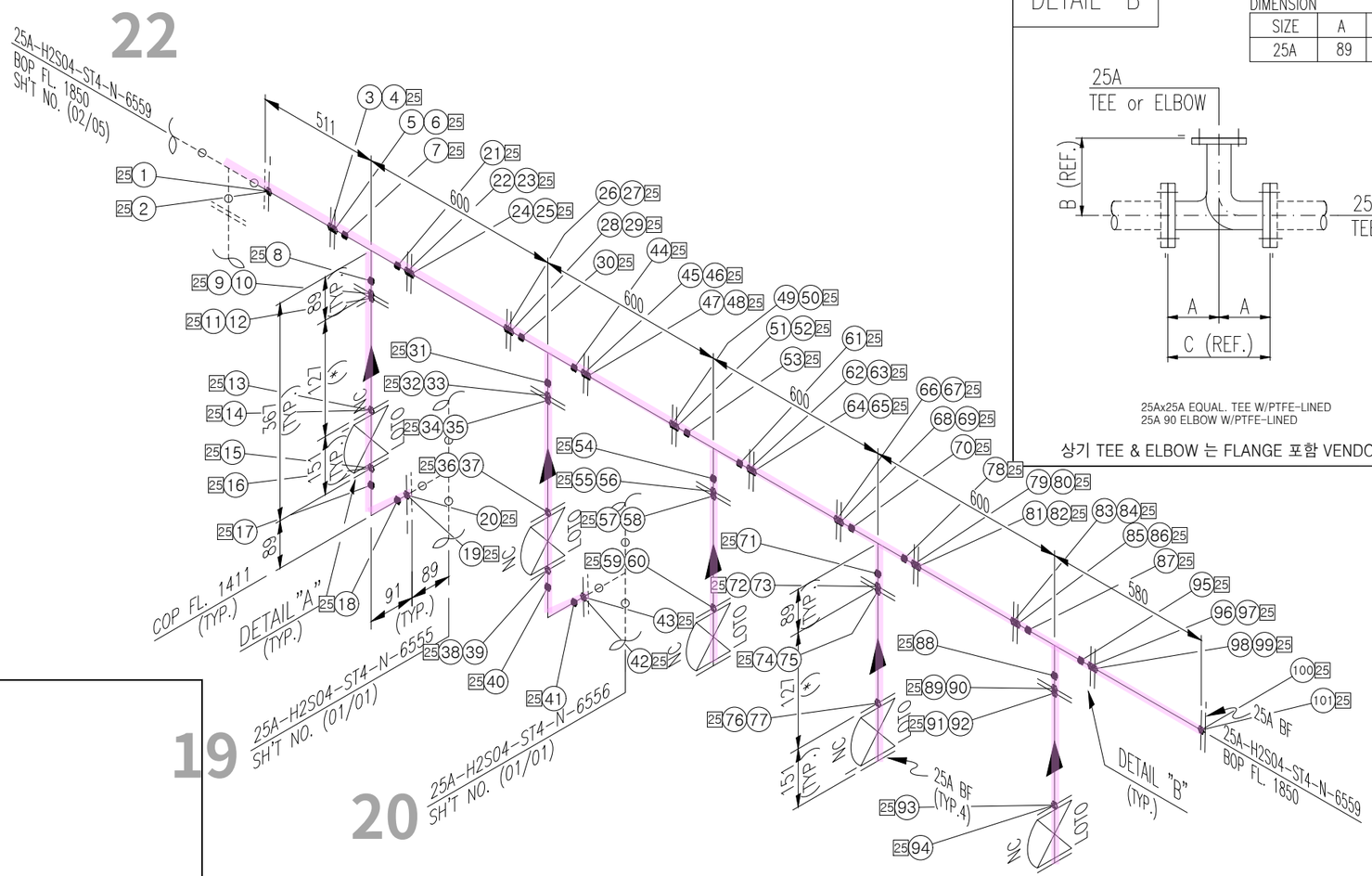
INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
	THICK.	–	DESIGN	6.7	60
			OPERATING	6.1	AMB

AREA W14
UNIT 4FA
REF. PLAN DWG. CJ-M15W14-PI-A30-040013-A
REF. P & ID CJ-M15W14-PR-A20-000145

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
1	25.03.31	REVISED AS MARKED	황래종	안남후	박동영
0	25.01.24	APPROVED FOR CONSTRUCTION	황래종	안남후	박동영

CONTRACTOR SK ecoplant
PROJECT NAME M15X Ph4 Project
COMMODITY H2S04
JOB NO. 22330D
LINE NO. 25A-H2S04-ST4-N-6559
SH'T NO. 02 / 05
REV. 1

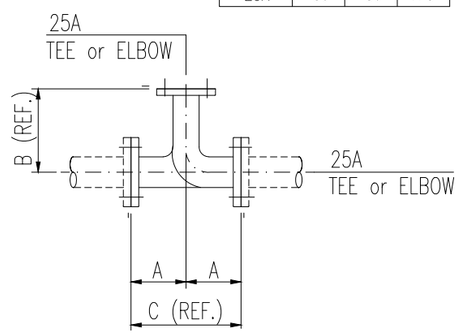
경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



DETAIL "B"

DIMENSION

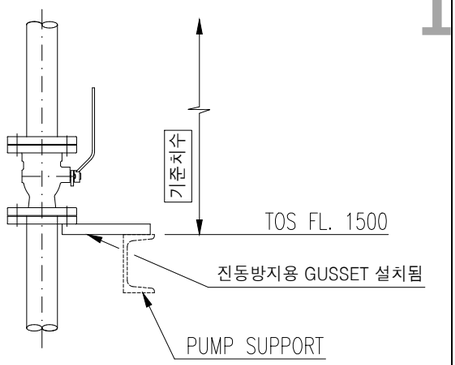
SIZE	A	B	C
25A	89	89	178



25Ax25A EQUAL. TEE W/PTFE-LINED
25A 90 ELBOW W/PTFE-LINED

상기 TEE & ELBOW 는 FLANGE 포함 VENDOR 공급품임.

DETAIL "A"



도면상의 치수는 상기 GUSSET 하부면 기준치수이니 참조할것.

<NOTE>
1. TRAY & ANGLE,CHANNEL SUPPORT가 적용되는
   배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

ITEM	SIZE	Q'TY	DESCRIPTION
		ST4	
P	25A	3.2M	PIPE, STS 304TP+PTFE Lined, ERW – SCH 10S
E90	25A	2EA	ELBOW 90 DEG, STSF304+PTFE Lined, ERW – SCH 10S
T	25A	5EA	TEE EQUAL, STSF304+PTFE Lined, ERW – SCH 10S
F	25A	41EA	FLG LAP JOINT, KS 10K, RF, STS304+PTFE Lined
FB	25A	4EA	FLG BLIND, KS 10K, RF, STS304+PTFE Lined
G	25A	26EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
B1	25A	26SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
VDLO	25A	6EA	KS 10K, RF, Teflon Lined SSC13, EPDM + PTFE Diaphrogm, BB, HW
			+LOTO
SH	25A	26EA	SHIELD

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT SK hynix

23

INSULATION	CODE	N
	THICK.	–

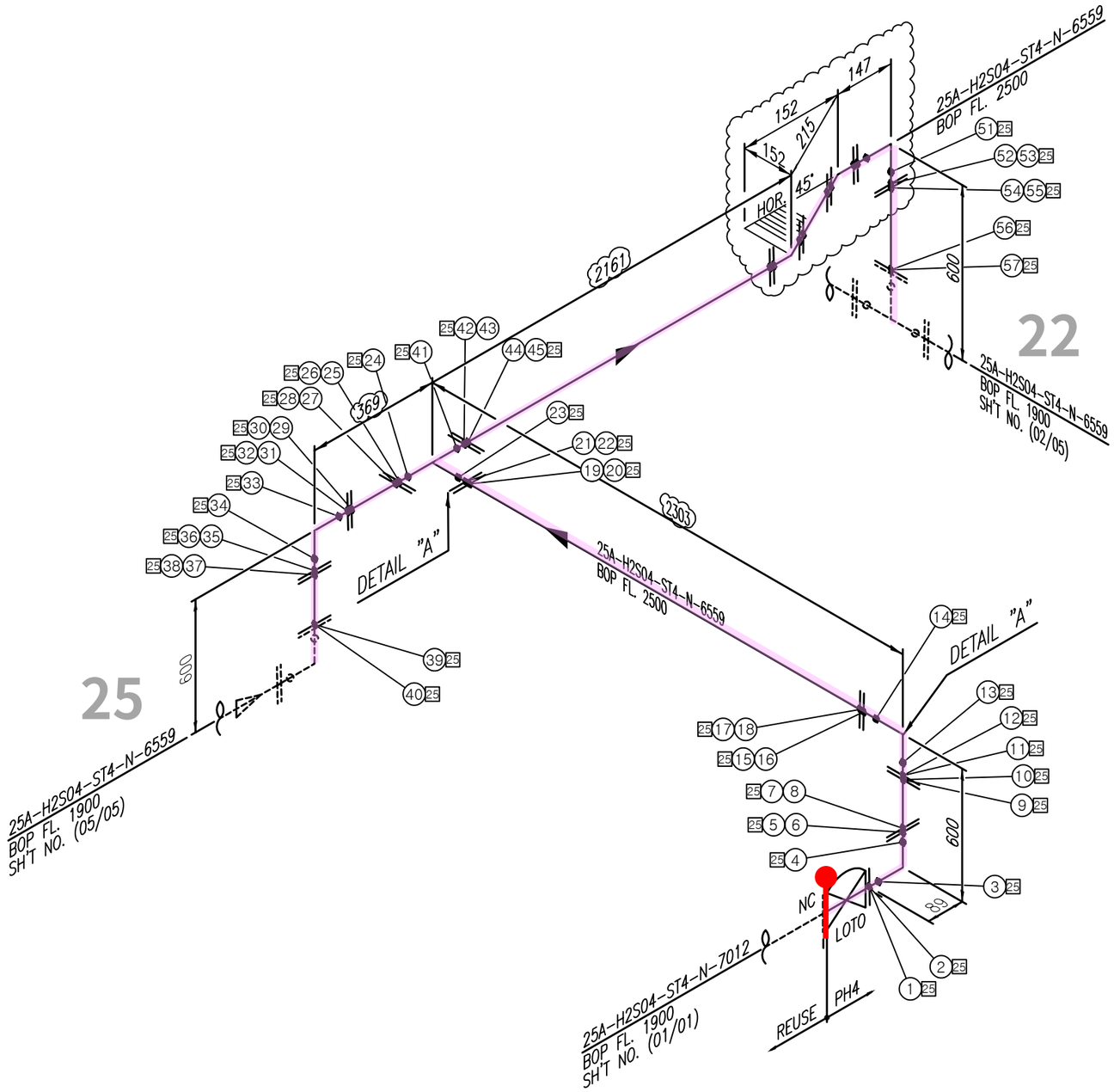
CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
DESIGN	6.7	60
OPERATING	6.1	AMB

AREA W14
UNIT 4FA
REF. PLAN DWG. CJ-M15W14-PI-A30-040013-A
REF. P & ID CJ-M15W14-PR-A20-000145

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래종	안남후	박동영

CONTRACTOR SK ecoplant
PROJECT NAME M15X Ph4 Project
COMMODITY H2S04
JOB NO. 22330D
LINE NO. 25A-H2S04-ST4-N-6559
SH'T NO. 03 / 05
REV. 0

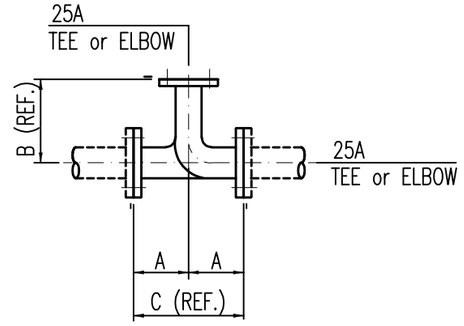
경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



DETAIL "A"

DIMENSION

SIZE	A	B	C
25A	89	89	178



25Ax25A EQUAL. TEE W/PTFE-LINED
25A 90 ELBOW W/PTFE-LINED

상기 TEE & ELBOW 는 FLANGE 포함 VENDOR 공급품임.

<NOTE>
1. (*) FIELD ADJUST

ITEM	SIZE	Q'TY	DESCRIPTION
		ST4	
P	25A	5.6M	PIPE, STS 304TP+PTFE Lined, ERW – SCH 10S
E90	25A	4EA	ELBOW 90 DEG, STSF304+PTFE Lined, ERW – SCH 10S
T	25A	1EA	TEE EQUAL, STSF304+PTFE Lined, ERW – SCH 10S
F	25A	29EA	FLG LAP JOINT, KS 10K, RF, STS304+PTFE Lined
G	25A	16EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
B1	25A	16SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
VDLO	25A	1EA	KS 10K, RF, Teflon Lined SSC13, EPDM + PTFE Diaphragm, BB, HW
			+LOTO
SH	25A	16EA	SHIELD
E45	25A	2EA	ELBOW 45 DEG, STSF304+PTFE Lined, ERW – SCH 10S

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
	THICK.	–	DESIGN	6.7	60
			OPERATING	6.1	AMB

AREA W14
UNIT 4FA
REF. PLAN DWG. CJ-M15W14-PI-A30-040013-A
REF. P & ID CJ-M15W14-PR-A20-000145

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래종	안남후	박동영

CLIENT SK hynix

CONTRACTOR SK ecoplant

PROJECT NAME M15X Ph4 Project

COMMODITY H2S04

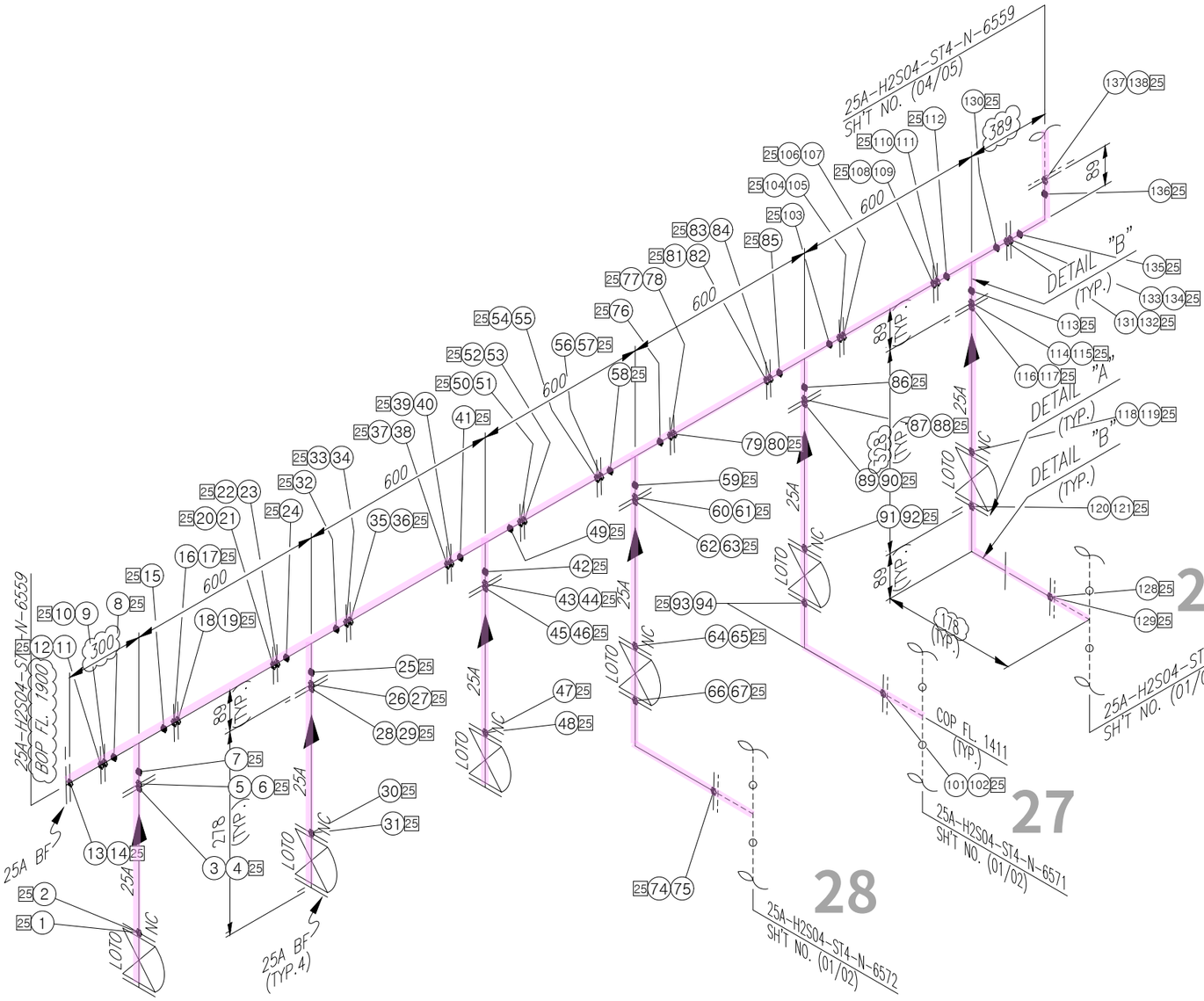
JOB NO. 22330D

LINE NO. 25A-H2S04-ST4-N-6559

SH'T NO. 04 / 05

REV. 0

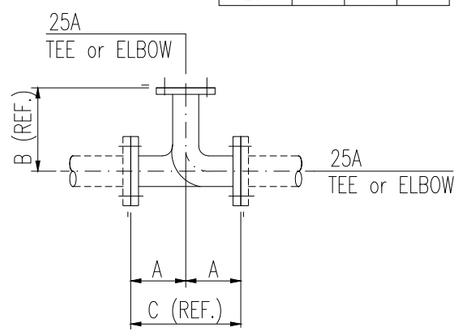
경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



DETAIL "B"

DIMENSION

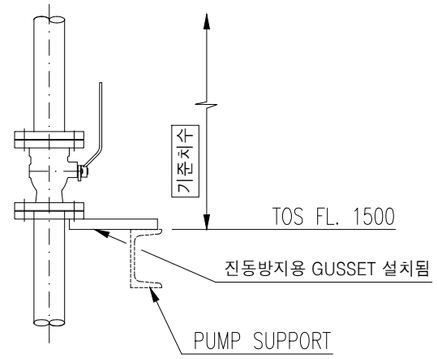
SIZE	A	B	C
25A	89	89	178



25Ax25A EQUAL. TEE W/PTFE-LINED
25A 90 ELBOW W/PTFE-LINED

상기 TEE & ELBOW 는 FLANGE 포함 VENDOR 공급품임.

DETAIL "A"



도면상의 치수는 상기 GUSSET 하부면 기준치수이니 참조할것.

<NOTE>
1. TRAY & ANGLE,CHANNEL SUPPORT가 적용되는 배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

ITEM	SIZE	Q'TY	DESCRIPTION
		ST4	
P	25A	3.9M	PIPE, STS 304TP+PTFE Lined, ERW – SCH 10S
E90	25A	4EA	ELBOW 90 DEG, STSF304+PTFE Lined, ERW – SCH 10S
T	25A	6EA	TEE EQUAL, STSF304+PTFE Lined, ERW – SCH 10S
F	25A	56EA	FLG LAP JOINT, KS 10K, RF, STS304+PTFE Lined
FB	25A	4EA	FLG BLIND, KS 10K, RF, STS304+PTFE Lined
G	25A	38EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
B1	25A	38SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
VDLO	25A	6EA	KS 10K, RF, Teflon Lined SSC13, EPDM + PTFE Diaphragm, BB, HW
			+LOTO
SH	25A	38EA	SHIELD

INSULATION	CODE	N
	THICK.	–

CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
DESIGN	6.7	60
OPERATING	6.1	AMB

AREA W14
UNIT 4FA
REF. PLAN DWG. CJ-M15W14-PI-A30-040013-A
REF. P & ID CJ-M15W14-PR-A20-000146

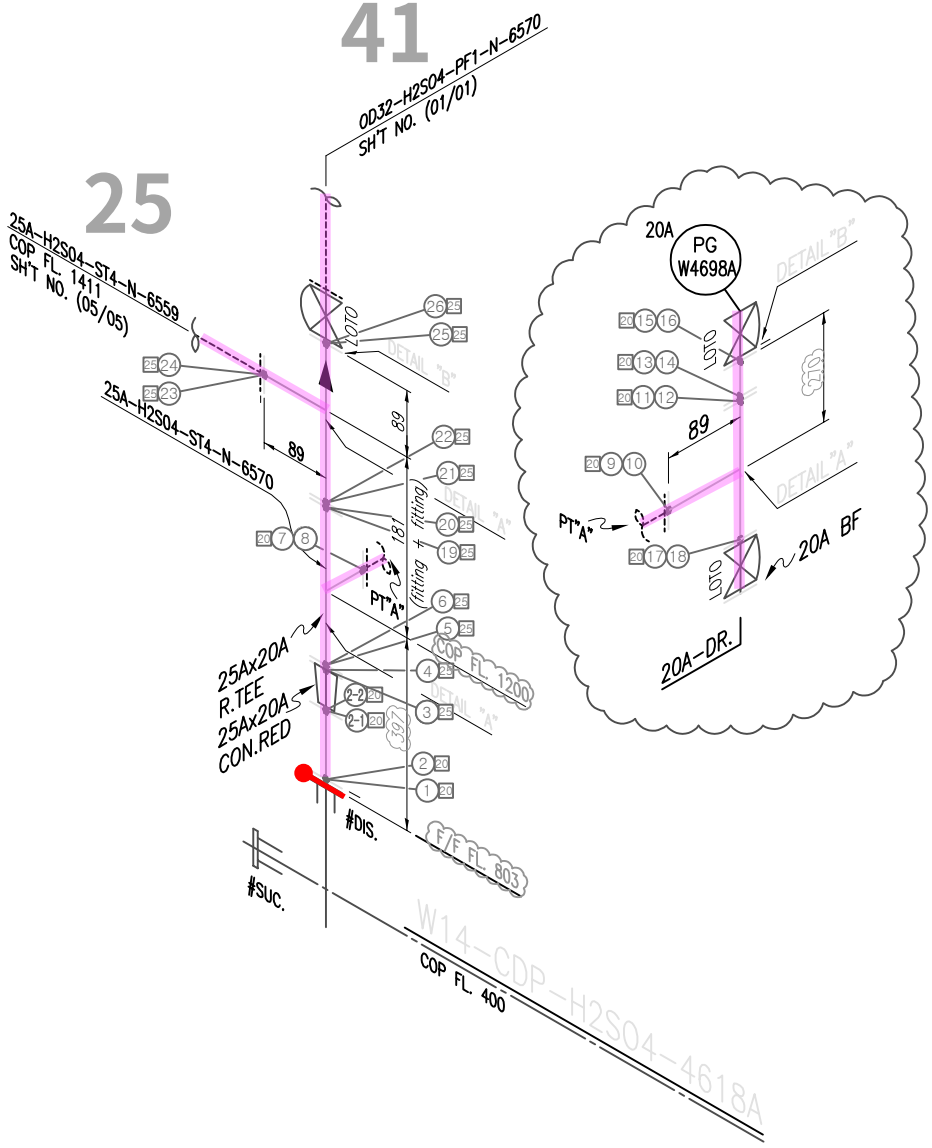
REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래종	안남후	박동영

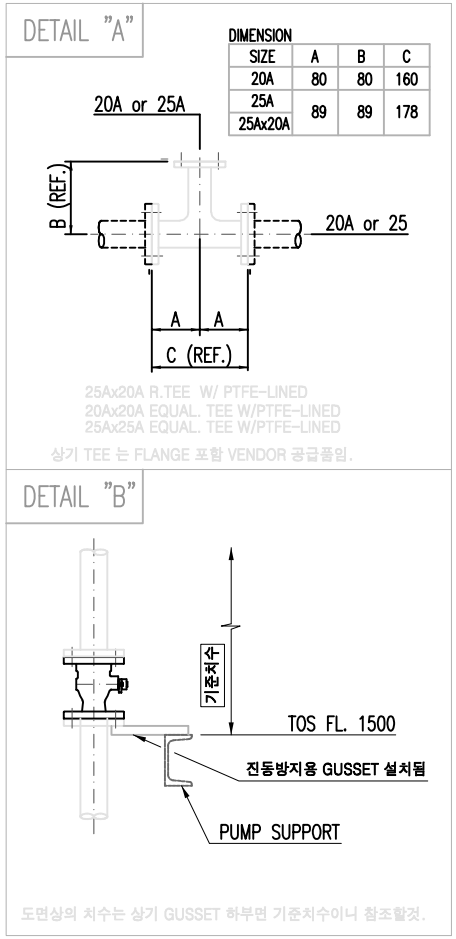
CLIENT SK hynix

25

CONTRACTOR SK ecoplant
PROJECT NAME M15X Ph4 Project
COMMODITY H2S04
JOB NO. 22330D
LINE NO. 25A-H2S04-ST4-N-6559
SH'T NO. 05 / 05
REV. 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.





<NOTE>
1. (*) FIELD ADJUST

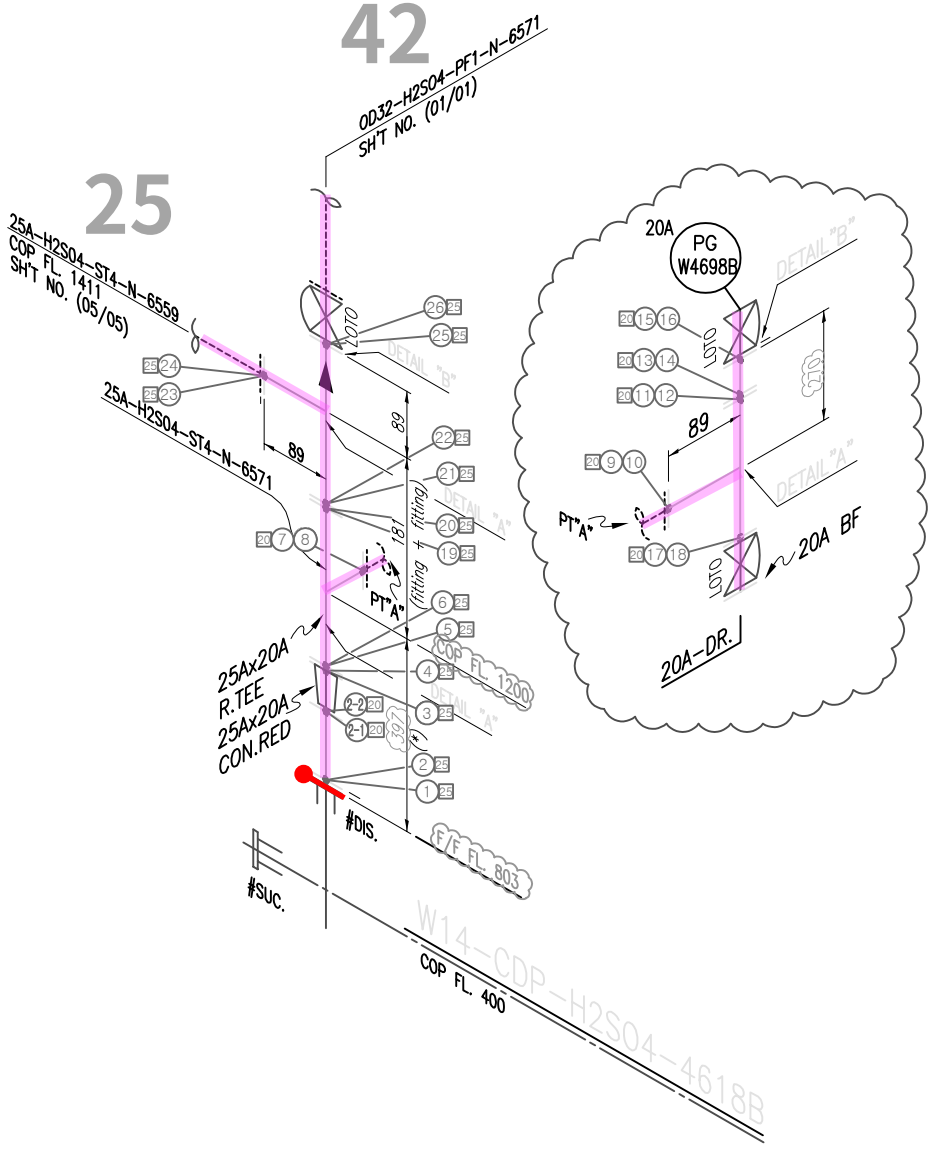
ITEM	SIZE	Q'TY	DESCRIPTION
		ST4	
P	25A	0.2M	PIPE, STS 304TP+PTFE Lined, ERW - SCH 10S, With FLG
P	20A	0.2M	PIPE, STS 304TP+PTFE Lined, ERW - SCH 10S, With FLG
T	25A	1EA	TEE EQUAL, STSF304+PTFE Lined, ERW - SCH 10S, With FLG
T	20A	1EA	TEE EQUAL, STSF304+PTFE Lined, ERW - SCH 10S, With FLG
TR	25Ax20A	1EA	TEE RED, STSF304+PTFE Lined, ERW - SCH 10S, With FLG
F	25A	7EA	FLG SO, KS 10K, RF, STS304+PTFE Lined
F	20A	8EA	FLG SO, KS 10K, RF, STS304+PTFE Lined
FB	20A	1EA	FLG BLIND, KS 10K, RF, STS304+PTFE Lined
G	25A	5EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
G	20A	6EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
B1	25A	4SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
B1	20A	5SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
VDLO	25A	1EA	KS 10K, RF, Teflon Lined SSC13, EPDM + PTFE Diaphragm, BB, HW
			+LOTO
VDLO	20A	2EA	KS 10K, RF, Teflon Lined SSC13, EPDM + PTFE Diaphragm, BB, HW
			+LOTO
SH	25A	4EA	SHIELD
SH	20A	5EA	SHIELD
PG	20A	1EA	PRESSURE GAUGE, KS 10K, RF
RE	25Ax20A	1EA	CON. RED, STSF304+PTFE Lined, ERW - SCH 10S

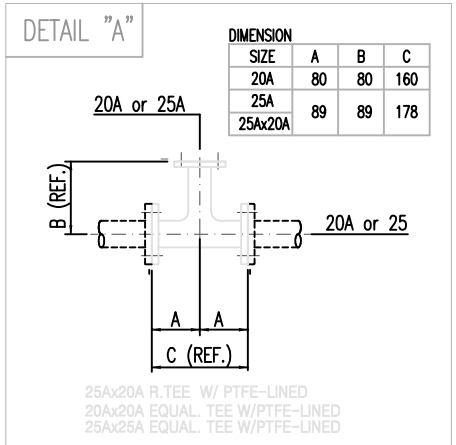
* PIPE : P/P1 * 90' ELBOW : E90 * 45' ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

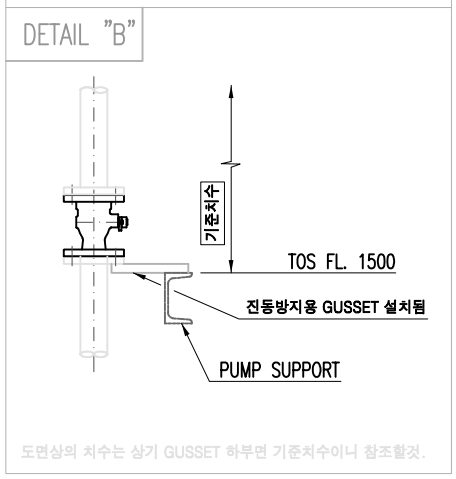
INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14				
	THICK.	-	DESIGN	1.0	60	UNIT	4FA				
			OPERATING	ATM	AMB	REF. PLAN DWG.	CJ-M15W14-PI-A30-040013-A				
								0	25.01.24	APPROVED FOR CONSTRUCTION	황래종 안남후 박동영
						REF. P. & ID	CJ-M15W14-PR-A20-000147	REV. NO.	DATE	DESCRIPTION	BY CHK APP'D

CONTRACTOR: SK ecoplant
COMMODITY: H2S04
LINE NO.: 25A-H2S04-ST4-N-6570
PROJECT NAME: M15X Ph4 Project
JOB NO.: 22330D
SH'T NO.: 01 / 01
REV.: 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.







<NOTE>
1. (*) FIELD ADJUST

SK ecoplant

ITEM	SIZE	Q'TY	DESCRIPTION
		ST4	
P	25A	0.2M	PIPE, STS 304TP+PTFE Lined, ERW - SCH 10S, With FLG
P	20A	0.2M	PIPE, STS 304TP+PTFE Lined, ERW - SCH 10S, With FLG
T	25A	1EA	TEE EQUAL, STSF304+PTFE Lined, ERW - SCH 10S, With FLG
T	20A	1EA	TEE EQUAL, STSF304+PTFE Lined, ERW - SCH 10S, With FLG
TR	25Ax20A	1EA	TEE RED, STSF304+PTFE Lined, ERW - SCH 10S, With FLG
F	25A	7EA	FLG SO, KS 10K, RF, STS304+PTFE Lined
F	20A	8EA	FLG SO, KS 10K, RF, STS304+PTFE Lined
FB	20A	1EA	FLG BLIND, KS 10K, RF, STS304+PTFE Lined
G	25A	5EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
G	20A	6EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
B1	25A	4SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
B1	20A	5SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
VDLO	25A	1EA	KS 10K, RF, Teflon Lined SSC13, EPDM + PTFE Diaphragm, BB, HW
			+LOTO
VDLO	20A	2EA	KS 10K, RF, Teflon Lined SSC13, EPDM + PTFE Diaphragm, BB, HW
			+LOTO
SH	25A	4EA	SHIELD
SH	20A	5EA	SHIELD
PG	20A	1EA	PRESSURE GAUGE, KS 10K, RF
RE	25Ax20A	1EA	CON. RED, STSF304+PTFE Lined, ERW - SCH 10S

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT: SK hynix

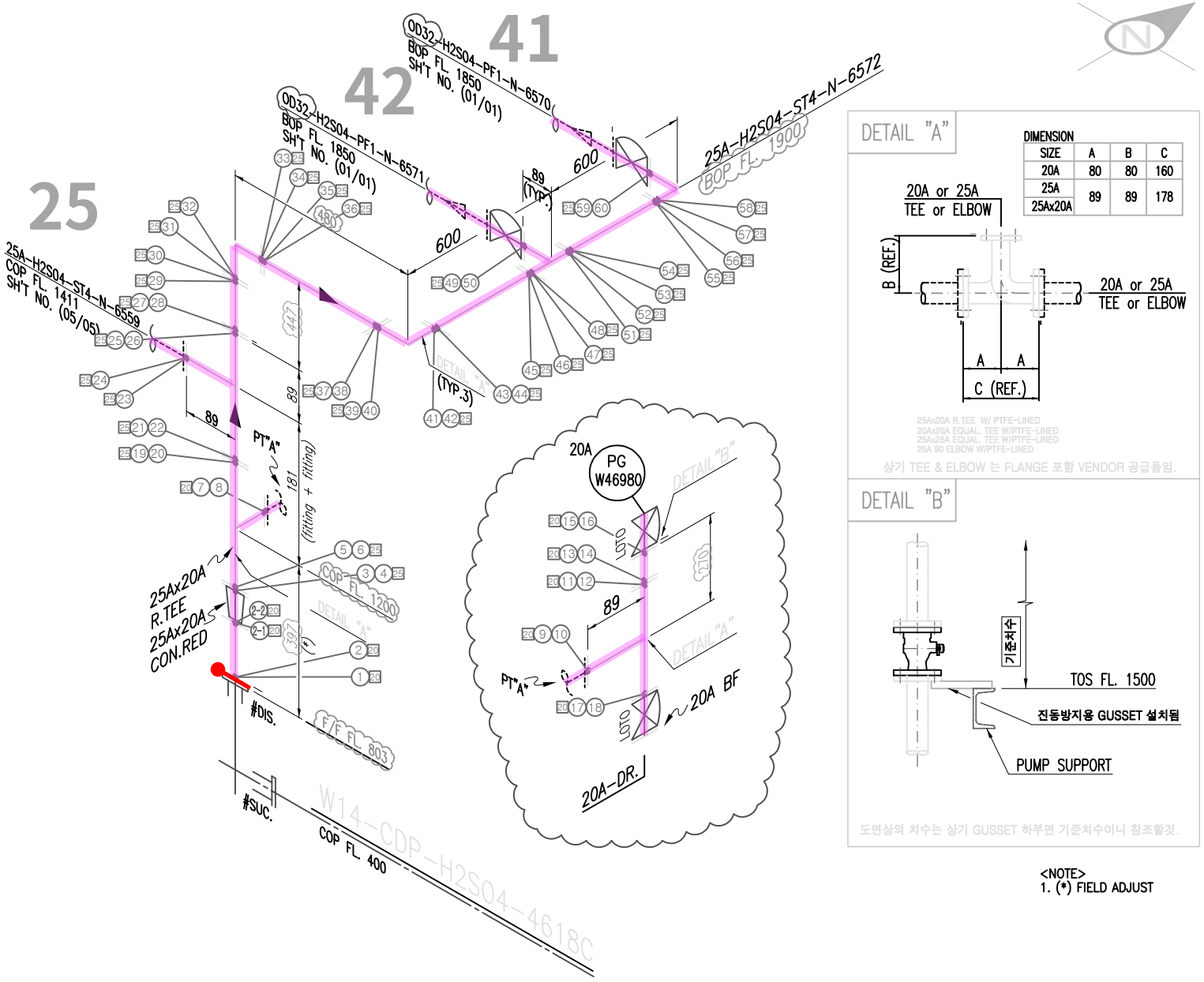
27

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14			
	THICK.	-	DESIGN	1.0	60	UNIT	4FA			
			OPERATING	ATM	AMB	REF. PLAN DWG.	CJ-M15W14-PI-A30-040013-A			
						REF. P & ID	CJ-M15W14-PR-A20-000147			

REV.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래종	안남후	박동영

CONTRACTOR: SK ecoplant
PROJECT NAME: M15X Ph4 Project
COMMODITY: H2S04
JOB NO.: 22330D
LINE NO.: 25A-H2S04-ST4-N-6571
SH'T NO.: 01 / 01
REV.: 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



DETAIL "A"

DIMENSION

SIZE	A	B	C
20A	80	80	160
25A 25Ax20A	89	89	178

25Ax20A R.TEE W/ PTFE-LINED
20Ax20A EQUAL. TEE W/PTFE-LINED
25Ax25A EQUAL. TEE W/PTFE-LINED
25A 90 ELBOW W/PTFE-LINED

상기 TEE & ELBOW 는 FLANGE 포함 VENDOR 공급품임.

DETAIL "B"

도면상의 치수는 상기 GUSSET 하부면 기준치수이니 참조할것.

<NOTE>
1. (*) FIELD ADJUST

SK ecoplant

ITEM	SIZE	Q'TY	DESCRIPTION
		ST4	
P	25A	1.6M	PIPE, STS 304TP+PTFE Lined, ERW - SCH 10S, With FLG
P	20A	0.2M	PIPE, STS 304TP+PTFE Lined, ERW - SCH 10S, With FLG
E90	25A	3EA	ELBOW 90 DEG, STSF304+PTFE Lined, ERW - SCH 10S, With FLG
T	25A	2EA	TEE EQUAL, STSF304+PTFE Lined, ERW - SCH 10S, With FLG
T	20A	1EA	TEE EQUAL, STSF304+PTFE Lined, ERW - SCH 10S, With FLG
TR	25Ax20A	1EA	TEE RED, STSF304+PTFE Lined, ERW - SCH 10S, With FLG
F	25A	23EA	FLG SO, KS 10K, RF, STS304+PTFE Lined
F	20A	7EA	FLG SO, KS 10K, RF, STS304+PTFE Lined
FB	20A	1EA	FLG BLIND, KS 10K, RF, STS304+PTFE Lined
G	25A	12EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
G	20A	6EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
B1	25A	12SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
B1	20A	6SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
VDLO	25A	2EA	KS 10K, RF, Teflon Lined SSC13, EPDM + PTFE Diaphragm, BB, HW
			+LOTO
VDLO	20A	2EA	KS 10K, RF, Teflon Lined SSC13, EPDM + PTFE Diaphragm, BB, HW
			+LOTO
SH	25A	13EA	SHIELD
SH	20A	5EA	SHIELD
PG	20A	1EA	PRESSURE GAUGE, KS 10K, RF
RE	25Ax20A	1EA	CON. RED, STSF304+PTFE Lined, ERW - SCH 10S

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT: SK hynix

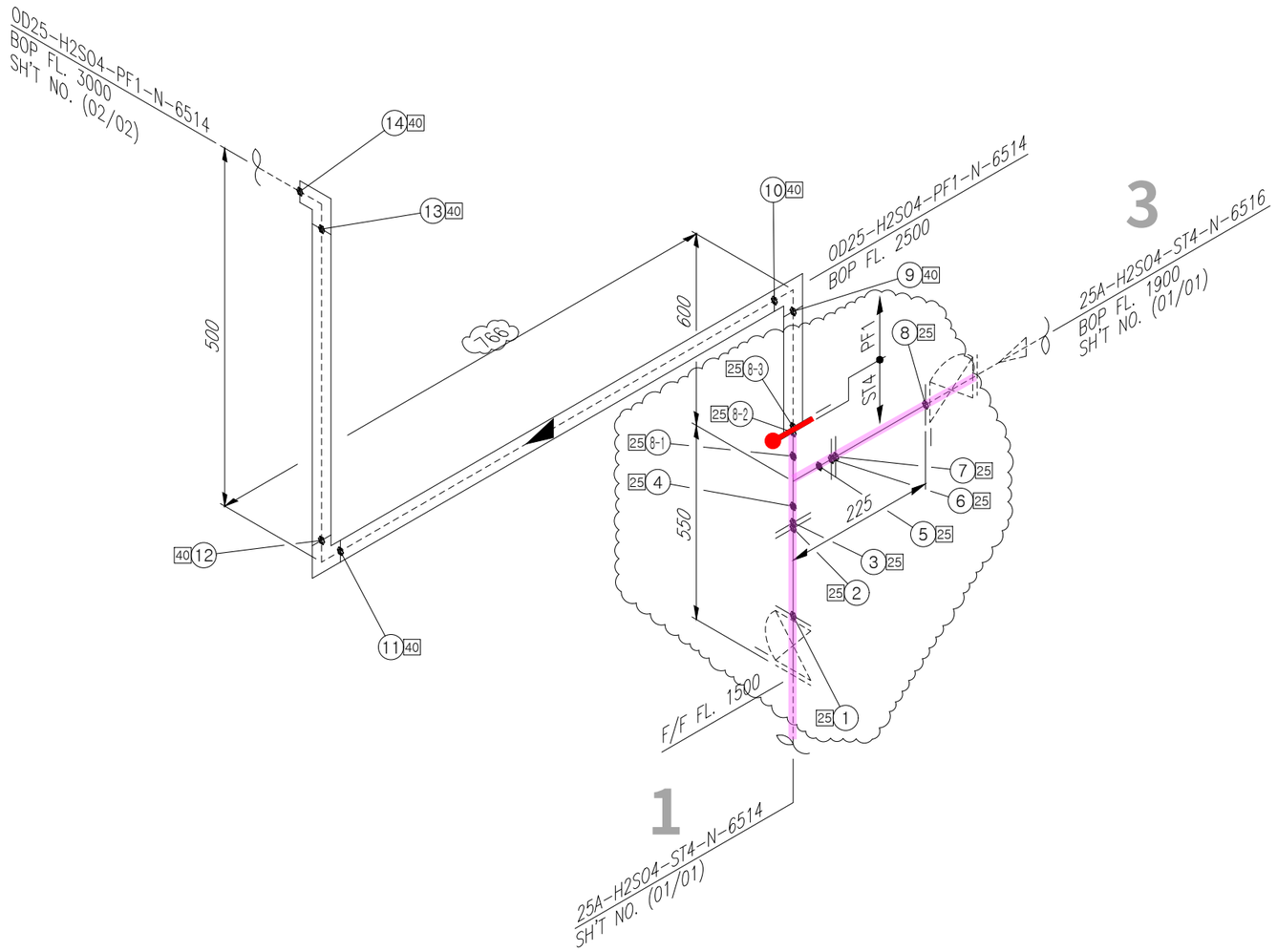
28

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14		
	THICK.	-	DESIGN	1.0	60	UNIT	4FA		
			OPERATING	ATM	AMB	REF. PLAN DWG.	CJ-M15W14-PI-A30-040013-A		
						REF. P & ID	CJ-M15W14-PR-A20-000147		

REV.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래종	안남후	박동영

CONTRACTOR: SK ecoplant
PROJECT NAME: M15X Ph4 Project
COMMODITY: H2S04
JOB NO.: 22330D
LINE NO.: 25A-H2S04-ST4-N-6572
SH'T NO.: 01 / 01
REV.: 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



<NOTE>
1. TRAY & ANGLE,CHANNEL SUPPORT가 적용되는 배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

SK ecoplant

ITEM	SIZE	Q'TY	DESCRIPTION
		PF1	
P	OD25	1.9M	PIPE, PFA TUBE, PE, FERRULE, 1.57MM TH'K
P3	40A	1.9M	PIPE, CLEAR PVC, PE, SOCKET BONDING, 4.0MM TH'K
E90A	40A	3EA	ELBOW 90 DEG, CLEAR PVC, PE, SOCKET BONDING, 4.0MM TH'K
F1	OD25	1EA	FLG, PFA STUB END, BACK RING(PTFE), LAP JOINT(STS 304)
		ST4	
P	25A	0.7M	PIPE, STS 304TP+PTFE Lined, ERW – SCH 10S
T	25A	1EA	TEE EQUAL, STSF304+PTFE Lined, ERW – SCH 10S
F	25A	4EA	FLG SO, KS 10K, RF, STS304+PTFE Lined
G	25A	4EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
B1	25A	4SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
SH	25A	3EA	SHIELD

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT SK hynix

29

INSULATION	CODE	N
	THICK.	–

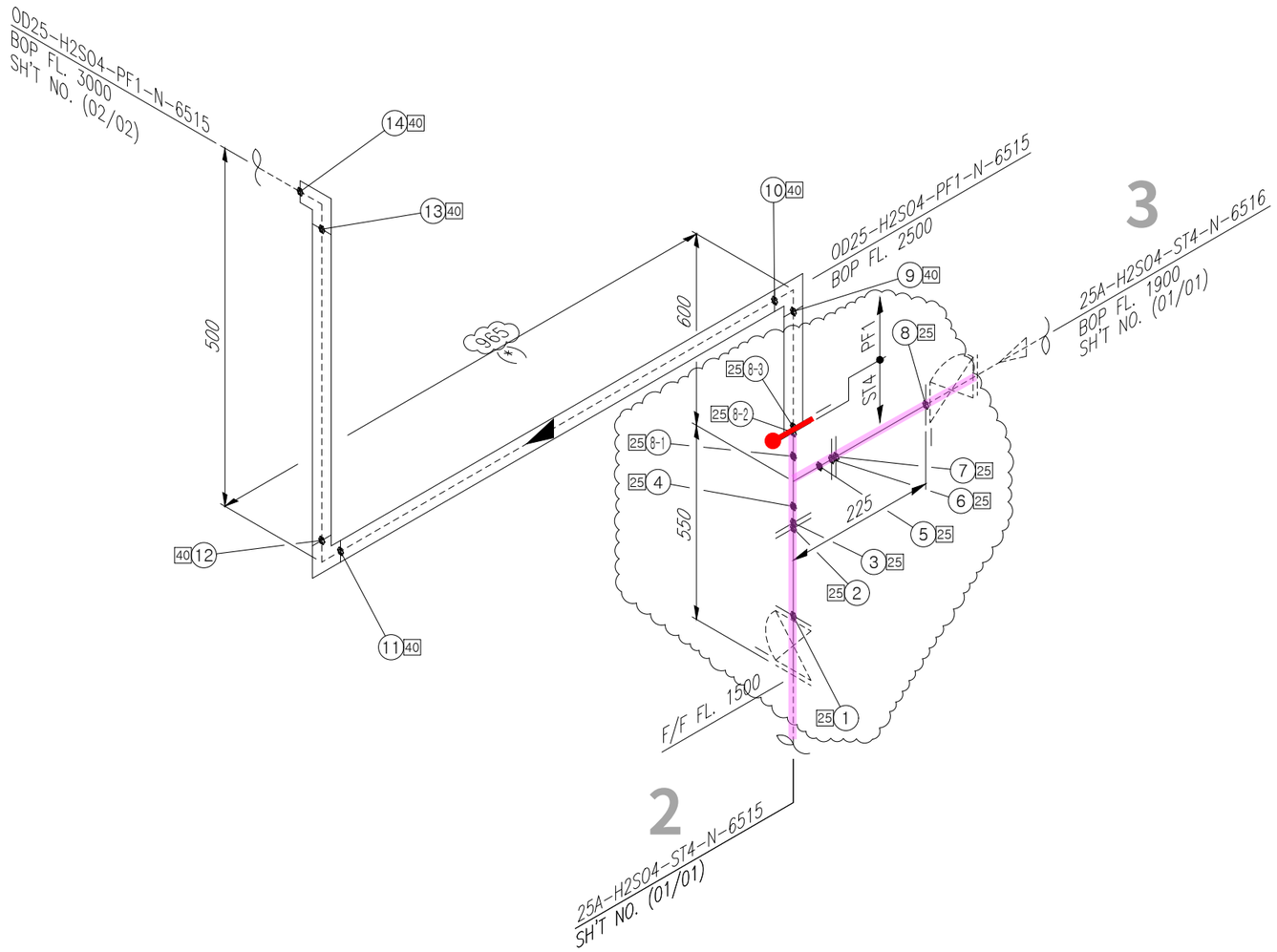
CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
DESIGN	6.7	60
OPERATING	6.1	AMB

AREA W14
UNIT 4FA
REF. PLAN DWG. CJ-M15W14-PI-A30-040012-A
REF. P & ID CJ-M15W14-PR-A20-000145

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래종	안남후	박동영

CONTRACTOR SK ecoplant
COMMODITY H2S04
LINE NO. 0D25-H2S04-PF1-N-6514
PROJECT NAME M15X Ph4 Project
JOB NO. 22330D
SH'T NO. 01 / 02
REV. 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



ITEM	SIZE	Q'TY	DESCRIPTION
		PF1	
P	OD25	2.1M	PIPE, PFA TUBE, PE, FERRULE, 1.57MM TH'K
P3	40A	2.1M	PIPE, CLEAR PVC, PE, SOCKET BONDING, 4.0MM TH'K
E90A	40A	3EA	ELBOW 90 DEG, CLEAR PVC, PE, SOCKET BONDING, 4.0MM TH'K
F1	OD25	1EA	FLG, PFA STUB END, BACK RING(PTFE), LAP JOINT(STS 304)
		ST4	
P	25A	0.7M	PIPE, STS 304TP+PTFE Lined, ERW – SCH 10S
T	25A	1EA	TEE EQUAL, STSF304+PTFE Lined, ERW – SCH 10S
F	25A	4EA	FLG SO, KS 10K, RF, STS304+PTFE Lined
G	25A	4EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
B1	25A	4SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
SH	25A	3EA	SHIELD

<NOTE>
1. TRAY & ANGLE,CHANNEL SUPPORT가 적용되는 배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N
	THICK.	–

CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
DESIGN	6.7	60
OPERATING	6.1	AMB

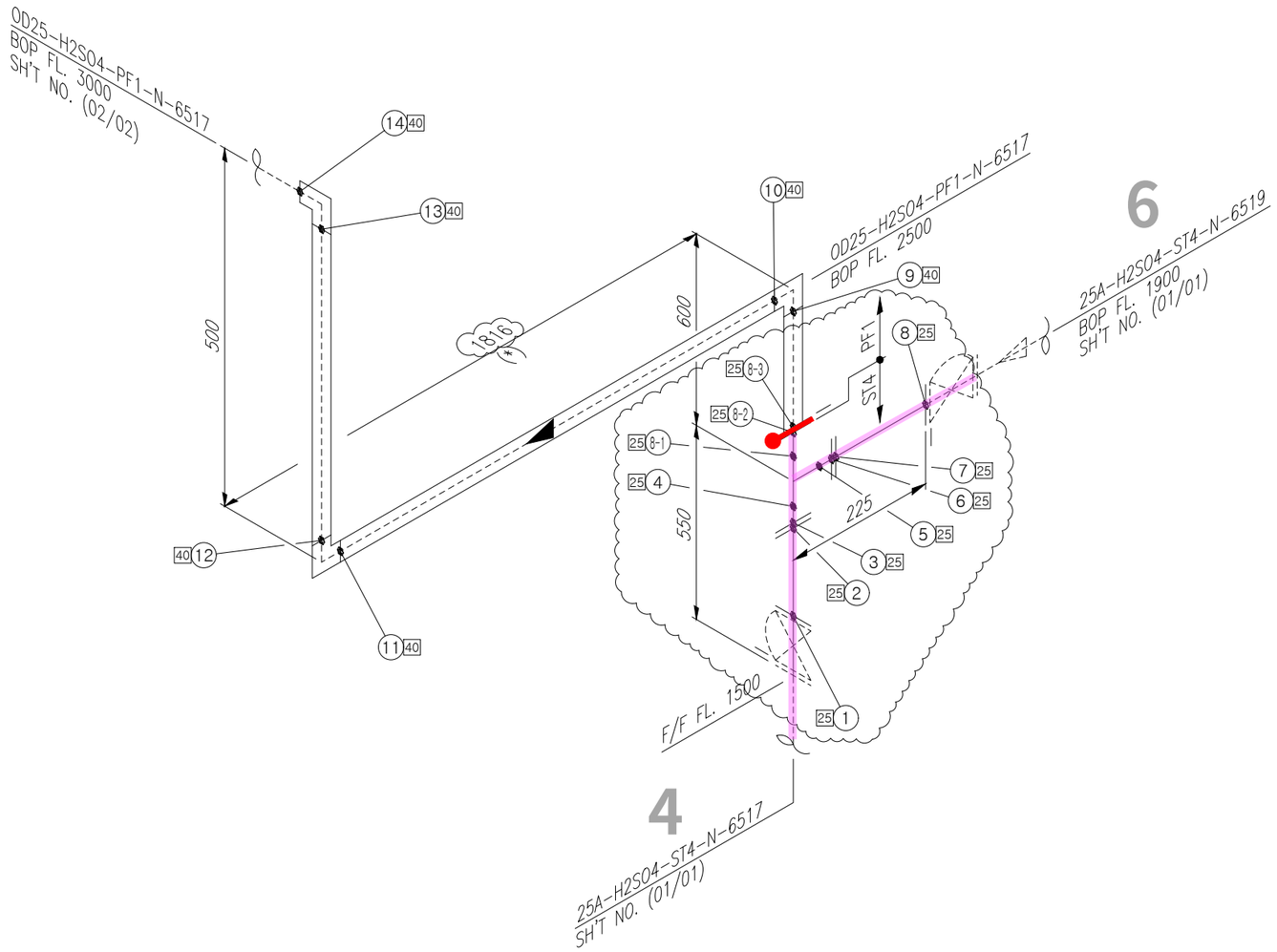
AREA W14
UNIT 4FA
REF. PLAN DWG. CJ-M15W14-PI-A30-040012-A
REF. P & ID CJ-M15W14-PR-A20-000145

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래종	안남후	박동영

CLIENT SK hynix
30

CONTRACTOR SK ecoplant
PROJECT NAME M15X Ph4 Project
COMMODITY H2SO4
JOB NO. 22330D
LINE NO. 0D25-H2SO4-PF1-N-6515
SH'T NO. 01 / 02
REV. 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



ITEM	SIZE	Q'TY	DESCRIPTION
		PF1	
P	OD25	3.0M	PIPE, PFA TUBE, PE, FERRULE, 1.57MM TH'K
P3	40A	3.0M	PIPE, CLEAR PVC, PE, SOCKET BONDING, 4.0MM TH'K
E90A	40A	3EA	ELBOW 90 DEG, CLEAR PVC, PE, SOCKET BONDING, 4.0MM TH'K
F1	OD25	1EA	FLG, PFA STUB END, BACK RING(PTFE), LAP JOINT(STS 304)
		ST4	
P	25A	0.7M	PIPE, STS 304TP+PTFE Lined, ERW – SCH 10S
T	25A	1EA	TEE EQUAL, STSF304+PTFE Lined, ERW – SCH 10S
F	25A	4EA	FLG SO, KS 10K, RF, STS304+PTFE Lined
G	25A	4EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
B1	25A	4SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
SH	25A	3EA	SHIELD

<NOTE>
1. TRAY & ANGLE,CHANNEL SUPPORT가 적용되는 배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N
	THICK.	–

CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
DESIGN	6.7	60
OPERATING	6.1	AMB

AREA W14
UNIT 4FA
REF. PLAN DWG. CJ-M15W14-PI-A30-040013-A
REF. P & ID CJ-M15W14-PR-A20-000145

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래종	안남후	박동영

CLIENT SK hynix

31

CONTRACTOR SK ecoplant

PROJECT NAME M15X Ph4 Project

COMMODITY H2SO4

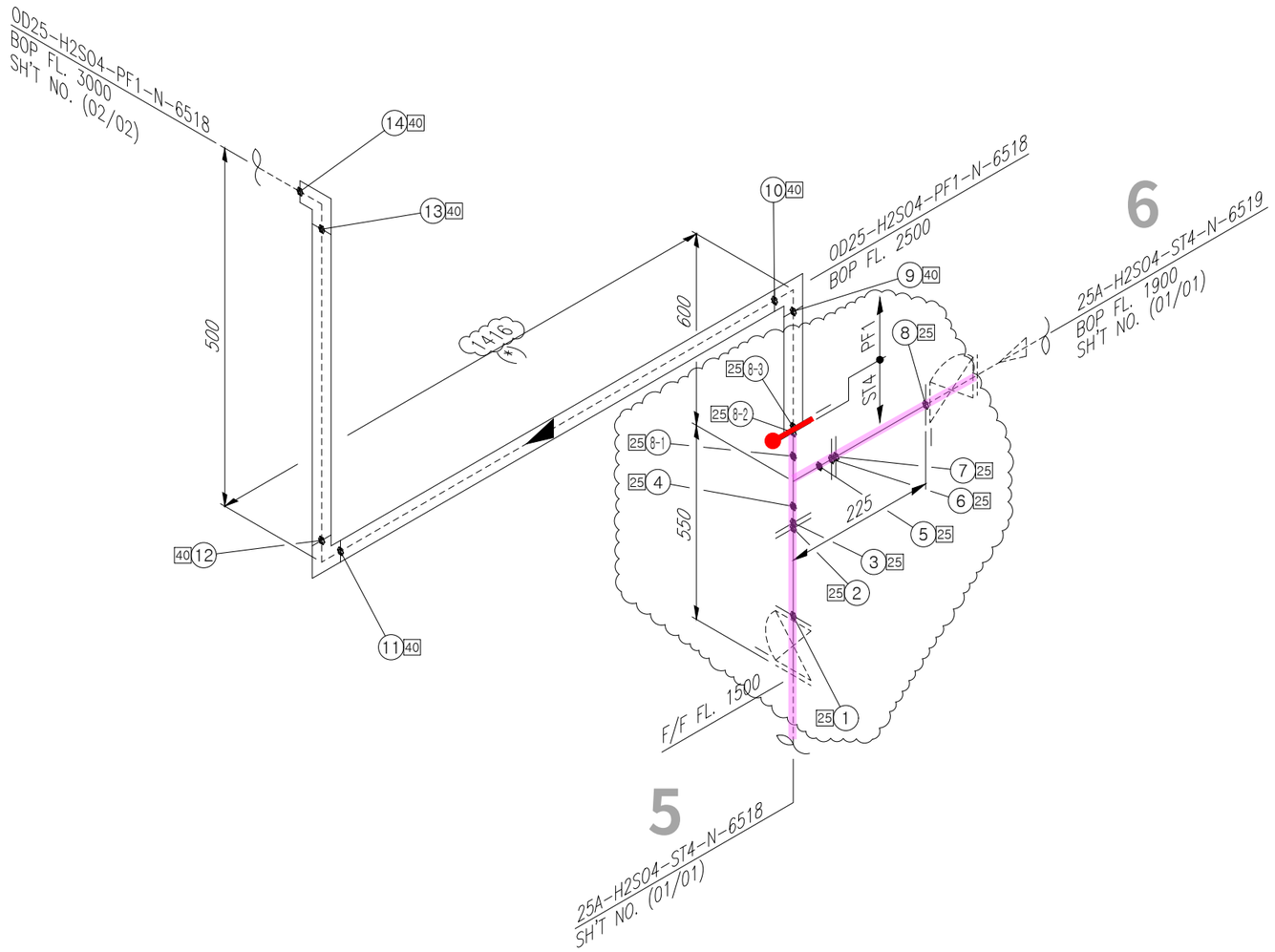
JOB NO. 22330D

LINE NO. 0D25-H2SO4-PF1-N-6517

SH'T NO. 01 / 02

REV. 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



ITEM	SIZE	Q'TY	DESCRIPTION
		PF1	
P	OD25	2.6M	PIPE, PFA TUBE, PE, FERRULE, 1.57MM TH'K
P3	40A	2.6M	PIPE, CLEAR PVC, PE, SOCKET BONDING, 4.0MM TH'K
E90A	40A	3EA	ELBOW 90 DEG, CLEAR PVC, PE, SOCKET BONDING, 4.0MM TH'K
F1	OD25	1EA	FLG, PFA STUB END, BACK RING(PTFE), LAP JOINT(STS 304)
		ST4	
P	25A	0.7M	PIPE, STS 304TP+PTFE Lined, ERW – SCH 10S
T	25A	1EA	TEE EQUAL, STSF304+PTFE Lined, ERW – SCH 10S
F	25A	4EA	FLG SO, KS 10K, RF, STS304+PTFE Lined
G	25A	4EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
B1	25A	4SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
SH	25A	3EA	SHIELD

<NOTE>
1. TRAY & ANGLE,CHANNEL SUPPORT가 적용되는 배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N
	THICK.	–

CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
DESIGN	6.7	60
OPERATING	6.1	AMB

AREA W14
UNIT 4FA
REF. PLAN DWG. CJ-M15W14-PI-A30-040013-A
REF. P & ID CJ-M15W14-PR-A20-000145

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래종	안남후	박동영

CLIENT SK hynix

32

CONTRACTOR SK ecoplant

PROJECT NAME M15X Ph4 Project

COMMODITY H2SO4

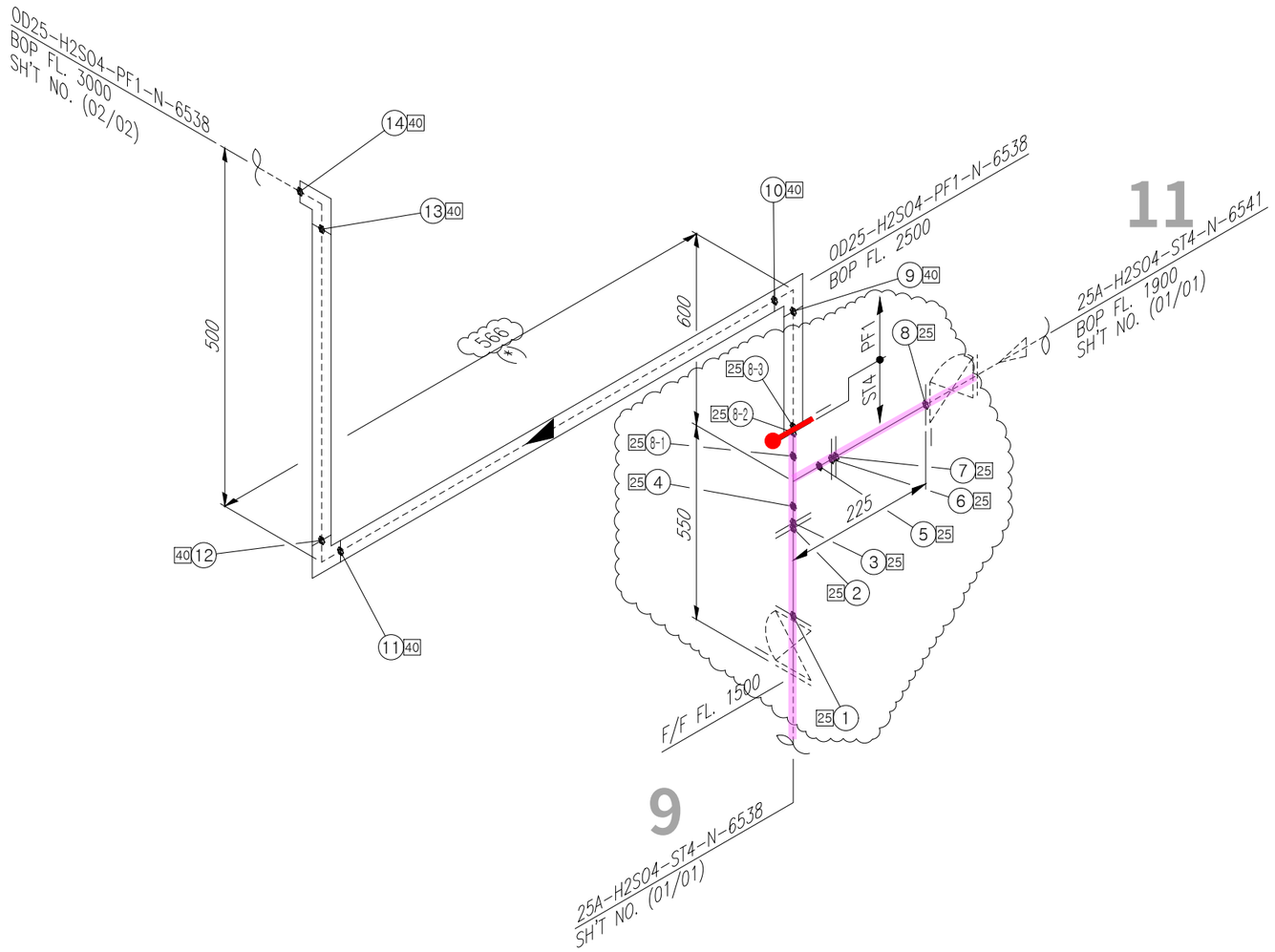
JOB NO. 22330D

LINE NO. 0D25-H2SO4-PF1-N-6518

SH'T NO. 01 / 02

REV. 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



ITEM	SIZE	Q'TY	DESCRIPTION
		PF1	
P	OD25	1.7M	PIPE, PFA TUBE, PE, FERRULE, 1.57MM TH'K
P3	40A	1.7M	PIPE, CLEAR PVC, PE, SOCKET BONDING, 4.0MM TH'K
E90A	40A	3EA	ELBOW 90 DEG, CLEAR PVC, PE, SOCKET BONDING, 4.0MM TH'K
F1	OD25	1EA	FLG, PFA STUB END, BACK RING(PTFE), LAP JOINT(STS 304)
		ST4	
P	25A	0.7M	PIPE, STS 304TP+PTFE Lined, ERW – SCH 10S
T	25A	1EA	TEE EQUAL, STSF304+PTFE Lined, ERW – SCH 10S
F	25A	4EA	FLG SO, KS 10K, RF, STS304+PTFE Lined
G	25A	4EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
B1	25A	4SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
SH	25A	3EA	SHIELD

<NOTE>
1. TRAY & ANGLE,CHANNEL SUPPORT가 적용되는 배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N
	THICK.	–

CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
DESIGN	6.7	60
OPERATING	6.1	AMB

AREA W14
UNIT 4FA
REF. PLAN DWG. CJ-M15W14-PI-A30-040013-A
REF. P & ID CJ-M15W14-PR-A20-000145

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래종	안남후	박동영

CLIENT SK hynix

33

CONTRACTOR SK ecoplant

PROJECT NAME M15X Ph4 Project

COMMODITY H2SO4

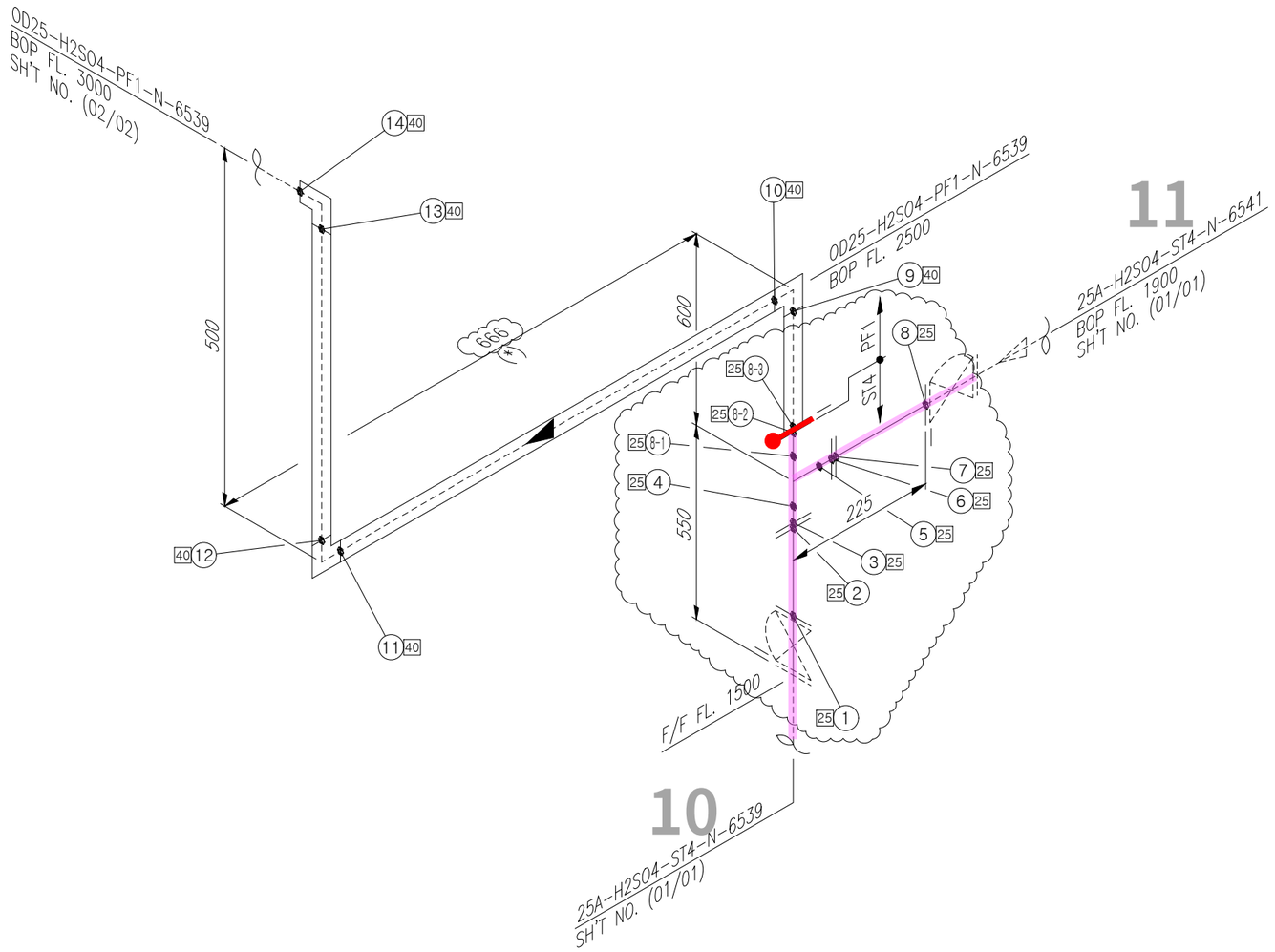
JOB NO. 22330D

LINE NO. 0D25-H2SO4-PF1-N-6538

SH'T NO. 01 / 02

REV. 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



<NOTE>
1. TRAY & ANGLE,CHANNEL SUPPORT가 적용되는 배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

ITEM	SIZE	Q'TY	DESCRIPTION
		PF1	
P	OD25	1.8M	PIPE, PFA TUBE, PE, FERRULE, 1.57MM TH'K
P3	40A	1.8M	PIPE, CLEAR PVC, PE, SOCKET BONDING, 4.0MM TH'K
E90A	40A	3EA	ELBOW 90 DEG, CLEAR PVC, PE, SOCKET BONDING, 4.0MM TH'K
F1	OD25	1EA	FLG, PFA STUB END, BACK RING(PTFE), LAP JOINT(STS 304)
		ST4	
P	25A	0.7M	PIPE, STS 304TP+PTFE Lined, ERW – SCH 10S
T	25A	1EA	TEE EQUAL, STSF304+PTFE Lined, ERW – SCH 10S
F	25A	4EA	FLG SO, KS 10K, RF, STS304+PTFE Lined
G	25A	4EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
B1	25A	4SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
SH	25A	3EA	SHIELD

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N
	THICK.	–

CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
DESIGN	6.7	60
OPERATING	6.1	AMB

AREA W14
UNIT 4FA
REF. PLAN DWG. CJ-M15W14-PI-A30-040013-A
REF. P & ID CJ-M15W14-PR-A20-000145

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래종	안남후	박동영

CLIENT SK hynix

CONTRACTOR SK ecoplant

PROJECT NAME M15X Ph4 Project

COMMODITY H2SO4

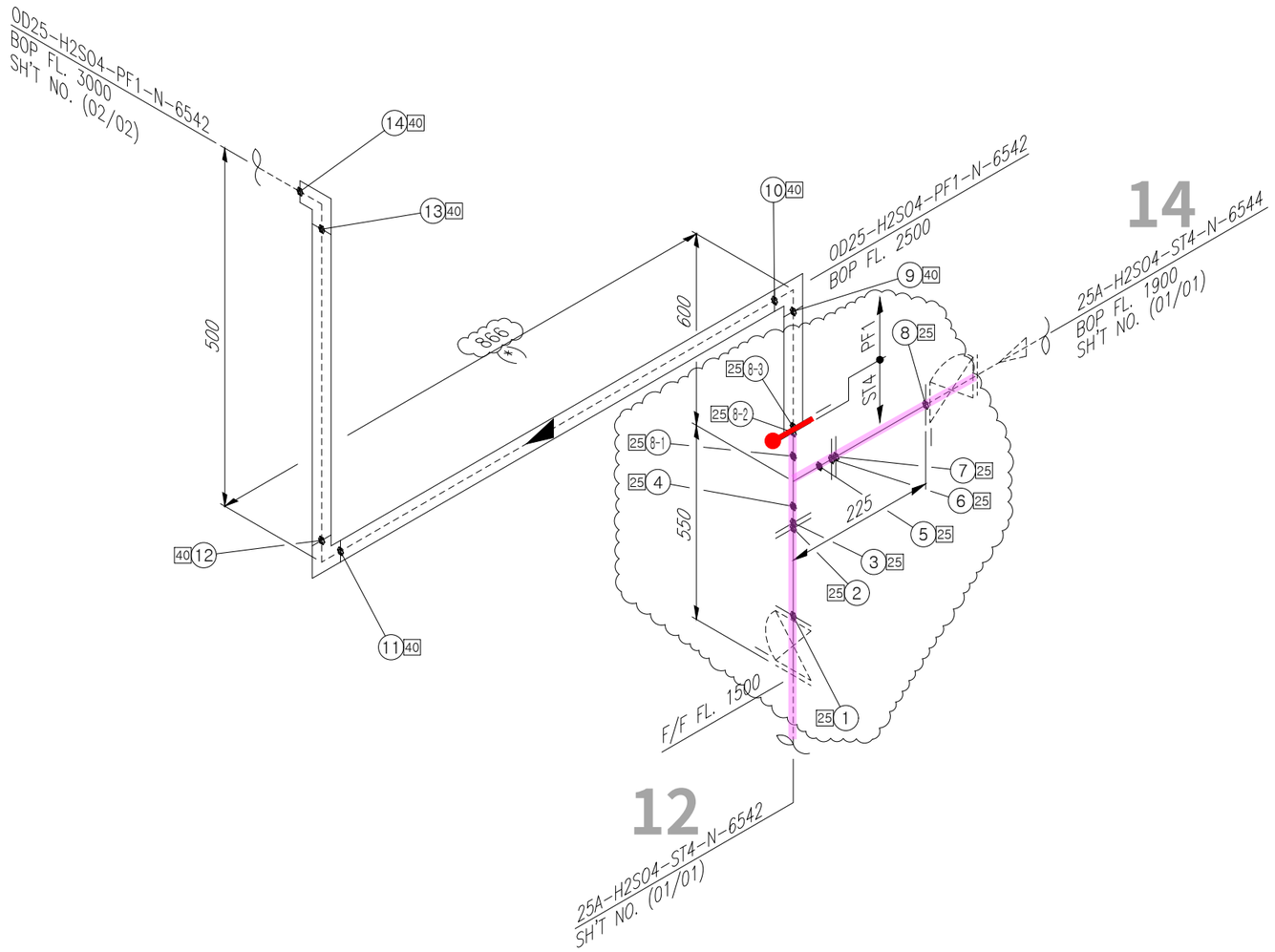
JOB NO. 22330D

LINE NO. 0D25-H2SO4-PF1-N-6539

SH'T NO. 01 / 02

REV. 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



<NOTE>

1. TRAY & ANGLE,CHANNEL SUPPORT가 적용되는 배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

ITEM	SIZE	Q'TY	DESCRIPTION
		PF1	
P	OD25	2.0M	PIPE, PFA TUBE, PE, FERRULE, 1.57MM TH'K
P3	40A	2.0M	PIPE, CLEAR PVC, PE, SOCKET BONDING, 4.0MM TH'K
E90A	40A	3EA	ELBOW 90 DEG, CLEAR PVC, PE, SOCKET BONDING, 4.0MM TH'K
F1	OD25	1EA	FLG, PFA STUB END, BACK RING(PTFE), LAP JOINT(STS 304)
		ST4	
P	25A	0.7M	PIPE, STS 304TP+PTFE Lined, ERW – SCH 10S
T	25A	1EA	TEE EQUAL, STSF304+PTFE Lined, ERW – SCH 10S
F	25A	4EA	FLG SO, KS 10K, RF, STS304+PTFE Lined
G	25A	4EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
B1	25A	4SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
SH	25A	3EA	SHIELD

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N
	THICK.	–

CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
DESIGN	6.7	60
OPERATING	6.1	AMB

AREA W14
UNIT 4FA
REF. PLAN DWG. CJ-M15W14-PI-A30-040013-A
REF. P & ID CJ-M15W14-PR-A20-000145

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래종	안남후	박동영

CLIENT SK hynix

35

CONTRACTOR SK ecoplant

PROJECT NAME M15X Ph4 Project

COMMODITY H2SO4

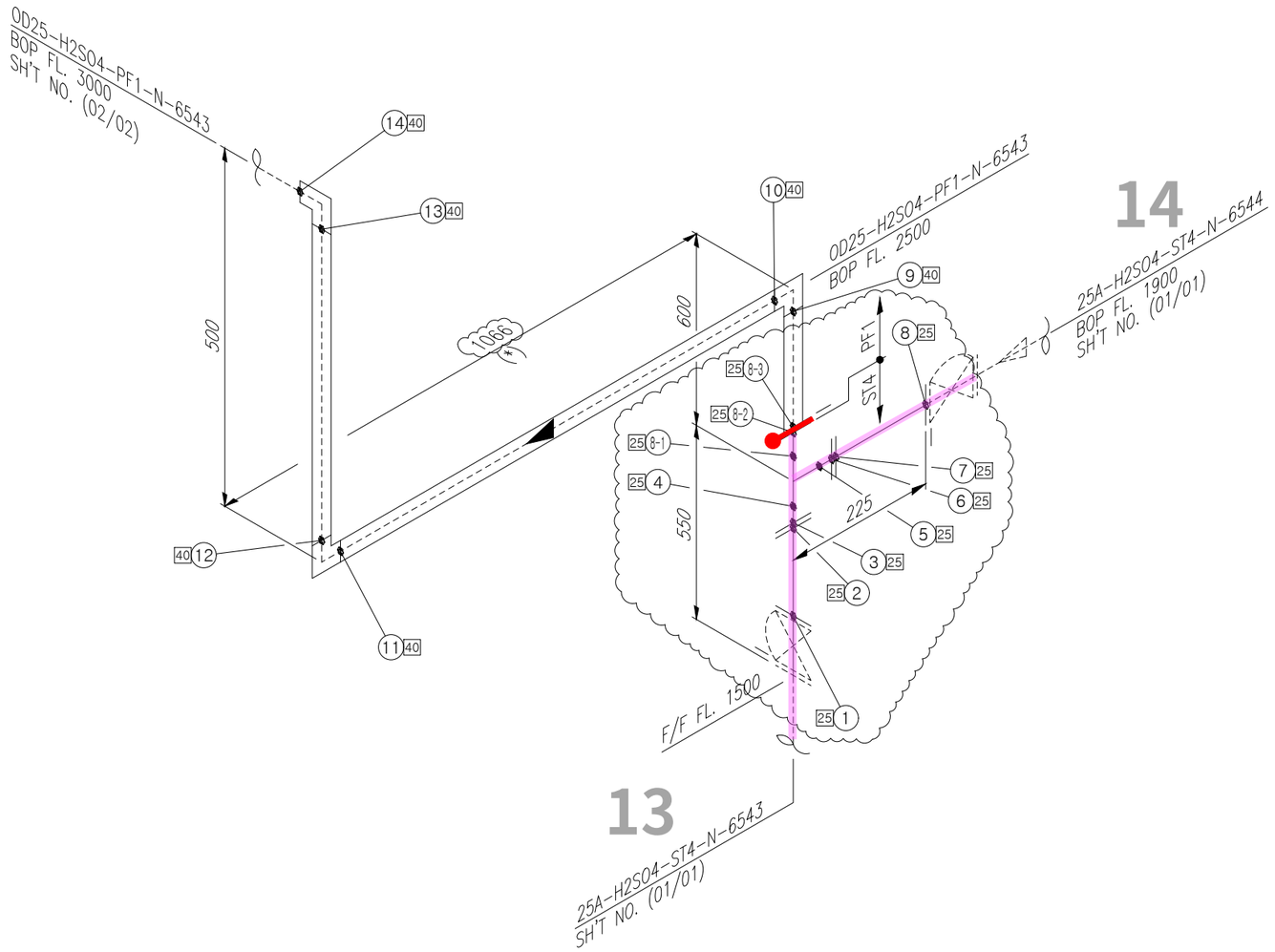
JOB NO. 22330D

LINE NO. 0D25-H2SO4-PF1-N-6542

SH'T NO. 01 / 02

REV. 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



<NOTE>

1. TRAY & ANGLE,CHANNEL SUPPORT가 적용되는 배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

SK ecoplant

ITEM	SIZE	Q'TY	DESCRIPTION
		PF1	
P	OD25	2.2M	PIPE, PFA TUBE, PE, FERRULE, 1.57MM TH'K
P3	40A	2.2M	PIPE, CLEAR PVC, PE, SOCKET BONDING, 4.0MM TH'K
E90A	40A	3EA	ELBOW 90 DEG, CLEAR PVC, PE, SOCKET BONDING, 4.0MM TH'K
F1	OD25	1EA	FLG, PFA STUB END, BACK RING(PTFE), LAP JOINT(STS 304)
		ST4	
P	25A	0.7M	PIPE, STS 304TP+PTFE Lined, ERW – SCH 10S
T	25A	1EA	TEE EQUAL, STSF304+PTFE Lined, ERW – SCH 10S
F	25A	4EA	FLG SO, KS 10K, RF, STS304+PTFE Lined
G	25A	4EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
B1	25A	4SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
SH	25A	3EA	SHIELD

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1 * STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7 * GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH * STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT SK hynix

36

INSULATION	CODE	N
	THICK.	–

CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
DESIGN	6.7	60
OPERATING	6.1	AMB

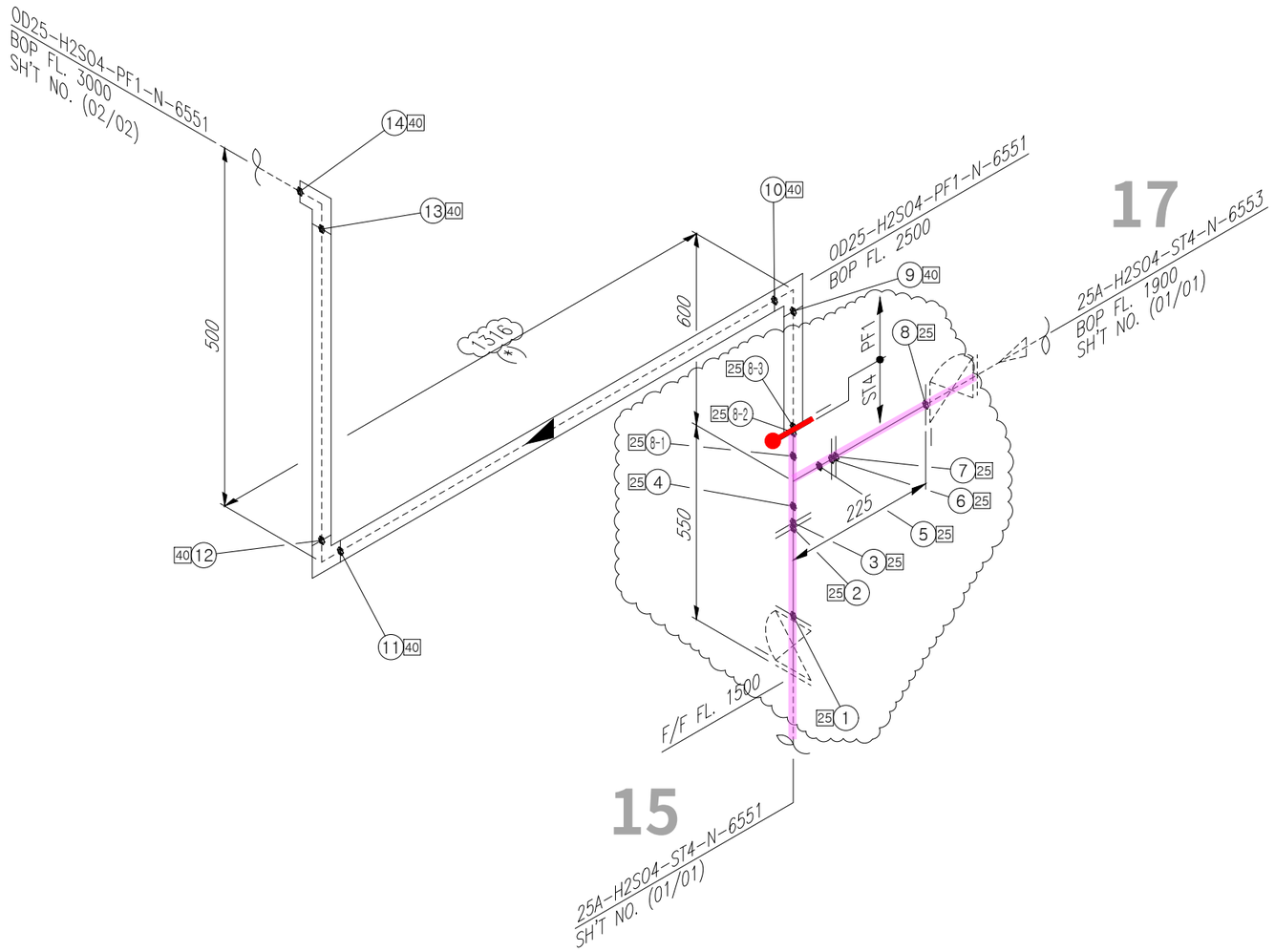
AREA W14
UNIT 4FA
REF. PLAN DWG. CJ-M15W14-PI-A30-040013-A
REF. P & ID CJ-M15W14-PR-A20-000145

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래종	안남후	박동영

CONTRACTOR SK ecoplant
COMMODITY H2SO4
LINE NO. OD25-H2SO4-PF1-N-6543

PROJECT NAME M15X Ph4 Project
JOB NO. 22330D
SH'T NO. 01 / 02
REV. 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



<NOTE>
1. TRAY & ANGLE,CHANNEL SUPPORT가 적용되는 배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

SK ecoplant

ITEM	SIZE	Q'TY	DESCRIPTION
		PF1	
P	OD25	2.5M	PIPE, PFA TUBE, PE, FERRULE, 1.57MM TH'K
P3	40A	2.5M	PIPE, CLEAR PVC, PE, SOCKET BONDING, 4.0MM TH'K
E90A	40A	3EA	ELBOW 90 DEG, CLEAR PVC, PE, SOCKET BONDING, 4.0MM TH'K
F1	OD25	1EA	FLG, PFA STUB END, BACK RING(PTFE), LAP JOINT(STS 304)
		ST4	
P	25A	0.7M	PIPE, STS 304TP+PTFE Lined, ERW – SCH 10S
T	25A	1EA	TEE EQUAL, STSF304+PTFE Lined, ERW – SCH 10S
F	25A	4EA	FLG SO, KS 10K, RF, STS304+PTFE Lined
G	25A	4EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
B1	25A	4SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
SH	25A	3EA	SHIELD

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1 * STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7 * GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH * STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT SK hynix

37

INSULATION	CODE	N
	THICK.	–

CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
DESIGN	6.7	60
OPERATING	6.1	AMB

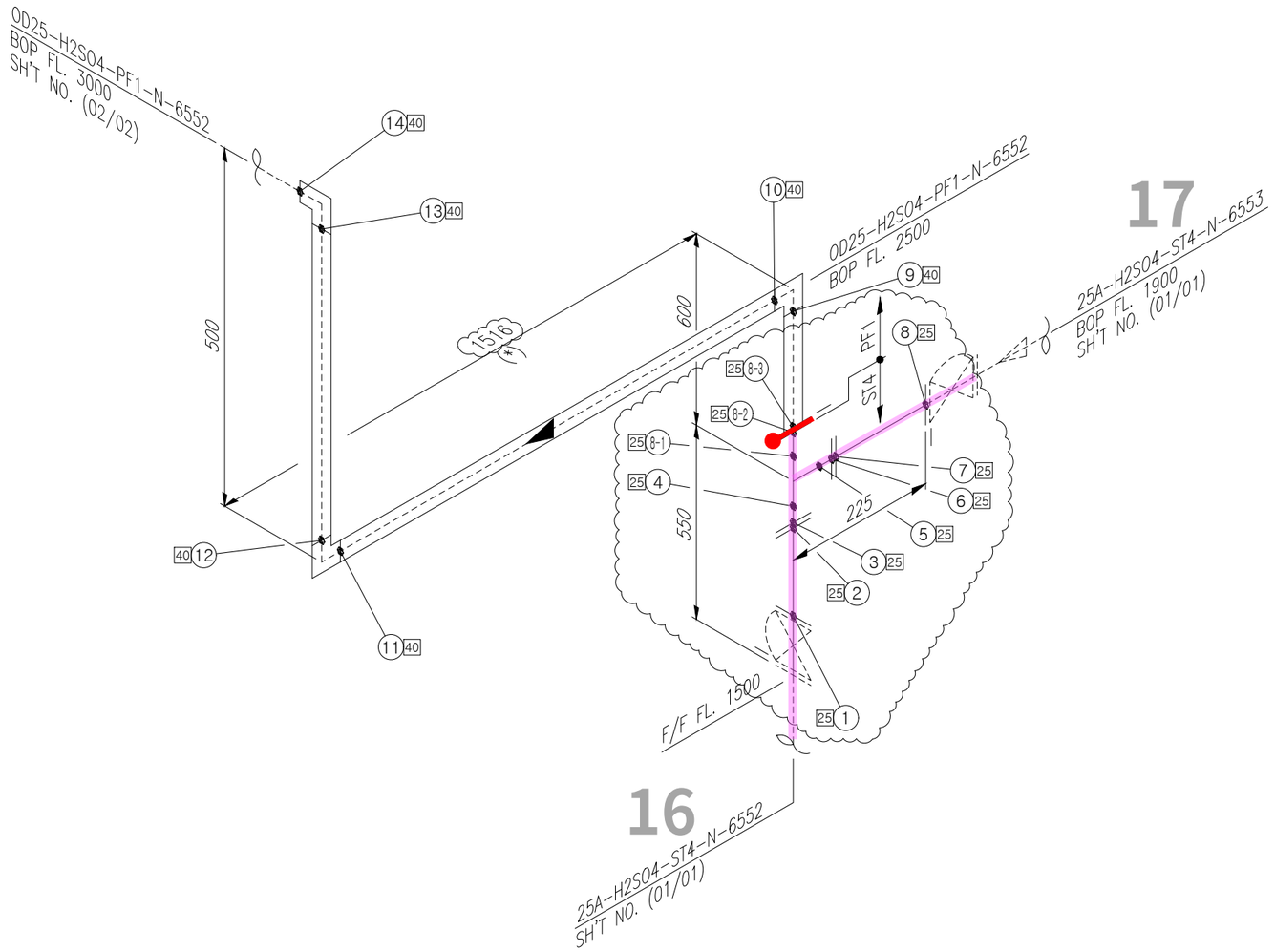
AREA W14
UNIT 4FA
REF. PLAN DWG. CJ-M15W14-PI-A30-040013-A
REF. P & ID CJ-M15W14-PR-A20-000145

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래종	안남후	박동영

CONTRACTOR SK ecoplant
COMMODITY H2S04
LINE NO. 0D25-H2S04-PF1-N-6551

PROJECT NAME M15X Ph4 Project
JOB NO. 22330D
SH'T NO. 01 / 02
REV. 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



<NOTE>
1. TRAY & ANGLE,CHANNEL SUPPORT가 적용되는 배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

ITEM	SIZE	Q'TY	DESCRIPTION
		PF1	
P	OD25	2.7M	PIPE, PFA TUBE, PE, FERRULE, 1.57MM TH'K
P3	40A	2.7M	PIPE, CLEAR PVC, PE, SOCKET BONDING, 4.0MM TH'K
E90A	40A	3EA	ELBOW 90 DEG, CLEAR PVC, PE, SOCKET BONDING, 4.0MM TH'K
F1	OD25	1EA	FLG, PFA STUB END, BACK RING(PTFE), LAP JOINT(STS 304)
		ST4	
P	25A	0.7M	PIPE, STS 304TP+PTFE Lined, ERW – SCH 10S
T	25A	1EA	TEE EQUAL, STSF304+PTFE Lined, ERW – SCH 10S
F	25A	4EA	FLG SO, KS 10K, RF, STS304+PTFE Lined
G	25A	4EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
B1	25A	4SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
SH	25A	3EA	SHIELD

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N
	THICK.	–

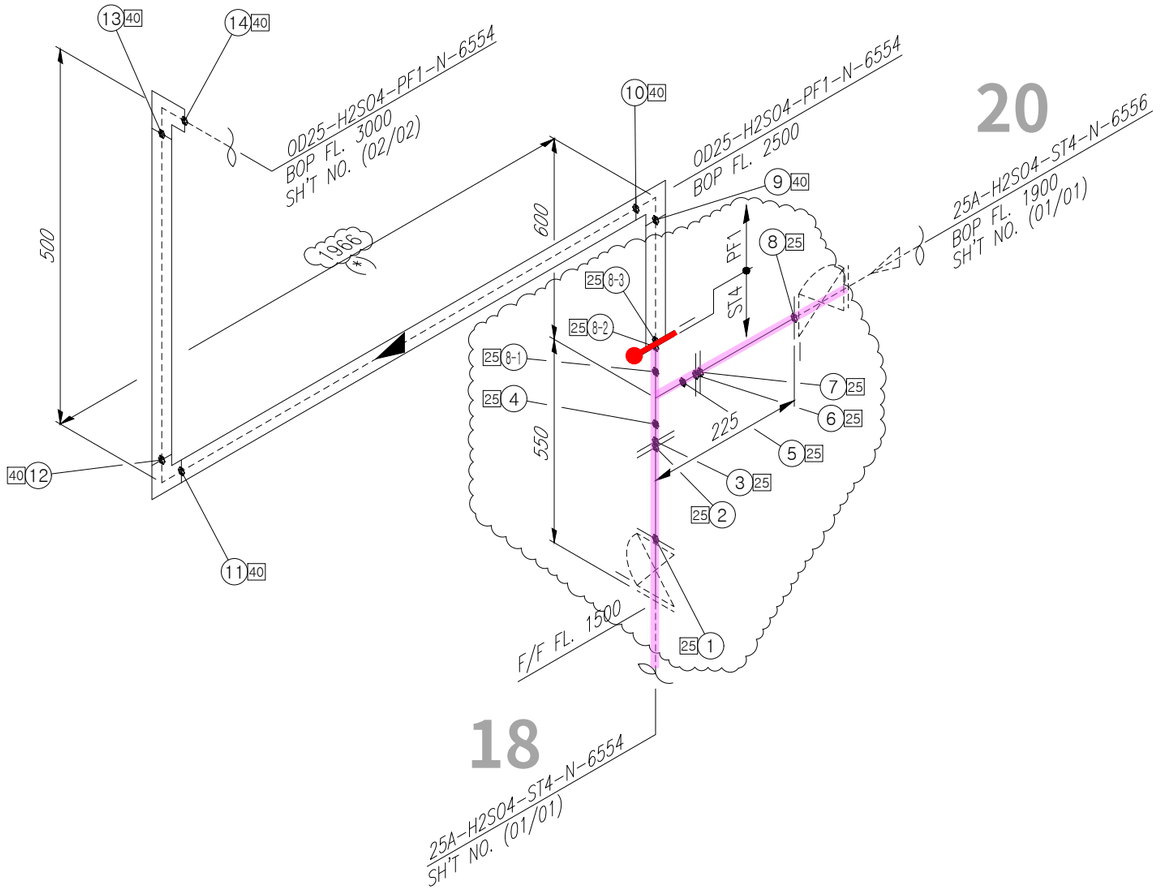
CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
DESIGN	6.7	60
OPERATING	6.1	AMB

AREA W14
UNIT 4FA
REF. PLAN DWG. CJ-M15W14-PI-A30-040013-A
REF. P & ID CJ-M15W14-PR-A20-000145

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래종	안남후	박동영

CLIENT SK hynix
38
CONTRACTOR SK ecoplant
PROJECT NAME M15X Ph4 Project
COMMODITY H2S04
JOB NO. 22330D
LINE NO. 0D25-H2S04-PF1-N-6552
SH'T NO. 01 / 02
REV. 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



<NOTE>
1. TRAY & ANGLE,CHANNEL SUPPORT가 적용되는 배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

SK ecoplant

ITEM	SIZE	Q'TY	DESCRIPTION
		PF1	
P	OD25	2.7M	PIPE, PFA TUBE, PE, FERRULE, 1.57MM TH'K
P3	40A	2.7M	PIPE, CLEAR PVC, PE, SOCKET BONDING, 4.0MM TH'K
E90A	40A	3EA	ELBOW 90 DEG, CLEAR PVC, PE, SOCKET BONDING, 4.0MM TH'K
F1	OD25	1EA	FLG, PFA STUB END, BACK RING(PTFE), LAP JOINT(STS 304)
		ST4	
P	25A	0.7M	PIPE, STS 304TP+PTFE Lined, ERW – SCH 10S
T	25A	1EA	TEE EQUAL, STSF304+PTFE Lined, ERW – SCH 10S
F	25A	4EA	FLG SO, KS 10K, RF, STS304+PTFE Lined
G	25A	4EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
B1	25A	4SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
SH	25A	3EA	SHIELD

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT SK hynix

39

INSULATION	CODE	N
	THICK.	–

CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
DESIGN	6.7	60
OPERATING	6.1	AMB

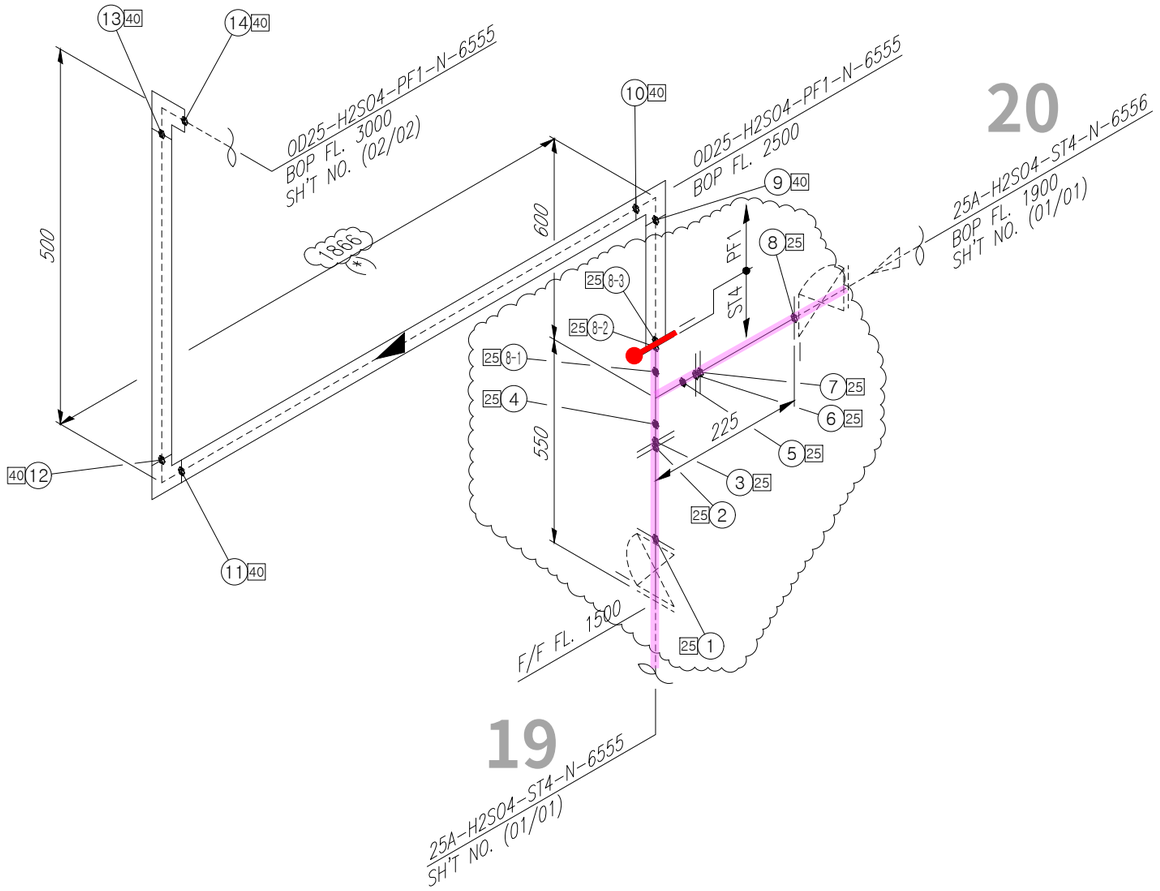
AREA W14
UNIT 4FA
REF. PLAN DWG. CJ-M15W14-PI-A30-040013-A
REF. P & ID CJ-M15W14-PR-A20-000145

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래종	안남후	박동영

CONTRACTOR SK ecoplant
COMMODITY H2S04
LINE NO. 0D25-H2S04-PF1-N-6554

PROJECT NAME M15X Ph4 Project
JOB NO. 22330D
SH'T NO. 01 / 02
REV. 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



<NOTE>
1. TRAY & ANGLE,CHANNEL SUPPORT가 적용되는 배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

ITEM	SIZE	Q'TY	DESCRIPTION
		PF1	
P	OD25	3.0M	PIPE, PFA TUBE, PE, FERRULE, 1.57MM TH'K
P3	40A	3.0M	PIPE, CLEAR PVC, PE, SOCKET BONDING, 4.0MM TH'K
E90A	40A	3EA	ELBOW 90 DEG, CLEAR PVC, PE, SOCKET BONDING, 4.0MM TH'K
F1	OD25	1EA	FLG, PFA STUB END, BACK RING(PTFE), LAP JOINT(STS 304)
		ST4	
P	25A	0.7M	PIPE, STS 304TP+PTFE Lined, ERW – SCH 10S
T	25A	1EA	TEE EQUAL, STSF304+PTFE Lined, ERW – SCH 10S
F	25A	4EA	FLG SO, KS 10K, RF, STS304+PTFE Lined
G	25A	4EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
B1	25A	4SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
SH	25A	3EA	SHIELD

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N
	THICK.	–

CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
DESIGN	6.7	60
OPERATING	6.1	AMB

AREA W14
UNIT 4FA
REF. PLAN DWG. CJ-M15W14-PI-A30-040013-A
REF. P & ID CJ-M15W14-PR-A20-000145

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래종	안남후	박동영

CLIENT SK hynix

40

CONTRACTOR SK ecoplant

PROJECT NAME M15X Ph4 Project

COMMODITY H2SO4

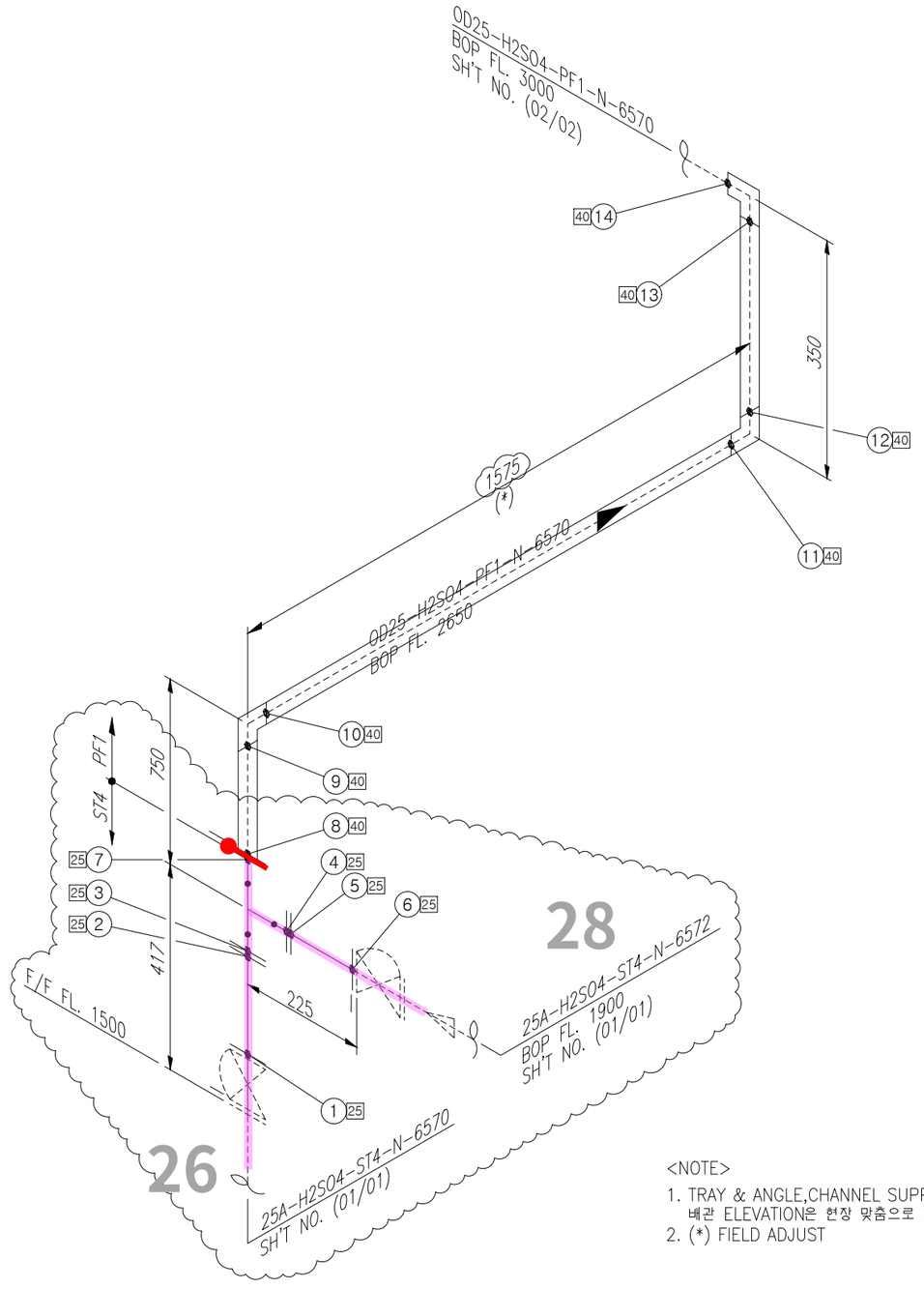
JOB NO. 22330D

LINE NO. 0D25-H2S04-PF1-N-6555

SH'T NO. 01 / 02

REV. 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



<NOTE>
1. TRAY & ANGLE,CHANNEL SUPPORT가 적용되는 배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

SK ecoplant

ITEM	SIZE	Q'TY	DESCRIPTION
		PF1	
P	OD25	2.7M	PIPE, PFA TUBE, PE, FERRULE, 1.57MM TH'K
P3	40A	2.7M	PIPE, CLEAR PVC, PE, SOCKET BONDING, 4.0MM TH'K
E90A	40A	3EA	ELBOW 90 DEG, CLEAR PVC, PE, SOCKET BONDING, 4.0MM TH'K
~~T1~~	~~OD25~~	1EA	~~TEE EQUAL, PE, UNION, BACK RING(PTFE)~~
~~TB~~	~~40A~~	1EA	~~TEE BOX~~
F1	OD25	1EA	FLG, PFA STUB END, BACK RING(PTFE), LAP JOINT(STS 304)
G1	OD25	1EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B1	OD25	1EA	HEX MACHINE BOLT & NYLON NUT :SM45C+HOT DIP GAL'V/W WASHERS
SC	40A	6EA	SOCKET, CLEAR PVC, PE, SOCKET BONDING, 4.0MM TH'K
		ST4	
P	25A	0.6M	PIPE, STS 304TP+PTFE Lined, ERW - SCH 10S, With FLG
T	25A	1EA	TEE EQUAL, STSF304+PTFE Lined, ERW - SCH 10S, With FLG
F	25A	4EA	FLG SO, KS 10K, RF, STS304+PTFE Lined
G	25A	5EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
B1	25A	5SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
SH	25A	5EA	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT SK hynix

41

INSULATION	CODE	N
	THICK.	-

CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
DESIGN	6.7	60
OPERATING	6.1	AMB

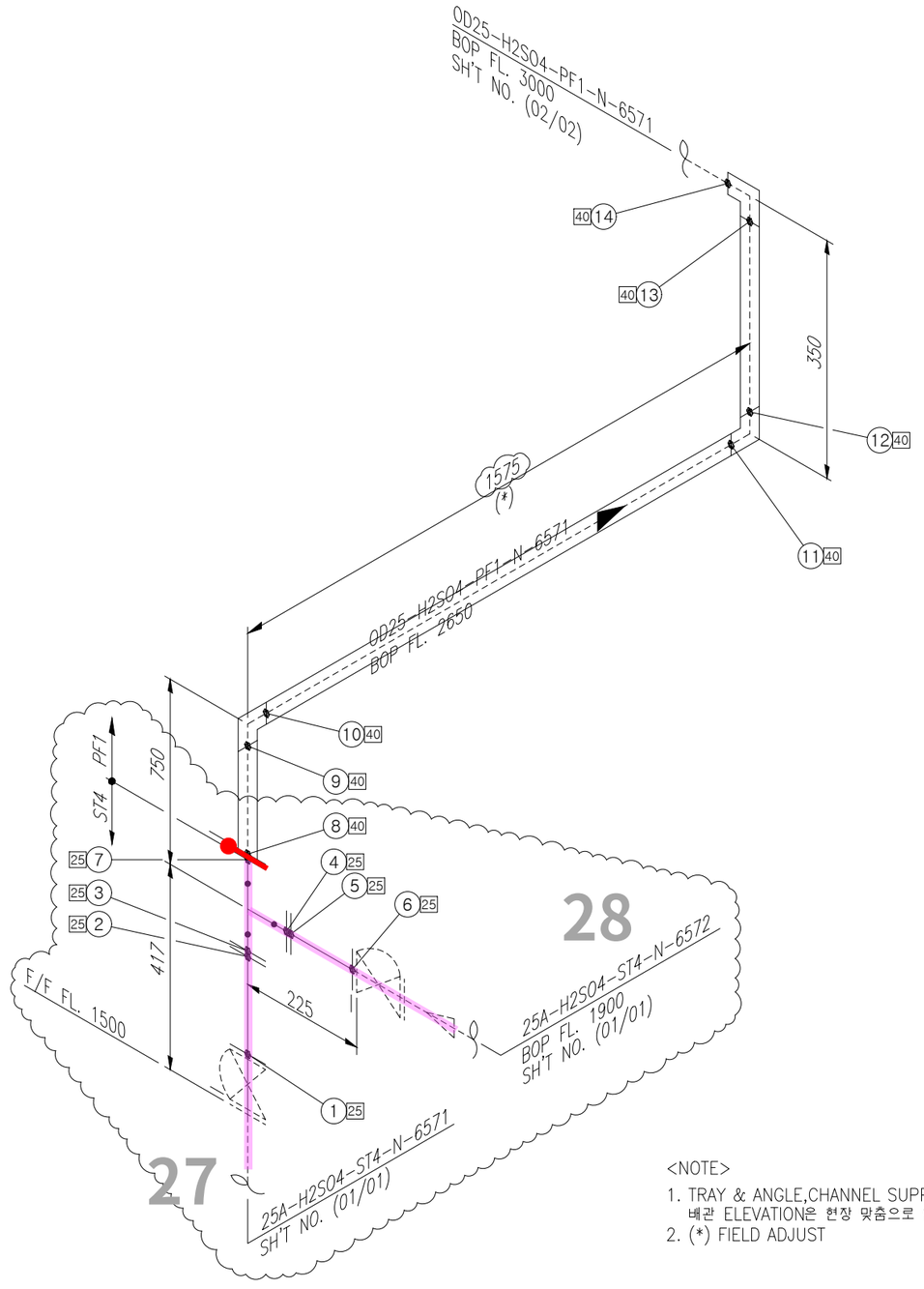
AREA W14
UNIT 4FA
REF. PLAN DWG. CJ-M15W14-PI-A30-040012-A
REF. P & ID CJ-M15W14-PR-A20-000145

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래종	안남후	박동영

CONTRACTOR SK ecoplant
COMMODITY H2SO4
LINE NO. 0D25-H2S04-PF1-N-6570

PROJECT NAME M15X Ph4 Project
JOB NO. 22330D
SH'T NO. 01 / 02
REV. 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



<NOTE>
1. TRAY & ANGLE,CHANNEL SUPPORT가 적용되는 배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

ITEM	SIZE	Q'TY	DESCRIPTION
		PF1	
P	OD25	2.7M	PIPE, PFA TUBE, PE, FERRULE, 1.57MM TH'K
P3	40A	2.7M	PIPE, CLEAR PVC, PE, SOCKET BONDING, 4.0MM TH'K
E90A	40A	3EA	ELBOW 90 DEG, CLEAR PVC, PE, SOCKET BONDING, 4.0MM TH'K
~~T1~~	~~OD25~~	1EA	~~TEE EQUAL, PE, UNION, BACK RING(PTFE)~~
~~TB~~	~~40A~~	1EA	~~TEE BOX~~
F1	OD25	1EA	FLG, PFA STUB END, BACK RING(PTFE), LAP JOINT(STS 304)
G1	OD25	1EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B1	OD25	1EA	HEX MACHINE BOLT & NYLON NUT :SM45C+HOT DIP GAL'V/W WASHERS
SC	40A	6EA	SOCKET, CLEAR PVC, PE, SOCKET BONDING, 4.0MM TH'K
		ST4	
P	25A	0.6M	PIPE, STS 304TP+PTFE Lined, ERW - SCH 10S, With FLG
T	25A	1EA	TEE EQUAL, STSF304+PTFE Lined, ERW - SCH 10S, With FLG
F	25A	4EA	FLG SO, KS 10K, RF, STS304+PTFE Lined
G	25A	5EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
B1	25A	5SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
SH	25A	5EA	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT SK hynix

42

INSULATION	CODE	N
	THICK.	-

CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
DESIGN	6.7	60
OPERATING	6.1	AMB

AREA	W14
UNIT	4FA
REF. PLAN DWG.	CJ-M15W14-PI-A30-040012-A
REF. P & ID	CJ-M15W14-PR-A20-000145

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래종	안남후	박동영

CONTRACTOR SK ecoplant

PROJECT NAME: M15X Ph4 Project

COMMODITY: H2SO4

JOB NO.: 22330D

LINE NO.: 0D25-H2S04-PF1-N-6571

SH'T NO.: 01 / 02

REV.: 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.